Time Is of the Essence: Experimental Evidence on the Effects of Job Application Timing*

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Abstract

This paper studies the role of job application timing in recruitment. Repurposing the correspondence experiment of Kline, Rose and Walters (2022), I show that, holding vacancy and resume content fixed, first-half experimental applications have a contact rate 1.16 percentage points higher, equivalent to 57 percent of the Black-White contact gap in the same setting. Each day earlier raises contact probability by 0.40 percentage points. The key mechanism is that firms often begin contacting while applications are still being received. Using randomized resume attributes, I instrument whether contact has already begun with the highest predicted contact probability among previously submitted applications. Stronger prior applications trigger earlier initiation, and IV estimates confirm a post-initiation contact penalty. The timing gradient is steeper in tighter labor markets. A sequential model separates applicant arrival rates from the shadow cost of delay and quantifies firms' tradeoff between earlier contact and the screening-value gains from waiting. Conditional on the firm contacting an experimental application, the highest-ranked application has not yet arrived in 51.6 percent of cases, yielding a 6.6 percent loss relative to the contact-initiating application's expected screening value. Together, the results provide the first large-scale causal evidence that firms' sequential contact decisions make queue position consequential within vacancies, revealing an employer-side margin through which labor market conditions shape who is considered.

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1 Introduction

Job applications arrive over time, but firms must decide when to stop waiting and begin contacting candidates. Waiting expands the applicant pool and may reveal a better match, while acting sooner shortens vacancy delay and can secure candidates before they become unavailable. The timing of contact is therefore part of the firm's recruiting policy: employers can respond to scarce applicants not only through wages, amenities, or recruiting effort, but by moving faster. Workers face the mirror image of this problem. Additional preparation may improve the targeting and content of an application, but it can also move the applicant behind the firm's contact decision. When firms act before the pool is complete, otherwise comparable applicants can receive different consideration simply because they arrive at different points in the vacancy's recruiting process. Application timing can therefore shape both who receives consideration and the quality of the pool from which firms ultimately select.

Recent work shows that workers' choices on when to search, apply, and accept offers can have persistent consequences for their employment outcomes (Bandiera, Jalal and Roussille, 2025; Cortés et al., 2023), while large-scale application data document how quickly workers respond to vacancies in modern hiring markets (Davis and de la Parra, 2024). This paper studies the complementary firm-side margin: how sequential screening makes relative application timing consequential. Identifying this margin is difficult because an application timestamp is endogenous to job seekers' information, motivation, constraints, and application quality, while firms' internal recruiting stage and organic applicant pool are usually unobserved. Aggregate data, in turn, compress these within-vacancy dynamics into vacancy durations, job-filling hazards, or matching-efficiency residuals.

To causally identify the application timing effect in the recruiting process, I repurpose the large-scale correspondence experiment of Kline, Rose and Walters (2022). The scale of the experiment is unusual: it generated 83,643 applications to 11,114 entry-level vacancies at 108 major U.S. firms. Each vacancy received up to eight fictitious applications with randomized characteristics, submitted in four pairs spaced one to two days apart, and the dates of employer contacts were recorded. About 24 percent of these applications received employer contact within 30 days, consistent with firms viewing the experimental applicants as credible can-

didates. The original experiment was designed to measure discrimination. I use its controlled arrival sequence for a different purpose: randomized applications to the same vacancy reveal how employer contact decisions evolve as the applicant pool accumulates. In this sense, the applications serve as experimental tracers of the vacancy's recruiting process. Exploiting this design, I identify the causal effect of relative arrival time and trace how contact initiation changes the consideration of subsequently arriving applications. To quantify the firm tradeoff underlying these effects, I then develop and estimate a sequential contact-initiation model and use it to conduct counterfactuals. To my knowledge, this is the first large-scale experimental study on the role of application timing in firms' recruiting process and its consequences for applicants.[1]

I first establish a sizable *early application advantage*. Applications submitted in the first half of the experimental sequence are 1.16 percentage points more likely to receive employer contact within 30 days. This is an increase of roughly 5 percent over the average contact rate. Each additional day before the final experimental submission raises contact probability by 0.40 percentage points, and the timing profile is approximately monotone over the experimental window. These estimates are stable across alternative timing definitions, sample restrictions, wave-specific submission protocols, and calendar controls. As a scale benchmark, the first-half effect is about 57 percent of the Black-White contact gap in the same experiment.

The timing effect raises the central mechanism question: why should arrival timing matter when resume content is randomized? Two families of mechanisms could explain this pattern: direct-timing mechanisms, in which earlier submissions receive mechanical priority or signal motivation, and pool-based mechanisms, in which timing matters through the set of applications available when firms act. The evidence points to a threshold-triggered transition into the contact phase: firms begin contacting once the best application received so far is sufficiently attractive to justify proceeding rather than waiting for more applications. I measure the attractiveness of experimental applications using an out-of-fold screening-value index that summarizes employers' revealed-preference ranking of randomized resume

[1] Throughout the paper, "earlier" means earlier relative to other experimental applications to the same vacancy. The experiment does not observe the exact vacancy posting date or the full organic applicant pool, so the estimates should not be interpreted as the return to applying immediately after a vacancy is posted.

content. The highest screening value among previously submitted experimental applications predicts both entry into the contact phase and earlier initiation, whereas their mean screening value does not.

Entry into the contact phase substantially changes the prospects of later applicants. Contacts occur quickly: conditional on contact within 30 days, 52 percent of contacts occur within three days after submission and 68 percent within six days. Nearly half of vacancies with at least one contacted experimental application continue to receive submissions after the first observed contact. An event-study comparison, controlling flexibly for submission order, shows a sharp decline in contact probability for applications arriving after contact initiation. Constructed contact histories imply a similar pattern: the contact rate is about 63 percent lower after at least one earlier experimental contact. Because resume attributes are randomized across submission positions, the best-so-far screening value provides an instrument for whether a later application arrives after contact initiation. The IV estimates support a negative post-initiation effect on subsequent contact.

Additional tests weigh against alternative mechanisms. The early advantage disappears when contact begins only after the full experimental sequence has arrived, weighing against a direct timing premium whether mechanical or signaling-based. The quality of future applications does not predict whether earlier applications are contacted, weighing against full-pool comparison. Later submissions are contacted later, making pure batch review difficult to reconcile with the bundle of evidence. Together, these results support a sequential threshold mechanism in which firms begin contacting before the experimental pool is complete and entry into the contact phase lowers the consideration of later applicants.

If contact initiation reflects the value and cost of waiting, the strength of the timing margin should vary with recruiting conditions. It does. The daily penalty for arriving later is steeper in tighter state-month labor markets, both for overall contact and for the probability of being the experimental contact initiator. This evidence is descriptive rather than a causal estimate of tightness, but it provides a disciplined micro-to-macro fact: when workers are scarce relative to vacancies, firms move through applicant queues more quickly and the return to reaching a vacancy before contact initiation rises.

I formalize these facts in a partial-equilibrium model of contact initiation. The experiment identifies the timing gradient and the contact outcomes of experimen-

tal applications, but it does not observe the organic applicants or the latent initiation decision when no experimental application is contacted. The model embeds the experimental applications in a broader applicant flow and separates two forces that the reduced form conflates. The first is the effective arrival intensity, λ, which governs how quickly additional applicant opportunities arrive. Following directed-search models, this arrival environment depends on market congestion and on a shifter, ξ, that summarizes hiring and separation conditions not captured by tightness alone (Birinci, See and Wee, 2025). The second is the shadow cost of delaying contact, κ, which can reflect vacancy delay, the risk that candidates become unavailable, and organizational pressure to move the recruiting process forward. The firm chooses when to initiate contact by comparing this cost with the expected screening-value gain from waiting for additional applications. The estimated model reproduces the front-loaded timing of experimental contact initiation and its variation with tightness.

I then use thze model to quantify firms' tradeoff between recruiting speed and the screening-value gains from waiting. Conditional on the firm contacting an experimental application, the highest-screening-value experimental application has not yet arrived at contact initiation in 51.6 percent of cases. The screening-value gap between the highest-valued experimental application overall and the highest-valued one available at initiation averages 6.6 percent of the screening value of the initiated application. A deliberately conservative back-of-the-envelope conversion places this baseline timing loss at approximately \$390-\$781 per vacancy-year.[2] The counterfactuals trace out a speed-loss frontier as firms adjust contact timing to changes in delay costs and applicant arrival conditions. Halving the shadow cost of delay leads firms to wait longer and lowers the relative timing loss from 6.6 to 6.2 percent; doubling it induces earlier contact and raises the loss to 7.1 percent. Timing losses also rise with labor-market tightness. In the preferred specification, the model attributes most of this increase to the arrival channel: tighter markets imply a thinner flow of additional applicant opportunities, reducing the option value of waiting and shifting contact initiation earlier, rather than primarily raising the shadow cost of delay.

[2]The conversion multiplies the relative screening-value shortfall by an entry-level wage benchmark, the estimated 63 percent post-initiation decline in contact probability, and a 20-40 percent contact-to-hire pass-through rate.

Taken together, this paper identifies a structural timing friction within a vacancy's matching process. The friction does not require a departure from firm private optimization: it arises because applicant arrivals are sequential and waiting is costly. Firms initiate contact when the expected screening-value gain from waiting for additional applications no longer exceeds the shadow cost of delay. The experimental design makes this tradeoff observable and quantifiable. Heterogeneity in delay costs and arrival environments generates different contact-initiation policies, which in turn create a private return to queue position for applicants and a potential arms race in monitoring and submission speed. At the aggregate level, this employer-side timing margin offers a microeconomic channel through which labor-market conditions affect matching efficiency by changing when firms act on their applicant pools. More accurate screening can improve the ranking of applications already received, but it need not eliminate this friction as long as applications arrive sequentially and firms compete on time. Even if application submission and initial screening become nearly instantaneous, the same timing logic may reappear in interviews, offers, or other downstream stages. Within-vacancy timing is therefore an economic margin of match formation, not merely an administrative feature of recruiting.

Contribution to the literature. This paper first contributes to a growing literature on the timing and dynamics of worker search. Recent work shows that workers' choices on when to search can have persistent employment consequences and contribute to gender gaps (Bandiera, Jalal and Roussille, 2025). Related evidence studies the timing of offer acceptance and early-career outcomes (Cortés et al., 2023). Large-scale application data document how quickly workers direct applications across vacancies (Davis and de la Parra, 2024), while field experiments show that application costs and dynamic selection shape the composition of applicant pools (Abebe, Caria and Ortiz-Ospina, 2021). This literature establishes that *when workers search, apply and accept offers* matters. I identify a complementary firm-side mechanism: holding application content fixed, relative arrival time changes employer consideration because firms act on an evolving applicant queue. The design therefore isolates the causal return to queue position from endogenous worker selection and identifies the firm behavior that produces this return.

Second, the paper contributes to research on employer screening, personnel

practices, and information acquisition. The closest work shows that testing, interview sequences, managerial discretion, employer attention, and information production shape which applicants firms hire and how effectively they select (Autor and Scarborough, 2008; Pallais, 2014; Bartoš et al., 2016; Hoffman, Kahn and Li, 2018; Cavounidis, Lang and Weinstein, 2024; Coffman, Kostyshak and Saygin, 2024; Radbruch and Schiprowski, 2025). This literature focuses primarily on the content, precision, production, and allocation of information. The threshold evidence and the stopping model in this paper add a distinct dynamic margin: how long firms continue gathering applications before acting, what feature of the accumulated pool triggers contact, and how that action changes the prospects of later arrivals. This margin also connects to broader research on employer search and recruiting strategies, including vacancy duration, recruiting intensity and standards, application queues, interviewing, and job filling. [3] The paper therefore links the information-acquisition problem studied in personnel economics to the sequential recruiting process emphasized in labor-search models.

Third, the paper speaks to the microfoundations of aggregate matching efficiency. In matching models, movements in job-filling rates conditional on unemployment and vacancies are often summarized by recruiting intensity or a residual matching-efficiency term (Petrongolo and Pissarides, 2001; Davis, Faberman and Haltiwanger, 2013; Barnichon and Figura, 2015; Birinci, See and Wee, 2025; Birinci, Mercan and See, 2025). Contact initiation is one employer-side process that can enter this residual: it determines when a vacancy begins acting on its applicant queue, and this paper shows that this timing margin varies with market conditions. The model's ξ is a contact-initiation analogue of matching efficiency and a micro margin through which the recruiting environment changes the conversion of applicant opportunities into employer contact. The paper therefore connects experimentally identified within-vacancy behavior to the economic forces emphasized in directed-search and queueing models (Wright et al., 2021; Wolthoff, 2018; Carrillo-Tudela, Gartner and Kaas, 2023).

Finally, the paper contributes to correspondence-experiment design and inter-

[3] See, among others, Barron, Bishop and Dunkelberg (1985); Barron, Berger and Black (1997); Mortensen and Pissarides (1994); Davis, Faberman and Haltiwanger (2012, 2013); Wolthoff (2018); Carrillo-Tudela, Gartner and Kaas (2023); Mueller et al. (2024). Related dynamic and granular models study worker-firm search and sorting over time (Bradley, 2022; Engbom, 2021; Jarosch, Nimczik and Sorkin, 2024; Cai, Gautier and Wolthoff, 2025).

pretation (Gaddis, 2018; Neumark, 2018; Kessler, Low and Sullivan, 2019; Kline, Rose and Walters, 2022). Audit studies generally treat applications as independent probes of a static employer response rule and focus on average differences across randomized characteristics. I show that repeated applications to the same vacancy can instead reveal and affect a dynamic employer state. Timing is therefore both a design consideration for correspondence experiments and a source of substantive evidence on within-vacancy recruiting behavior.

Structure of the paper. The remainder of the paper is organized as follows. Section 2 describes the experiment and complementary data. Section 3 defines the timing measures and establishes the early application advantage. Section 4 studies contact initiation and evaluates alternative mechanisms. Section 5 relates the timing gradient to labor market tightness and motivates the model. Section 6 develops and estimates the contact-initiation model and reports the counterfactuals. Section 7 concludes.

2 Empirical Context and Data

This paper uses the large-scale resume correspondence experiment of Kline, Rose and Walters (2022). The original experiment was designed to study discrimination among large U.S. employers. I use a different feature of the design: each sampled vacancy received multiple randomized applications submitted over several days. I refer to these applications as experimental tracers because their staggered arrival and observed contact outcomes reveal how the employer's recruiting process evolves as applications accumulate within a vacancy.

This section summarizes the features of the experiment that are needed for the analysis. I first describe the experimental setting, then discuss how the timing of the experimental applications maps into a vacancy's recruiting window, and finally introduce the complementary labor-market data used later in the paper. Appendix G provides details on data sources, matching, and variable construction.

2.1 Experimental Setting

The experiment sampled large U.S. employers from the 2018 Fortune 500 list. Following Kline, Rose and Walters (2022), the sampling frame was restricted to companies with distinct brands, proprietary hiring portals, broad geographic reach, and a sufficient volume of entry-level vacancies to support the audit design. The resulting sample includes 108 firms across a wide range of industries, collectively employing roughly 15 million workers in 2020.

Vacancies were sampled over five waves from October 2019 to April 2021. For each employer, the original experiment targeted 125 entry-level vacancies, with 25 vacancies per wave. A majority of firms (72) participated in all waves. Some firms missed Wave 1 because of COVID-19 disruptions, while others were excluded from later waves because of hiring portal changes and technological barriers. For each employer, sampled vacancies corresponded to establishments in counties not previously sampled for that employer. Firms entered the sampling queue in randomized order within each wave, and vacancies were sampled from the firms' own hiring portals.

The vacancies were entry-level positions that did not require a bachelor's or advanced degree, substantial prior experience, or a specialized license. For each sampled vacancy, the experiment submitted up to eight applications with randomized resume characteristics. Applications were sent in four pairs. Within each pair, one application carried a distinctively Black name and the other a distinctively White name. Gender, age, education, employment history, and other resume characteristics were randomized as in Kline, Rose and Walters (2022). Some vacancies stopped accepting applications before all eight submissions were completed. The final sample contains 83,643 applications to 11,114 vacancies, and 88.4% of vacancies received the full set of eight experimental applications.

The outcome is employer contact within 30 days of submission. Each fictitious applicant had a unique phone number and email address, which were monitored to record employer callbacks, interview requests, or other positive employer responses. The average contact rate is about 24%, making the setting well suited to studying heterogeneity in employer contact behavior.[4]

[4] As in most correspondence experiments, the outcome is employer contact rather than interviews, offers, hires, or wages. Accordingly, the paper identifies effects on the contact and contact-initiation margins, not on later stages of the hiring pipeline.

2.2 Experimental Timing within a Vacancy

The timing of the experimental submissions is central to this paper. Kline, Rose and Walters (2022) scraped firms' hiring portals daily and, within each wave, sampled the most recent job posting in a new county for each firm, randomizing ties. Applications were then submitted over a short window. In Wave 1, pairs were typically sent every other day; beginning in Wave 2, when most applications were submitted automatically, one pair was typically sent per day. Overall, 94% of applications were submitted within eight days of the first experimental application to the vacancy.

This design places the experimental submissions within an active application window and relatively close to the posting's appearance on the employer portal. At the same time, the experiment does not observe the exact vacancy posting date, the full organic applicant pool, or the employer's recruiting stage when the first experimental application arrives. The estimand in this paper is therefore the effect of arriving earlier *within the experimental submission window*, not the return to applying immediately after a vacancy is first posted.[5]

This use of the experiment differs from the usual correspondence experiment design, which focuses on average differences in contact rates across randomized applicant characteristics. Here, the repeated, timed applications trace how employer contact decisions change as the experimental sequence unfolds, revealing within-vacancy screening dynamics that are usually unobserved.

2.3 Complementary Market Data

To study how the application timing gradient varies with recruiting conditions and to parameterize the recruiting environment in the model, I merge experimental vacancies to state-month labor-market indicators. The main market condition measure is local labor market tightness from the Job Openings and Labor Turnover Survey (JOLTS), defined as vacancies per unemployed worker in the state of the vacancy and the month in which the first experimental application to that vacancy is submitted. I also use additional JOLTS series including hires, quits, layoffs and discharges, and job-opening rates to characterize broader recruiting conditions at

[5]Throughout the paper, "earlier" means earlier relative to other experimental applications to the same vacancy.

the onset of the experiment.

Additional covariates used in appendix heterogeneity exercises come from industry, state, and firm-level data sources. These include Occupational Information Network (O*NET) measures of job content, industry employment composition, time-varying state labor market conditions, and firm-level measures constructed from external workforce and job-posting data (Revelio Labs, 2026). I use these variables primarily for descriptive external-validity checks. Appendix G describes the construction of all complementary covariates. With the experimental timing variation and outcome defined, Section 3 estimates how employer contact changes with an application's relative arrival time within the vacancy.

3 Estimating the Early Application Advantage

This section estimates the causal effect of arriving earlier within the experimental submission sequence on the probability of employer contact. In observational data, application timing is confounded by applicant quality, information, motivation, and preparation. In the experiment, resume characteristics are randomized across multiple applications to the same vacancy, and the applications are submitted at different points in the experimental sequence. The analysis therefore compares applications to the same vacancy that differ in relative arrival time but have randomized resume content.

I begin by defining the timing measures and empirical specifications, and then document the magnitude, shape, and robustness of the early application advantage.

3.1 Timing Measures and Empirical Strategy

For each application, I observe the submission date and realized within-vacancy submission order. I construct three main timing measures. First, let $Order_i \in \{1, \ldots, 8\}$ denote application i's realized order of submission within vacancy $j(i)$. The distribution of $Order_i$ and the prevalence of vacancies with fewer than 8 submissions are summarized in Table A.1.

Second, define the baseline binary early indicator

$$early_i^1 = \mathbf{1}\left(Order_i \leq 4\right),$$

which classifies application i as early if it arrives among the first four submissions to the vacancy $j(i)$, with applications to vacancies that received at most four experimental submissions coded as late.[6] In the full sample, 41,112 (49.15%) submissions are coded as early.

Third, I construct a calendar-time measure based on distance to the final experimental submission. Let $D2L_i$ denote the number of days between application i's submission date and the last experimental submission to the same vacancy. Due to occasional operational delays and vacancy closures, the distribution is right-skewed with 99.3% of all submissions having no more than 10 days of separation from the last submission for each vacancy. I therefore use a top-coded measure $D2L_i^t \in \{0, 1, \ldots, 6, 7\}$, where the final category denotes seven or more days before the last experimental submission.[7] Sections 4-6 use the forward analogue $D2F_i^t$, days since the first experimental submission, to study contact initiation over the observed experimental window.

Let Y_i be the indicator for whether application i receives employer contact within 30 days. The baseline specification is

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{3.1}$$

where T_i is either $early_i^1$ or $D2L_i^t$, $\mathbf{X}_i$ contains randomized resume characteristics, and $\eta_{j(i)}$ is a vacancy fixed effect. Identification comes from randomized assignment of resume content to experimental applications submitted at different times within the same vacancy. Appendix Table A.2 confirms that the constructed timing measures are balanced with respect to randomized resume characteristics.

To visualize the timing profile, I also estimate flexible nonparametric specifica-

[6] For vacancies that received at most four experimental applications (836 vacancies, 7.52%; 2,321 submissions, 2.77%), the experiment ended before a full early/late split could be observed. Appendix Section A.2 reports robustness to alternative definitions: $early_i^2$, which defines early submissions as those not among the last four realized submissions, and $early_i^3$, which mirrors $early_i^1$ with the designed pair order instead of the realized submission order.

[7] By design, the eight applications were planned to be submitted in four pairs, either every other day in Wave 1 or daily in Waves 2-5. Following the intended design window, values of $D2L_i$ above 6 are coded as 7. Appendix Figure A.1 plots the distribution of the raw measure $D2L_i$, and Appendix Figure A.2 plots the distribution of the top-coded measure $D2L_i^t$.

tions by submission order and by days to the last experimental submission:

$$Y_i = \alpha + \sum_{k=2}^{8} \beta_k \mathbf{1}\{Order_i = k\} + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{3.2}$$

$$Y_i = \alpha + \sum_{d=1}^{7} \delta_d \mathbf{1}\{D2L_i^t = d\} + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i. \tag{3.3}$$

In all specifications, standard errors are clustered at the vacancy level to allow for within-vacancy dependence and heterogeneous treatment effects (Abadie et al., 2023).

3.2 Evidence of Early Application Advantage

Table 1 reports the baseline estimates. Early submissions are 1.16 percentage points more likely to receive employer contact within the same vacancy. Relative to the average contact rate of about 24%, this is a roughly 5% increase in contact probability. As a scale benchmark, the effect is about 57% of the Black-White contact gap in this experiment. The calendar-time specification gives a daily measure of this early application advantage. One additional day earlier relative to the last experimental submission raises the probability of contact by 0.40 percentage points. The estimates are nearly identical in the complete eight-application sample, which is used in mechanism exercises below.

The estimates establish a statistically and economically meaningful early application advantage within vacancies, conditional on randomized resume content. Figure 1 shows the shape of the timing gradient with respect to the constructed relative application timing measures $Order_i$ and $D2L_i^t$. Figure 1a plots contact probability gaps by realized submission order relative to the first submission, estimated from Specification (3.2). Within a vacancy, the first submission is the most likely to be contacted. The order effect is weakly monotone: later submissions are not more likely to receive contact than earlier submissions, with a significant decline between the fourth and fifth submissions. Figure 1b plots the nonparametric relationship with days to the last experimental submission, $D2L_i^t$. Contact probability rises approximately linearly with the number of days between submission and the final experimental application.

The early application advantage is not driven by differential returns to random-

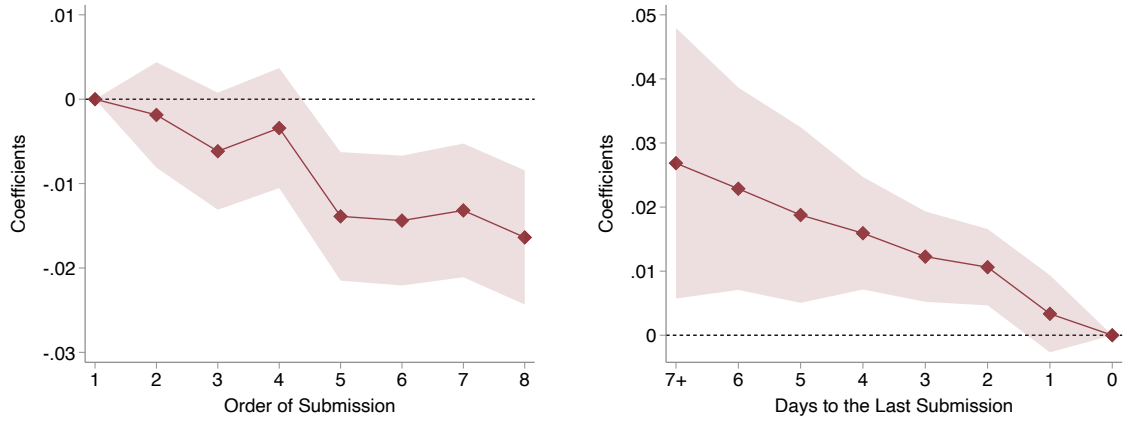
Table 1: Baseline Early Application Advantage

	(1) Full	(2) Complete	(3) Full	(4) Complete
$early_i^1$	0.0116	0.0112		
	(0.0024)	(0.0024)		
$D2L_i^t$			0.0040	0.0038
			(0.0008)	(0.0008)
Black	-0.0205	-0.0210	-0.0205	-0.0210
	(0.0018)	(0.0019)	(0.0018)	(0.0019)
Female	-0.0021	-0.0022	-0.0021	-0.0022
	(0.0022)	(0.0023)	(0.0022)	(0.0023)
Over 40	-0.0043	-0.0047	-0.0043	-0.0047
	(0.0020)	(0.0021)	(0.0020)	(0.0021)
Pol. club	0.0059	0.0060	0.0060	0.0061
	(0.0051)	(0.0052)	(0.0051)	(0.0052)
Acad. club	0.0049	0.0046	0.0050	0.0047
	(0.0051)	(0.0052)	(0.0051)	(0.0052)
LGBTQ club	-0.0003	-0.0010	-0.0002	-0.0009
	(0.0036)	(0.0037)	(0.0036)	(0.0037)
Typical pron.	-0.0077	-0.0081	-0.0078	-0.0082
	(0.0051)	(0.0052)	(0.0051)	(0.0052)
Neutral pron.	-0.0086	-0.0092	-0.0085	-0.0091
	(0.0051)	(0.0052)	(0.0051)	(0.0052)
Assoc. degree	0.0021	0.0018	0.0021	0.0018
	(0.0020)	(0.0021)	(0.0020)	(0.0021)
Constant	0.2475	0.2523	0.2453	0.2501
	(0.0025)	(0.0025)	(0.0027)	(0.0028)
Vacancy FEs	✓	✓	✓	✓
N	83,643	78,632	83,643	78,632

Note: This table presents the effects of early submission on the contact probability. The outcome is employer contact within 30 days. Columns (1) and (2) estimate Equation (3.1) with $T_i = early_i^1$; columns (3) and (4) estimate the same specification with $T_i = D2L_i^t$. Columns titled "Full" include all vacancies, while columns titled "Complete" only include vacancies with 8 experimental submissions. The constructions of $early_i^1$ and $D2L_i^t$ are described in Section 3.1. All regressions include vacancy fixed effects and the full set of randomized resume characteristics. Standard errors in parentheses are clustered at the vacancy level.

ized resume characteristics. Appendix Table A.7 interacts $early_i^1$ with the full set of randomized resume attributes. A joint test cannot reject equality of the timing effect across these characteristics ($F = 0.58$). Appendix Figures A.8 and A.9 show that average race and gender contact gaps are also stable across submission order

Figure 1: Contact Gaps by Relative Application Timing



(a) Submission Order (b) Days to Last Submission

Notes: The figures plot the contact gaps by relative application timing measures defined in Section 3.1. Panel A plots coefficients $\hat{\beta}_k$, $k = 2, \ldots, 8$, from Specification (3.2), with the first submission omitted. Panel B plots coefficients $\hat{\delta}_d$, $d = 1, \ldots, 7$, from Specification (3.3), with $D2L_i^t = 0$ omitted. Both specifications include vacancy fixed effects and the full set of randomized resume controls. Standard errors are clustered at the vacancy level. Shaded areas show 95% confidence intervals. The sample includes all experimental applications.

and $D2L_i^t$. These results indicate that average race and gender contact gaps do not evolve systematically with application timing. They do not rule out group-specific state dependence after contact initiation: a demographic gap may remain stable over submission time even if earlier contacts affect later applications asymmetrically. Section 4 turns to this dynamic question using contact history measures.

Taken together, the estimates establish a robust early application advantage within the experimental submission window. In the full sample, applications in the first half of the experimental sequence are 1.16 percentage points more likely to receive contact, and each additional day before the final experimental submission raises contact probability by 0.40 percentage points. The estimates are nearly identical in the complete eight-application sample. Appendix Section A.2 reports robustness to alternative timing definitions, designed pair order, wave-specific submission protocols, incomplete-vacancy restrictions, and calendar controls.

These results establish the timing fact that motivates the rest of the paper, but they do not explain why relative arrival time matters. Section 4 therefore turns to employer-side recruiting dynamics and asks whether the advantage arises because firms enter the contact phase before later applications arrive. The mechanism analysis restricts to complete eight-application vacancies so that the full experimental

sequence is observed.

4 Why Timing Matters: Early Contact Initiation and Sequential Screening

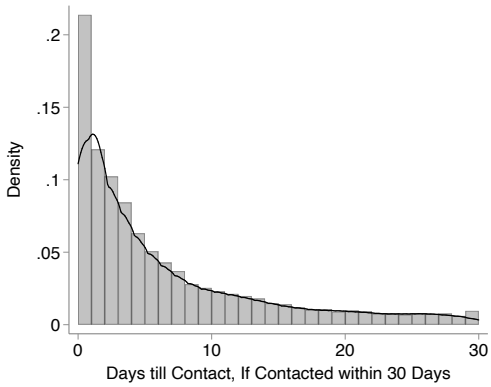
Section 3 documents a robust early application advantage: within the same vacancy and holding randomized resume characteristics fixed, applications submitted earlier in the experimental sequence are more likely to receive contact. A natural explanation is that recruiters do not wait for the full experimental pool to arrive. Instead, they often initiate the contact phase early, so applications arriving after contact initiation face a different recruiting environment. This section evaluates that mechanism. I first document that contacts occur quickly and that contact probabilities fall after contact initiation. I then use a screening-value index, constructed from out-of-fold predictions of recruiter contact, to show that contact initiation responds to the best applicant observed so far. Finally, I use this variation to instrument the post-initiation state and present several tests against alternative explanations.

4.1 Contacts Happen Fast

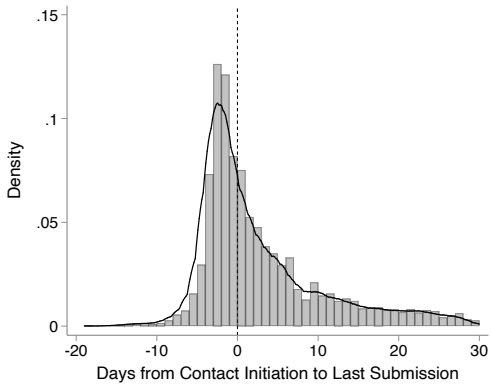
Figure 2 summarizes the timing of recruiter contact. Figure 2a plots the distribution of days between submission and contact, conditional on contact within 30 days. Contacts occur quickly: among the 20,102 contacted applications, the mean time from submission to contact is 6.04 days. A large share of contacts occurs almost immediately: 21.55% occur on the submission day, 52.28% occur within three days, and 67.91% occur within six days.

These response times are short relative to the experimental cadence, under which the sequence of experimental submissions to a vacancy typically spans several days. As a result, later experimental submissions often arrive after the recruiter has already made at least one contact decision. Figure 2b shows this directly: among the 4,036 vacancies with at least one contacted experimental application, 1,899 receive experimental submissions after observed contact initiation.

Figure 2: Contact Timing Relative to Experimental Application Submissions



(a) Contacted Application Distribution by Days from Submission to Contact



(b) Vacancy Distribution by Days from Contact Initiation to Last Submission

Notes: The figures plot the relative timing of contacts with respect to experimental application submissions. Panel 2a plots the distribution of experimental applications by the number of days from submission to recruiter contact, conditional on contact within 30 days. Panel 2b plots, among vacancies with at least one contacted experimental application, the distribution of the signed number of days from contact initiation to the last experimental submission. Contact initiation is defined as the first observed contact with any experimental application submitted to the vacancy. Both panels use an Epanechnikov kernel with bandwidth one day.

4.2 Contact Probabilities Fall after Contact Initiation

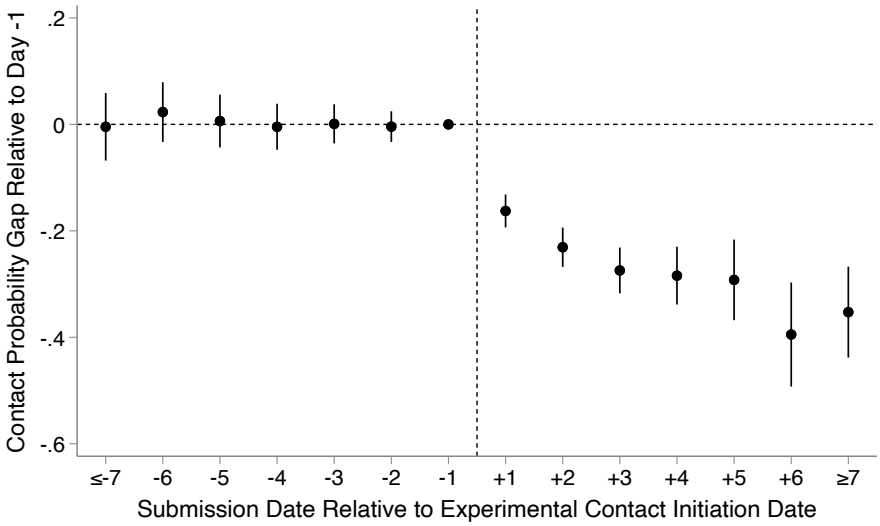
I next examine whether the contact probability changes once the recruiting process enters the contact phase. For each experimental application i to vacancy j, let $e_{j(i)}$ be the signed number of days from the first observed contact date among experimental applications submitted to vacancy j to the submission date of application i. I estimate the event-time specification

$$Y_i = \sum_{k \in \mathcal{K}, k \neq -1} \beta_k \cdot \mathbf{1}\left(e_{j(i)} = k\right) + \boldsymbol{\gamma}' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{4.1}$$

where Y_i is an indicator for recruiter contact within 30 days, $\eta_{j(i)}$ is a vacancy fixed effect, and $\mathbf{X}_i$ includes randomized resume controls, submission-order fixed effects, and submission-day-of-week fixed effects. Standard errors are clustered at the vacancy level. The event-time support is $\mathcal{K} = \{-7, -6, \cdots, -1, 1, \cdots, 6, 7\}$. The endpoint $k = -7$ includes applications with event time $e_{j(i)} \leq -7$, and the endpoint $k = 7$ includes applications with event time $e_{j(i)} \geq 7$. Applications submitted on the contact-initiation date are excluded because the data record dates

but not the exact time of day.[8]

Figure 3: Contact Probability Pre and Post Contact Initiation



Notes: This figure plots the coefficients from linear probability models of recruiter contact on event-time indicators (4.1), where event time is defined as the application submission date minus the first experimental contact date for the vacancy. The omitted category is applications submitted one day before contact initiation. The sample is restricted to vacancies with at least one experimental contact. Applications submitted on the first contact date are excluded because within-day ordering is not observed. Each point reports the estimated coefficient β_k from Equation (4.1). The regression includes vacancy fixed effects, submission-order fixed effects, day-of-week fixed effects and resume controls. Vertical lines denote 95% confidence intervals based on standard errors clustered at the vacancy level.

Figure 3 shows a sharp change in contact probabilities around contact initiation. Experimental applications submitted before contact initiation have relatively stable contact probabilities, while applications submitted after contact initiation are substantially less likely to be contacted. Because the specification includes submission-order fixed effects, the post-initiation decline is not simply a comparison of earlier and later positions in the experimental sequence. The event-study evidence suggests that contact initiation shifts the recruiting environment faced by subsequent applications.

I also construct contact history measures for each submission i that summarize whether i arrives after previous experimental contacts to the same vacancy. Let $day(i)$ denote the calendar day of submission i. The constructed contact history of i, NoC_i, counts all experimental submissions to the same vacancy that are con-

[8]This event-study is descriptive. The event date is defined by the first experimental contact, so lead coefficients should not be interpreted as pre-trend tests.

tacted *at least one day* before $day(i)$. Since the data only record contact dates but not exact times of day, it is not possible to order submissions and contacts precisely within a day. To accommodate this limitation, I also construct an upper-bound measure, NoC_i^{alt}, that counts other experimental applications contacted no later than $day(i)$, including same-day contacts. Table A.8 illustrates these constructions with an example.[9]

For each experimental application, $NoC_i > 0$ indicates that it arrives after at least one earlier experimental contact. In Table A.11, I compare the contact probability and the full set of resume characteristics of applications submitted before and after contact initiation. Contact probabilities fall from 26.41% before initiation to 9.74% after initiation, a 63% decline, consistent with the event-study evidence in Figure 3.

Finally, I re-estimate the baseline timing specifications from Section 3.1 (Equation (3.1)), adding the constructed contact history measure NoC_i as a control:

$$Y_i = \alpha_0 + \beta \cdot timing_i + \iota \cdot \mathbf{1}\left(NoC_i > 0\right) + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{4.2}$$

where $timing_i$ is one of the timing measures from Section 3.1. Table A.12 reports the results. Across specifications, contact history is strongly negatively associated with subsequent contact, and controlling for it absorbs and reverses the baseline early application advantage. Because $\mathbf{1}\left(NoC_i > 0\right)$ is itself an outcome of earlier recruiter decisions, I treat these regressions as descriptive evidence. I next use randomized variation in earlier applicants' screening values to study contact initiation more directly.

4.3 What Triggers Contact Initiation?

The results so far show that the recruiting environment changes significantly after contact initiation. However, contact initiation is an endogenous choice of recruiters. I therefore take advantage of the randomization design and ask what triggers contact initiation. A sequential stopping model predicts that initiation

[9] Intuitively, NoC_i treats application i as if it were submitted at the beginning of $day(i)$, while NoC_i^{alt} treats it as if it were submitted at the end of $day(i)$. Thus NoC_i is a lower bound and NoC_i^{alt} is an upper bound for the realized contact history of application i, with $NoC_i \leq NoC_i^{alt}$. Table A.9 summarizes these measures by submission order, and Table A.10 reports their correlations with various timing measures used in Section 3.1.

should respond to the best application observed so far: once an application crosses the recruiter's contact threshold, the contact phase starts. A competing learning-about-the-pool interpretation predicts that the average quality of prior applications should be more informative. To distinguish these mechanisms, I first construct a screening-value index for each experimental application.

4.3.1 Measuring Applicant Screening Value

I construct an out-of-fold screening-value index that predicts employer contact using high-dimensional resume and vacancy characteristics, including detailed fictitious employment histories. The prediction model is trained on the first experimental application submitted to each vacancy, so the training sample reflects the employer's initial-screening environment and is not mechanically affected by within-vacancy application timing. I use the resulting predicted contact probability as a revealed-preference measure of employer-perceived screening value: a resume with a higher index is one that the model predicts would be more likely to receive contact if it arrived at the start of the recruiting spell. [10]

Two features of the index are important for the mechanism tests below. First, the index is constructed out of fold, so it is not an in-sample fitted value for the observation being studied. Second, when I use the index to form within-vacancy measures such as best-so-far quality, I standardize it within each vacancy. Thus the identifying variation comes from the randomized assignment of resume quality to earlier versus later submission slots rather than from vacancy-level differences in baseline contact propensity.

Tables A.13 and A.14 provide validation evidence for the index. The index is monotone in observed contact rates, strongly predicts within-vacancy differentiation in contact probabilities, and is balanced with respect to within-vacancy timing by design. Appendix B describes the model construction, cross-fitting procedure, features, and robustness across alternative algorithms in detail.

[10] Conceptually, the recruiter applies an unobserved screening rule that maps resume and vacancy characteristics into a latent evaluation index v_{ij}. Under weak regularity conditions, the conditional contact probability is monotonically increasing in this evaluation index. The prediction model flexibly estimates this conditional probability and uses it as a *revealed-preference* summary of how recruiters value observable resume content in this setting.

4.3.2 Best-So-Far Application Predicts Contact Initiation

For each application i, I define Q_i^{best} as the highest within-vacancy standardized screening value among applications to the same vacancy that arrived at least one day before i. I also define Q_i^{mean} as the corresponding mean screening value among those prior applications. Conditional on vacancy fixed effects, these variables are driven by the exogenously randomized assignment of screening values to earlier versus later submission slots. I first estimate

$$\mathbf{1}\left(NoC_i > 0\right) = \alpha_0 + \pi \cdot Q_i^{best} + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{4.3}$$

where $\mathbf{1}\left(NoC_i > 0\right)$ indicates that at least one experimental contact occurred at least one day before the submission of i, $\mathbf{X}_i$ includes randomized resume characteristics, submission-order fixed effects, and submission-day-of-week fixed effects, and $\eta_{j(i)}$ denotes vacancy fixed effects. I estimate this specification on the last four experimental applications to each complete vacancy, so that the prior application pool is nontrivial.

Table 2: Prior Screening Value and Contact Initiation

	(1)	(2)	(3)	(4)
	All Vacancies		Vacancies w. ≥ 1 Contact	
Q_i^{best}	0.0078		0.0235	
	(0.0039)		(0.0098)	
Q_i^{mean}		0.0040		0.0134
		(0.0079)		(0.0180)
Resume Controls	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓
Day-of-Week FEs	✓	✓	✓	✓
N	39,300	39,300	16,142	16,142

Note: This table estimates Equation (4.3). The dependent variable is $\mathbf{1}(NoC_i > 0)$, an indicator that at least one experimental application to the same vacancy was contacted at least one day before application i was submitted. Q_i^{best} is the highest within-vacancy standardized screening-value index among applications submitted at least one day before i; Q_i^{mean} is the corresponding mean. The sample consists of the last four experimental applications to complete vacancies with eight experimental applications; observations with no application submitted at least one day earlier are excluded because Q_i^{best} and Q_i^{mean} are undefined. Columns (1) and (2) include all complete vacancies; columns (3) and (4) restrict to vacancies with at least one contacted experimental application. All regressions include vacancy fixed effects, submission-order fixed effects, submission-day-of-week fixed effects, and randomized resume controls. Standard errors in parentheses are clustered at the vacancy level.

Table 2 shows that the best prior screening value predicts entry into the contact phase. A higher Q_i^{best} increases the probability that contact initiation has already occurred before application i arrives, and the relationship is stronger among vacancies with at least one experimental contact. By contrast, the mean prior screening value is not predictive. This distinction is informative: a threshold-crossing model emphasizes the best application observed so far, while a learning-about-the-pool mechanism would naturally make the average quality of prior applications relevant. Table A.15 provides a complementary horse race where both Q_i^{best} and Q_i^{mean} are used to predict application i's own contact probability. Both variables are negatively signed when included separately, reflecting that stronger prior applications reduce later contact probabilities, but only Q_i^{best} remains predictive in the joint specification. Together, the evidence favors a threshold/stopping interpretation over the competing mechanism of learning about the quality distribution.

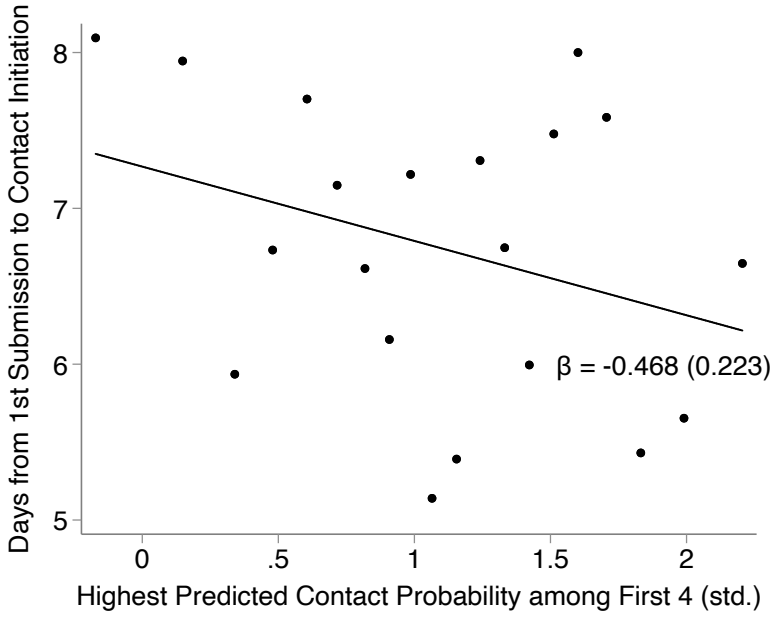
Lastly, I directly test whether randomized early quality predicts the timing of contact initiation at the vacancy level. For each vacancy j, I calculate the highest within-vacancy standardized screening value among the first four experimental applications and the number of days between the first experimental submission and contact initiation. Consistent with evidence discussed above, Figure 4 shows that vacancies randomly assigned a stronger early best applicant initiate contact earlier.

The evidence in this subsection points to a specific trigger for contact initiation: the best applicant observed so far. Because the assignment of resume quality to submission slots is randomized, the screening value of earlier applications provides quasi-random variation in whether the employer has entered the contact phase by the time a later application arrives. I use this variation in the next subsection to instrument the post-initiation state and causally identify the impact of contact initiation on the contact probability of subsequent applications.

4.4 Instrumenting Contact Initiation with Best-So-Far

The contact history regressions show that applications submitted after contact initiation are less likely to be contacted, but the constructed contact history measure is endogenous. To provide a more causal test, I use Q_i^{best} as an instrument for $\mathbf{1}(NoC_i > 0)$, the indicator that the application arrives after contact initiation. The

Figure 4: Highest Early Screening Value and Contact Initiation Timing



Notes: This figure plots the relationship between best early screening value and contact-initiation timing at the vacancy level. The y-axis is the number of days between the first experimental submission and the first observed contact among experimental applications. The x-axis is the highest within-vacancy standardized screening-value index among the first four experimental applications. The sample consists of complete vacancies with eight experimental applications and at least one observed experimental contact. Vacancy-level observations are grouped into 20 equal-sized bins. The fitted line comes from a vacancy-level regression with state-by-wave fixed effects. The annotation reports the estimated slope, with its standard error in parentheses; the standard error is clustered at the state-by-wave level.

first stage is Equation (4.3). The second stage is

$$Y_i = a_0 + b \cdot \hat{\mathbf{1}}\left(NoC_i > 0\right) + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{4.4}$$

where Y_i is the indicator of contact within 30 days after submission. For both stages of the IV estimation, I include the standardized screening value of submission i as a control, in addition to resume characteristics and fixed effects in Equation (4.3).

The IV-2SLS estimates in Table 3, together with the first-stage results in Table 2, strengthen the interpretation of the contact-history results. Both the OLS and IV-2SLS estimates indicate that applications arriving after contact initiation are less likely to receive employer contact. Because the IV identifies a local margin among vacancies whose contact initiation status is shifted by strong earlier applications, I emphasize the signs and qualitative implications rather than their exact magnitudes. These results help address concerns that the contact-history measures

Table 3: Contact Probability and Contact Initiation, IV-2SLS Estimation

	(1)	(2)	(3)	(4)
	All Vacancies		Vacancies w. ≥ 1 Contact	
	IV-2SLS	OLS	IV-2SLS	OLS
$\mathbf{1}(NoC_i > 0)$	-1.2642	-0.1204	-1.1077	-0.1275
	(0.7380)	(0.0206)	(0.5950)	(0.0217)
Kleibergen-Paap F	5.218		7.602	
Kleibergen-Paap LM	5.236		7.654	
Anderson-Rubin test p value	0.025		0.023	
Resume Controls	✓	✓	✓	✓
Screening Value of i	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓
Day-of-Week FEs	✓	✓	✓	✓
N	39,300	39,300	16,142	16,142

Note: This table reports IV-2SLS and OLS estimates for the effect of arriving after contact initiation. The outcome in all columns is an indicator for contact within 30 days. In Columns (1) and (3), $\mathbf{1}(NoC_i > 0)$ is instrumented with Q_i^{best}, the highest within-vacancy standardized screening-value index among applications submitted at least one day before application i. Columns (2) and (4) report the corresponding OLS estimates. All columns control for application i's own screening-value index, randomized resume characteristics, vacancy fixed effects, submission-order fixed effects, and submission-day-of-week fixed effects. The sample consists of the last four experimental applications to complete vacancies with eight experimental applications; observations with no application submitted at least one day earlier are excluded because Q_i^{best} is undefined. Columns (3) and (4) restrict the sample to vacancies with at least one contacted experimental application. Standard errors in parentheses are clustered at the vacancy level.

merely reflect endogenous recruiter behavior, and support the interpretation that contact initiation causally lowers the contact probability of subsequent applications.

4.5 Alternative Explanations

The evidence so far supports a threshold-based mechanism: stronger best-so-far applications are more likely to trigger contact initiation, and entry into the contact phase lowers subsequent contact probabilities. This mechanism drives the early application advantage observed in Section 3.2. I now consider alternative explanations for the early application advantage.

First, early timing itself may be intrinsically valuable. Recruiters may interpret early submissions as signals of preparation, enthusiasm, or motivation. Alterna-

tively, applicant tracking systems may mechanically prioritize earlier timestamps. If such a direct timing premium were the main mechanism, earlier submissions should remain advantaged even when the employer does not initiate contact until after the entire experimental pool has arrived. I test this prediction by splitting contacted vacancies according to whether contact initiation occurs before the final experimental submission and re-estimating Equation (3.1) separately for the two subsamples.

Table 4: Restriction Test by Contact Initiation Timing

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies		Early Contact Init.		Late Contact Init.	
$early$	0.0273		0.0933		-0.0523	
	(0.0058)		(0.0076)		(0.0085)	
$D2L^t$		0.0096		0.0326		-0.0243
		(0.0020)		(0.0025)		(0.0031)
Resume Controls	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	32,288	32,288	17,616	17,616	14,672	14,672

Note: This table estimates the baseline timing specifications (3.1) separately by whether contact initiation occurs before the final experimental submission. Columns (1)-(2) use all complete vacancies with at least one contacted experimental application. Columns (3)-(4) restrict to vacancies in which at least one experimental submission arrives after contact initiation. Columns (5)-(6) restrict to vacancies in which contact initiation occurs only after all experimental submissions have arrived. The variable $early$ indicates that the application is in the first half of the experimental submission sequence; $D2L^t$ is the days-to-last-submission timing measure from Section 3.1. All regressions include vacancy fixed effects and randomized resume controls. Standard errors in parentheses are clustered at the vacancy level.

The results in Table 4 contradict a direct timing-premium explanation. In vacancies where contact initiation occurs after all experimental submissions arrive, the early application advantage disappears and even reverses sign. Early timing therefore does not appear to carry a positive premium once the contact-initiation channel is effectively shut down.

Second, Section 4.3.2 supports a threshold stopping structure of contact initiation, but the contact initiation timing could also reflect the incentive to wait to compare the full pool of applications before contacting applicants. This full-pool comparison mechanism predicts that earlier applications' contact probabilities should depend on the quality of later applications. In particular, a strong future application should reduce the probability that an earlier application is contacted. I test

this prediction among early applications by regressing an indicator for whether submission i is contacted on the highest screening value among subsequent submissions to the same vacancy, controlling for i's own characteristics and timing.

Table 5: Early Contact Decisions and Screening Value of Later Submissions

	(1)	(2)	(3)	(4)
	All Vacancies		Vacancies w. ≥ 1 Contact	
$Q_i^{best,post}$	0.0053		0.0122	
	(0.0048)		(0.0118)	
$Q_i^{mean,post}$		0.0027		0.0044
		(0.0099)		(0.0233)
Resume Controls	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓
Day-of-Week FEs	✓	✓	✓	✓
N	39,281	39,281	16,143	16,143

Note: This table tests whether later applications' screening values predict earlier applications' contact probabilities. The dependent variable is an indicator for contact within 30 days. The sample consists of the first four experimental applications to complete vacancies with eight experimental applications. Observations without valid values of $Q_i^{best,post}$ or $Q_i^{mean,post}$ are excluded. $Q_i^{best,post}$ is the highest within-vacancy standardized screening-value index among applications submitted after application i; $Q_i^{mean,post}$ is the corresponding mean. Columns (1)-(2) include all complete vacancies; columns (3)-(4) restrict to vacancies with at least one contacted experimental application. All regressions include vacancy fixed effects, submission-order fixed effects, submission-day-of-week fixed effects, and randomized resume controls. Standard errors in parentheses are clustered at the vacancy level.

Table 5 shows that later applications' screening values do not predict whether earlier applications are contacted. This evidence is difficult to reconcile with a model in which employers primarily wait to compare the full experimental pool before contacting.

Finally, I consider whether the evidence is better described by batch contact rather than sequential contact initiation. In Figure A.10, I plot the distribution of vacancies by the average number of contacts per day for vacancies with at least one experimental contact. Some contacts are clustered on the same day, which is consistent with partial batching in recruiter workflows, but, for 80% of these vacancies, the average number of experimental contacts per contact day is at most two. However, batch contact is not the same as batch review: the experiment observes contact dates, not internal review dates. Moreover, two facts point toward sequential processing. First, the event-study evidence above excludes same-day submis-

sions and still shows a post-initiation penalty. Second, Figure A.11 shows that applications submitted later are generally contacted later, consistent with arrival-order or near-arrival-order processing. Thus, while the data do not rule out partial batching, the evidence is inconsistent with a pure batch-review model and remains consistent with a sequential contact-initiation framework.

4.6 Summary of Mechanism Evidence

The evidence presented in this section consistently points to a specific mechanism behind the early application advantage. Timing does not appear to matter because early applications are intrinsically higher quality, because early timestamps carry a direct positive premium, or because employers wait to compare the full experimental pool. Instead, firms begin the recruiting pipeline quickly and recruiters often initiate contact before the experimental application pool is complete; applications arriving after initiation face lower contact probabilities; and contact initiation responds to the best applicant observed so far. These facts motivate the sequential screening model introduced in Section 6.

The same contact history framework can also be used to study demographic-group-specific recruiting dynamics. Tables A.18 and A.19 present descriptive evidence that contact histories affect subsequent contact differently by race and gender. Due to data limitations, I leave the interpretation of these group-specific dynamics as an important direction for future work.

The next section zooms out to ask how the strength of this front-loaded screening process varies with local labor market conditions, setting empirical targets for the model.

5 Market Conditions and Application Timing

Section 4 shows that the early application advantage is closely tied to contact initiation: recruiters often begin contacting before the experimental pool has arrived, and applications submitted after this contact phase begins face lower contact probabilities. This section asks whether the strength of this timing gradient varies with local labor market conditions. I focus on labor market tightness because it maps directly into the recruiting tradeoff emphasized by the mechanism: when vacancies

are plentiful relative to available workers, delaying contact may be more costly, making firms more likely to initiate contact early.

The evidence in this section is descriptive. Market conditions are not experimentally assigned, and the experiment spans a period in which local labor markets evolved rapidly. The goal is therefore not to estimate a causal effect of tightness. Instead, I use market-level heterogeneity to assess whether the early application advantage moves in the direction predicted by the contact-initiation mechanism, and to provide pre-model facts that the quantitative framework in Section 6 should rationalize.

I estimate the specification

$$\begin{aligned} Y_i =& \alpha_0 + \delta\, D2F_i^t + \phi\left(D2F_i^t \times Z_{m(i),t(i)}\right) + \phi^{\text{pre}}\left(D2F_i^t \times \bar{Z}_{m(i),t(i)-3:t(i)-1}\right) \\ &+ \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i. \end{aligned} \tag{5.1}$$

Here, $D2F_i^t$ is the number of calendar days between application i's submission date and the first experimental submission date to the same vacancy, top-coded at seven days. Larger values of $D2F_i^t$ therefore correspond to later experimental submissions.[11] δ captures the baseline daily early application advantage, while ϕ captures how this timing gradient varies with the contemporaneous market condition $Z_{m(i),t(i)}$. The market $m(i)$ is the state of the vacancy and $t(i)$ is the month of the vacancy's first experimental submission. $\bar{Z}_{m(i),t(i)-3:t(i)-1}$ is the average of the same market condition over the preceding three months; its interaction with $D2F_i^t$ flexibly controls for persistent local conditions prior to the month of the experimental submissions.

I consider two dependent variables. The first is an indicator for whether application i receives employer contact within 30 days, used in Section 3. The second is the probability that application i is the contact initiator for the vacancy. Because contact is observed at the date level, if multiple experimental applications are contacted on the first contact date, I assign equal weight to each such application so that the initiator probabilities sum to one within a vacancy. All specifications include resume controls $\mathbf{X}_i$ and vacancy fixed effects $\eta_{j(i)}$. Since the market variable varies at the state-by-month level, I report standard errors clustered at both the

[11] The days-to-first-submission measure $D2F_i^t$ is almost perfectly negatively correlated with the days-to-last-submission measure $D2L_i^t$ used in Equations (3.1) and (3.3). Controlling for vacancy fixed effects, regressing $D2F_i^t$ on $D2L_i^t$ yields an estimated coefficient of -0.9748 (s.e. 0.0018).

vacancy level and the state-by-month level. The latter are the more conservative benchmark for inference on ϕ.

Table 6 reports the results when $Z_{m(i),t(i)}$ is log labor-market tightness, defined as vacancies per unemployed worker in the vacancy's state and month. The negative coefficient on $D2F_i^t$ indicates that later experimental submissions are less likely to receive contact and less likely to be the contact initiator, consistent with the early application advantage results in Section 3.2. More importantly, the coefficient for the interaction with tightness is also negative in both columns. Thus, in tighter markets, the daily penalty for arriving later is larger: the early application advantage is steeper, and later applications are less likely to be the first experimental application contacted. The estimates are robust when standard errors are clustered at the vacancy level or at the state-by-month level.

Table 6: Early Application Advantage and Contact Initiation by Market Tightness

	(1) **1** (Contact)	(2) Pr (Contact Initiator)
$D2F_i^t$	-0.0026	-0.0135
	(0.0010)	(0.0006)
	[0.0010]	[0.0008]
$D2F_i^t \times \log(\text{tightness}_{m(i),t(i)})$	-0.0078	-0.0291
	(0.0057)	(0.0040)
	[0.0056]	[0.0049]
Resume Controls	✓	✓
Prior Market Condition	✓	✓
Vacancy FEs	✓	✓
N	78,632	78,632

Note: This table reports estimates of Model (5.1). $D2F_i^t$ is the number of calendar days between application i's submission date and the first experimental submission date to the same vacancy, top-coded at seven. Larger values correspond to later submissions. Column (1) uses an indicator for employer contact within 30 days as the dependent variable. Column (2) uses the probability that application i is the contact initiator for the vacancy. If multiple experimental applications are contacted on the first contact date, each receives weight equal to the reciprocal of the number contacted on that date, so the initiator weights sum to one within the vacancy. $\text{tightness}_{m(i),t(i)}$ is vacancies per unemployed worker in the vacancy's state and month of first experimental submission. All regressions include vacancy fixed effects, resume controls, and the interaction of $D2F_i^t$ with the three-month pre-period average of log tightness. Standard errors clustered at the vacancy level are shown in parentheses. Standard errors clustered at the state-by-month level are shown in brackets. The sample consists of vacancies with eight experimental applications.

A large literature in directed and competitive search emphasizes that firms and workers interact through both matching probabilities and queues (Wolthoff, 2018;

Wright et al., 2021; Carrillo-Tudela, Gartner and Kaas, 2023): recruiting policies determine not only whether a firm fills a vacancy, but also how quickly it does so and how selective it can be. In this sense, the timing margin is part of firms' recruiting technology. When labor markets are tight, firms face a thinner queue of applications and greater job-filling pressure. The finding in Table 6 that the daily timing gradient is steeper in tighter state-month labor markets is therefore consistent with a recruiting-pressure interpretation: recruiters have stronger incentives to initiate contact quickly, and applications arriving after contact initiation are penalized more sharply. I do not interpret this heterogeneity as causal evidence that tightness changes firm behavior in isolation. Rather, I use it as a disciplined descriptive pattern that the model should rationalize.

Tables A.20 and A.21 report analogous specifications using additional state-month labor-market series from JOLTS: hires rate, quits rate, layoffs and discharges rate, and job-opening rate. These series capture distinct dimensions of the recruiting environment that may affect the opportunity cost of delaying contact. The estimates point in the same general direction for contact initiation, but they are noisier for the overall contact outcome. Because these aggregate indicators are correlated and not randomly assigned, I report them both separately and jointly as descriptive robustness exercises rather than as separate causal tests. The additional JOLTS series are not used in the main reduced-form argument. Instead, they inform the macro-state specification in the model, where I combine them into a parsimonious recruiting-state shifter that allows the contact-initiation policy to vary with broader labor-market conditions not captured by tightness alone.

Taken together, the heterogeneity evidence connects the within-vacancy mechanism of contact initiation to broader labor-market conditions. The early application advantage is more pronounced when vacancy-filling pressure is higher. These facts motivate the quantitative framework in the next section, which formalizes the tradeoff between initiating contact quickly and waiting for additional applicants.

6 Modeling Contact Initiation and Firms' Tradeoff

The mechanism evidence points to a dynamic employer-side screening process behind the early application advantage observed in Section 3. Contacts occur quickly, often before the experimental pool is complete. Applications submitted after a va-

cancy enters the contact phase face lower contact probabilities, and contact initiation responds to the best application observed so far rather than to the average quality of prior applications. The timing gradient is also steeper in tighter labor markets, suggesting that the value of waiting for additional applicants varies with recruiting conditions. These facts suggest that firms do not simply wait to compare all experimental applications before initiating contact. Instead, they manage applicant queues through a contact-initiation rule: waiting preserves the option value of additional applicants, while moving quickly advances the recruiting process.

I use a model to formalize this contact-initiation margin and to quantify the associated tradeoff between faster contact initiation and the screening value of the applicant pool considered at initiation.[12] The reduced-form estimates identify how contact probabilities vary with experimentally assigned application timing, but they do not reveal the firm-side policy rule that generates those patterns. The model fills this gap by treating the experimental applications as tracers embedded in a broader applicant flow and by describing the firm's decision as a sequential screening problem.

The model is intentionally partial equilibrium. It focuses on the recruiting subgame conditional on a posted vacancy and a given recruiting environment, abstracting from the complete hiring pipeline and the general-equilibrium feedback from one firm's contact timing. Market conditions enter only through the vacancy's expected applicant inflow and the shadow cost of delaying contact. This scope matches the experimental variation: the design observes how tracer applications are processed within a vacancy, but not the equilibrium process that generated the vacancy or the applicant pool.

The model plays three roles. First, it maps experimental applications to a recruiting process with unobserved organic applicants, so that observed tracer outcomes can be interpreted as signals of a latent contact-initiation policy. Second, it separates two forces that reduced-form timing gradients conflate: the expected flow of additional applicants and the shadow cost of delaying contact. Third, it delivers counterfactual objects that cannot be recovered from the reduced-form re-

[12] I use "screening value" to refer to the employer's revealed-preference ranking of applications based on resume and vacancy observables. It is proxied empirically by the out-of-fold prediction of employer contact introduced in Section 4.3.1. This object is not a direct measure of realized productivity, match surplus, wages, or retention. I use the phrase "speed-quality tradeoff" as shorthand for the tradeoff between earlier contact initiation and this screening-value measure.

sults alone, such as the probability that the highest-screening-value experimental application has not yet arrived at initiation and the associated screening-value loss.

This section proceeds as follows. Section 6.1 defines the recruiting subgame, its scope, and the key economic objects. Section 6.2 characterizes the firm's contact-initiation rule and its comparative statics. Section 6.3 maps the model to the experimental data, and discusses identification, estimation, and model fit. Section 6.4 uses the estimated model to quantify the speed-quality tradeoff and the effect of changes in recruiting pressure. Section 6.5 discusses implications for timing frictions and applicant-side incentives.

6.1 Scope and Environment

Recruiting unfolds in within-vacancy time. Let d denote elapsed time within a vacancy's recruiting spell. In the empirical implementation, vacancy posting dates are not observed, so I normalize $d = 0$ to the first experimental submission within each vacancy. The model should therefore be interpreted as describing the recruiting process over the experimental observation window, not the complete applicant pool from the posting date.

The experimental resumes are tracers inserted into a broader applicant flow. The firm observes both organic and experimental applications, but the econometrician observes only the experimental applications. As a result, the first contacted applicant in the vacancy need not be one of the experimental resumes, and contact initiation may occur through either an experimental tracer or an unobserved organic applicant. Section 6.3 formalizes the mapping between the latent contact-initiation process and the observed tracer outcomes.

Applications for vacancy j arrive sequentially with arrival intensity λ_j.[13] Each reviewed application generates a scalar screening value v_{ij}, which summarizes the firm's expected payoff from entering the contact phase with application i as the current best candidate. This value is a revealed-preference screening value, not realized productivity or total match surplus.[14]

Let d_{ij} denote the arrival day of application i to vacancy j. The key state vari-

[13] I call λ_j an effective arrival intensity because organic applications are not observed directly. The parameter summarizes the model-implied rate at which additional applicant opportunities arrive.

[14] Appendix C.1 discusses the normalization of this latent value scale and its connection to the machine-learning (ML) screening index used in the counterfactual analysis.

able is the best screening value observed so far,

$$v_j^{\max}(d) \equiv \max\{v_{ij} : d_{ij} \leq d\}, \tag{6.1}$$

with the convention that the maximum is undefined before the first reviewed application arrives. This state variable is motivated by the mechanism evidence: contact initiation responds to the best application observed so far, while the average quality of prior applications is not predictive of entering the contact phase.

Delaying contact initiation carries a flow shadow cost κ_j. I interpret κ_j as a composite cost of waiting, not as a behavioral friction or evidence that firms are not optimizing. It may reflect vacancy carrying costs, the risk that promising applicants become unavailable, recruiter capacity constraints, or organizational pressure to move the hiring pipeline forward. The experiment does not separately identify these components. A high κ_j means that the firm behaves as if waiting for additional applicants is costly relative to initiating contact with the current pool.

Given this environment, the firm decides when to enter the contact phase. The firm has recall over applications submitted so far: when it enters the contact phase, it can contact any application that has arrived by that date.[15] It can initiate contact using the best screened applicant in the current pool, obtaining value $v_j^{\max}(d)$. Alternatively, it can wait, pay the shadow cost of delay, and preserve the option value of future arrivals. The next subsection formalizes this tradeoff and derives the reservation rule used in estimation.

6.2 Firm Problem and Contact-Initiation Rule

I now formalize the contact-initiation decision for a single vacancy. Fix a vacancy j and condition on its recruiting environment. To simplify notation, I suppress the vacancy subscript throughout this subsection. In the tractable environment used for estimation, applications arrive according to a Poisson process with rate $\lambda > 0$. Each arrival is reviewed and generates a screening value $v \sim F$, independently across arrivals conditional on the vacancy state. The firm discounts at a daily rate $\delta > 0$ and faces a shadow cost of delay $\kappa \geq 0$.

[15] The model focuses on the initiation margin. The mechanism evidence shows a large decline, but not a collapse to zero, in later contact probabilities. Firms may continue contacting, interviewing, or screening candidates after first contact. Appendix C.3 shows that allowing a monotone post-contact continuation value preserves the threshold structure of the initiation decision.

The firm chooses when to enter the contact phase. If it initiates contact at stopping time τ, it obtains the best screening value of the current pool at that time and pays the flow shadow cost of delay until τ. The firm's problem is

$$\max_{\tau} \mathbb{E}\left[e^{-\delta\tau} v^{\max}(\tau) - \int_0^{\tau} e^{-\delta s}\kappa \mathrm{d}s\right], \tag{6.2}$$

where $\delta > 0$ is the daily discount rate. Equation (6.2) expresses the central trade-off: Waiting preserves the option value of future arrivals, while initiating contact sooner avoids the shadow cost of delay.

Because no new information arrives between applications, it is without loss to evaluate the firm's decision at arrival dates. Let $W(v)$ denote the value of the firm immediately after an arrival when the best screening value observed so far is v. If the firm continues, the next inter-arrival time is exponential with rate λ, and the next screening value v' is drawn from F. The Bellman equation is

$$W(v) = \max\left\{v, -\frac{\kappa}{\lambda+\delta} + \frac{\lambda}{\lambda+\delta}\mathbb{E}\left[W\left(\max\{v, v'\}\right)\right]\right\}. \tag{6.3}$$

The first term inside the maximum is the value of entering the contact phase with the current best application. The second term is the continuation value from waiting until the next application arrival, net of the expected delay cost. The solution has a threshold form.

Proposition 6.1 (Contact-initiation threshold)**.** *Suppose applications arrive according to a Poisson process with intensity $\lambda > 0$, screening values $v' \sim F$ are i.i.d. with finite first moment, the flow shadow cost of delay is $\kappa \geq 0$, and the discount rate is $\delta > 0$. Then there exists a unique reservation value $r \in \mathbb{R}$ such that the firm initiates contact if and only if the current best value satisfies $v^{\max} \geq r$.*

The cutoff follows because the payoff from stopping rises one-for-one with the current best value, whereas the value of continuing rises less than one-for-one: a better current option reduces the incremental value of waiting for a still-better draw. The net gain from waiting is therefore strictly decreasing in the best-so-far value and crosses zero once. Appendix C.2.2 provides the full proof.

At the reservation value, the firm is indifferent between entering the contact

phase and waiting for another draw. The cutoff is characterized by

$$\delta r + \kappa = \lambda \mathbb{E}\left[(v - r)^+\right], \qquad (x)^+ \equiv \max\{x, 0\}. \tag{6.4}$$

The left-hand side is the marginal cost of waiting: delaying contact postpones the current payoff and incurs the flow shadow cost. The right-hand side is the option value of waiting for additional applicants, measured by the expected gain from a future draw that exceeds the current reservation value.

Let

$$p \equiv \Pr(v \geq r) = 1 - F(r) \tag{6.5}$$

denote the probability that an arriving application crosses the contact-initiation threshold. By thinning, threshold-crossing arrivals occur at rate λp, so the contact-initiation time satisfies

$$\tau \sim \text{Exp}(\lambda p), \qquad \mathbb{E}[\tau] = \frac{1}{\lambda p}. \tag{6.6}$$

This expression links the firm problem to the experimental contact-timing moments: contact initiation depends both on how quickly applications arrive and on how selective the firm is.

The reservation rule implies two useful comparative statics. Differentiating Equation (6.4) gives

$$\frac{\partial r}{\partial \kappa} = -\frac{1}{\delta + \lambda p} < 0, \qquad \frac{\partial r}{\partial \lambda} = \frac{\delta r + \kappa}{\lambda(\delta + \lambda p)} > 0. \tag{6.7}$$

A higher shadow cost of delay lowers the reservation value and induces the firm to enter the contact phase with a weaker current pool. A higher arrival intensity raises the option value of waiting and makes the firm more selective. The overall hazard of contact initiation, λp, reflects both forces: more expected arrivals increase the rate at which opportunities arrive, but they also increase the value of waiting for a better draw. The estimation below uses the experimental tracer outcomes to discipline these two channels separately.

The tractable model abstracts from several features of real recruiting. Appendix C.4 allows arrivals to vary with posting age, and Appendix C.5 discusses selective and batch review. These extensions modify the exact mapping from primitives to

the cutoff, but preserve the central insight: contact initiation is governed by the tradeoff between the option value of additional applications and the shadow cost of waiting.

6.3 Parameterization, Experimental Mapping and Estimation

I estimate the model using the same contact-initiation objects introduced in Section 5. For vacancy j, market conditions are measured in the state of the vacancy and the month in which the first experimental application is submitted. Application timing within a vacancy is measured by $d_{ij} = D2F_i^t \in \{0, 1, \ldots, 7\}$, with the final bin top-coded at seven or more days. The model does not assume at most one experimental application per day: if multiple tracers are submitted in the same $D2F_i^t$ bin, the model aggregates their threshold-crossing probabilities within that bin. The empirical contact initiator outcome therefore follows Section 5: if one experimental application is contacted on the first experimental contact date, it receives weight one. If multiple experimental applications are contacted on that date, the initiator weight is split equally across them.

For a parameter vector Θ, the model yields two objects for each vacancy. First, it assigns each experimental application an initiation weight $\omega_{ij}(\Theta)$, the model-implied probability that application i is the experimental contact initiator for vacancy j. Second, it assigns residual mass $\omega_j^{\text{none}}(\Theta)$, the probability that contact initiation is not assigned to any experimental tracer. This residual state includes cases in which an organic applicant initiates contact first, no experimental application is contacted during the observation window, or the relevant contact event is otherwise unobserved. The none state is therefore an empirical residual for the experimental tracer design, not a primitive outcome of the full hiring process. Appendix D.1 gives the exact construction of the empirical and model-implied initiation weights.

The estimated parameters fall into three blocks. The first governs effective arrival intensity, λ_j, which summarizes the rate at which additional applicant opportunities arrive and potentially preempt initiation on the experimental tracers. The second governs the shadow cost of delaying contact, κ_j, which determines how costly it is for the firm to wait for additional applicants rather than enter the contact phase. The third governs the recruiting-state shifter ξ_j, which summarizes

broader market conditions following the congestion logic of urn-ball and directed-search application technologies (Wright et al., 2021; Birinci, See and Wee, 2025). As in Section 5, tightness provides the main reduced-form measure of recruiting pressure. Additional JOLTS series enter only through ξ_j.[16] Appendix D.2 reports the full parameterization and parameter list.

I estimate the model using the method of simulated moments. The moment vector has two families. The first consists of level moments: the distribution of experimental contact initiators across $D2F_i^t$ bins and the residual none state. These moments discipline the frequency and timing with which contact initiation is assigned to experimental tracers. The second is the tightness-gradient moment from Column (2) of Table 6, which disciplines how the contact-initiation policy shifts with recruiting pressure. Let g^{data} denote the empirical moment vector and $g^{\text{sim}}(\Theta)$ the corresponding model-implied vector. The estimator minimizes

$$Q(\Theta) = \left[g^{\text{sim}}(\Theta) - g^{\text{data}}\right]' W \left[g^{\text{sim}}(\Theta) - g^{\text{data}}\right].$$

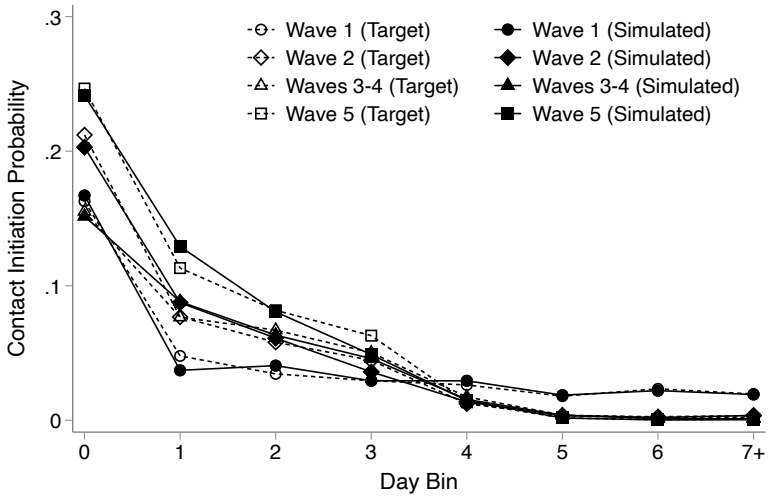
The preferred specification uses diagonal weights based on bootstrap moment variances. This avoids placing excessive weight on small differences among highly correlated day-bin moments. Appendix D.3 describes the full moment vector and bootstrap procedure, and Appendix D.4 reports the full-covariance and wave-specific robustness specifications.

The model is disciplined by contact-initiation moments, and its interpretation should be correspondingly limited. The decomposition between effective arrival intensity and the shadow cost of delay is model-implied under the maintained sequential contact-initiation structure, instead of a nonparametric decomposition. The experiment does not observe the full organic applicant pool or downstream interviews, offers, hires, and wages. The model therefore identifies a contact-initiation policy on the experimental margin, not a complete hiring technology. Because the model's key stopping object is the relative ranking and threshold-crossing behavior of applications, the contact-timing moments identify the contact-initiation policy on a normalized screening-value scale: they discipline the timing and frequency with which initiation is assigned to experimental tracers, and hence

[16] Because the JOLTS series are correlated aggregate indicators of recruiting conditions, I do not interpret the individual loadings as separate causal effects. The relevant object is the fitted recruiting-state shifter ξ_j.

the implied threshold-crossing probabilities.[17] Appendix D.5 discusses the identification strategy and reports the parameter estimates.

Figure 5: Level Moments, Simulation versus Empirical Targets



Notes: The figure plots the empirical and simulated level moments: the contact-initiation probability mass by day bin and wave group. The model-implied moments use the macro-disciplined specification with diagonal weighting. Empirical targets are marked with hollow markers and dashed lines. Model-implied moments are shown with solid markers and solid lines. Both the empirical calculations and model simulations use vacancies with the full set of eight experimental applications.

Figure 5 compares the empirical and model-implied level moments. The model reproduces the front-loaded distribution of experimental contact initiation across the displayed day bins. Appendix D.6 reports detailed fit diagnostics and robustness checks. I use the fitted model in the next subsection to quantify the screening-value cost of faster contact initiation.

6.4 Counterfactuals: Early Contact Initiation and Screening Value

I use the estimated model to quantify the screening-value cost of early contact initiation. The model delivers initiation weights over experimental applications

[17] The ML screening index introduced in Section 4.3.1 serves as a noisy revealed-preference proxy for screening value. Estimating the screening-value distribution directly from this ML index would require treating the index scale as cardinal and free of measurement error. Unreported estimation exercises using the ML index for the quality distribution yield fragile results. I therefore use the ML screening index post-estimation to quantify screening-value losses, rather than treating it as an active primitive in the stopping problem. Appendix C.1 discusses this normalization.

and initiation dates. The screening-value index introduced in Section 4.3.1 ranks experimental applications by employers' revealed preferences. Combining these objects allows me to ask how often contact begins before the highest-screening-value experimental application has arrived, and how large the resulting screening-value shortfall is.[18]

Timing-loss measures. Let I_j denote the eight experimental applications to vacancy j, and let A_{ij} denote the ML screening-value index for application i. Let $A_j^{\max} \equiv \max_{i \in I_j} A_{ij}$ be the highest screening value among all experimental applications, and let $A_j(d) \equiv \max_{i \in I_j : d_{ij} \leq d} A_{ij}$ be the highest screening value in the current pool by initiation day d. If initiation is assigned to tracer i on day d, the gap between the highest-screening-value tracer and the tracer assigned initiation weight decomposes as

$$A_j^{\max} - A_{ij} = \underbrace{A_j^{\max} - A_j(d)}_{\text{timing component}} + \underbrace{A_j(d) - A_{ij}}_{\text{within-pool value gap}} . \tag{6.8}$$

The first component is the gap between the full-pool maximum and the maximum available at initiation. It is positive when the highest-screening-value experimental application has not yet arrived when contact begins. The second is the gap between the maximum available at initiation and the screening value of the tracer assigned initiation weight.[19]

The counterfactual analysis focuses on the timing component. I report two loss measures: The *Missed-Best Probability* is the probability that the highest-screening-value experimental application has not yet arrived when contact initiation occurs. The *Relative Timing Loss (RTL)* is the expected timing component in Equation (6.8), scaled by the expected screening value of the application receiving initiation weight. Both objects use the model-implied initiation weights and condition on

[18] The loss measures cannot be recovered from reduced-form contacts alone because the experiment observes only tracer outcomes. When no experimental application is the initiator, the data do not reveal whether an organic applicant initiated contact, or whether the vacancy remained inactive. The model supplies the missing initiation weights over experimental applications and the residual none state, allowing the screening-value losses to be averaged over the same latent contact-initiation process used in estimation.

[19] The within-pool value gap is positive when the tracer assigned initiation weight has a lower estimated screening-value index than the highest-index tracer available by day d. It is an accounting object, not a measure of employer screening inefficiency.

initiation being assigned to an experimental tracer. Appendix E.1 discusses the full construction.

Timing loss and the shadow cost of delay κ. The timing-loss measures are the quality-side counterpart to the shadow cost of delay, κ_j. In the model, firms initiate contact early because waiting is costly: κ_j summarizes the ex ante value of moving the contact process forward rather than waiting for additional applications. The timing-loss measures quantify the other side of the same stopping problem: the ex post screening value forgone when contact begins before later experimental applications arrive. Early initiation is privately optimal whenever the expected screening-value gain from waiting is smaller than the shadow cost of delay. The timing losses should therefore be interpreted as break-even screening-value benchmarks for κ_j: they characterize the quality-side gain from waiting that firms forgo when moving faster, not the net welfare effect of delaying contact.

Baseline and one-day shift counterfactuals. Table 7 reports the baseline timing losses with bootstrap 95% confidence intervals. In the preferred specification, contact initiation is assigned to an experimental tracer with probability 0.4079. Conditional on experimental initiation, the Missed-Best Probability is 0.5156 and the Relative Timing Loss is 0.0656. Appendix Table E.1 reports the same baseline objects by wave group.

Table 7 also reports one-day shift benchmarks that provide a scale for the speed-quality tradeoff. They mechanically shift the model-implied experimental initiation day distribution one day earlier or later, holding fixed the probability that initiation is assigned to an experimental tracer and the within-day allocation of initiation weights.[20] The exercises do not re-solve the firm's stopping problem and should not be interpreted as policy experiments. Instead, they show how much screening value would be gained or lost under these one-day shift rules along the contact-initiation margin. A one-day earlier shift raises the Missed-Best Probability by 0.1175 and the Relative Timing Loss by 0.0217, while a one-day later shift

[20] Conditional on experimental initiation, the expected initiation-day bin is 1.213 at baseline. The one-day earlier shift reduces the conditional expected initiation-day bin by 0.450 to 0.763, while the one-day later shift increases it by 0.719 to 1.932. The final bin is top-coded at seven. The realized shifts are smaller than one because initiation weight remains in place at the boundaries or when the destination bin contains no experimental application.

Table 7: Baseline Timing Losses and One-Day Shift Benchmarks

	$\bar{P}^{\exp}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	0.4079	0.5156	0.0656
	[0.3985, 0.4177]	[0.5069, 0.5234]	[0.0642, 0.0674]
One Day Earlier		0.6330	0.0873
		[0.6270, 0.6379]	[0.0859, 0.0889]
Δ from Baseline		0.1175	0.0217
		[0.1144, 0.1209]	[0.0211, 0.0223]
One Day Later		0.3442	0.0363
		[0.3370, 0.3514]	[0.0354, 0.0373]
Δ from Baseline		-0.1713	-0.0292
		[-0.1733, -0.1691]	[-0.0301, -0.0285]

Note: This table reports the model-implied probability of experimental contact initiation and the two timing-loss measures defined in Section 6.4, computed on vacancies with eight experimental submissions. $\bar{P}^{\exp}(\widehat{\Theta})$ is the sample-average probability that initiation is assigned to one of the experimental applications. Missed-Best Probability is the probability that the highest-screening-value experimental application has not yet arrived when contact initiation occurs. Relative Timing Loss is the timing component in Equation (6.8), scaled by the expected screening value of the application receiving initiation weight. The one-day benchmarks shift each vacancy's experimental initiation weights to the adjacent day bin whenever that bin lies between 0 and 7 and contains an experimental application; otherwise, the weights remain in their original bin. The probability of experimental initiation is held fixed, and weights are allocated within each destination bin using the baseline application shares. Brackets report 500-draw vacancy-clustered bootstrap 95% confidence intervals. The screening-value index is described in Section 4.3.1 and Appendix B.

lowers them by 0.1713 and 0.0292.

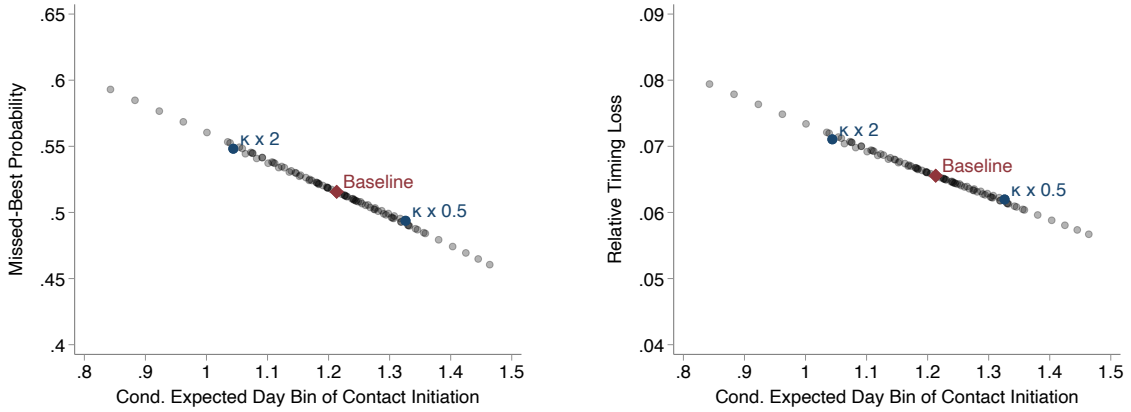
Speed-loss frontier across counterfactuals. Figure 6 plots contact timing and timing losses across counterfactual changes in the shadow cost of delay, labor-market tightness, and aggregate recruiting conditions. For each environment, I recompute the model-implied contact rule and initiation weights while holding the experimental application schedules and screening-value index fixed.

The highlighted scenarios illustrate the response to changes in the shadow cost of delay. Halving κ_j raises the conditional expected initiation-day bin from 1.213 to 1.326 and lowers Relative Timing Loss from 0.0656 to 0.0620. Doubling κ_j lowers the expected bin to 1.044 and raises Relative Timing Loss to 0.0711. Appendix Table E.2 reports these scenarios and selected tightness counterfactuals. Across the plotted counterfactuals, the fitted relationship has a slope of approximately -0.036 in Relative Timing Loss units per expected initiation-day bin.[21] The one-

[21]The slope is the coefficient from an OLS regression of Relative Timing Loss on the conditional expected initiation-day bin, including a constant and giving each point in Figure 6 equal weight.

day benchmarks in Table 7 mechanically shift initiation weights while preserving each vacancy's probability of experimental initiation. The parameter counterfactuals allow both the contact rule and that probability to adjust, changing the relative weights of vacancies in the conditional timing and loss measures. The speed-loss frontier thus quantifies the screening-value cost of firms' optimal contact timing across recruiting environments.

Figure 6: Contact Initiation and Timing Loss across Counterfactuals



(a) Missed-Best Probability

(b) Relative Timing Loss

Notes: The figures plot timing-loss measures against the conditional expected initiation-day bin for vacancies with eight experimental applications under the preferred specification. Counterfactuals vary the shadow cost of delay, labor-market tightness, or the aggregate hires, quits, and layoffs and discharges rates entering the recruiting-state shifter ξ_j. Each point reports the total model response after recomputing the contact rule and initiation weights, holding experimental application schedules and the screening-value index fixed. The red diamond marks the baseline. The labeled blue dots halve or double κ_j for every vacancy. Expected timing and the loss measures condition on experimental initiation, with the final day bin top-coded at seven. Missed-Best Probability and Relative Timing Loss are defined in Section 6.4 and Appendix E.1.

Back-of-the-envelope wage equivalent. As a scale benchmark, I translate Relative Timing Loss into wage-equivalent units with a conservative adjustment,

$$\text{Wage-equivalent timing loss} = \text{RTL} \times \bar{w} \times \Delta p_{\text{contact}} \times \pi_{H|C}, \tag{6.9}$$

where the wage benchmark $\bar{w}$ is set to \$47,150,[22] $\Delta p_{\text{contact}}$ is the estimated relative post-initiation decline in contact probability, 63.12%, from Table A.11, and $\pi_{H|C}$ is a

For comparison, dividing the loss changes under the earlier and later mechanical shift rules by their realized changes in expected timing gives slopes of approximately -0.048 and -0.041, respectively.

[22]The wage benchmark is the 2024 median annual wage for occupations whose typical entry-level education is a high school diploma or equivalent (U.S. Bureau of Labor Statistics, 2025).

contact-to-hire pass-through rate between 0.20 and 0.40.[23] $\Delta p_{\text{contact}}$ recognizes that contact initiation does not fully stop later recruiting. $\pi_{H|C}$ adjusts for the fact that the experiment observes contacts rather than hires.

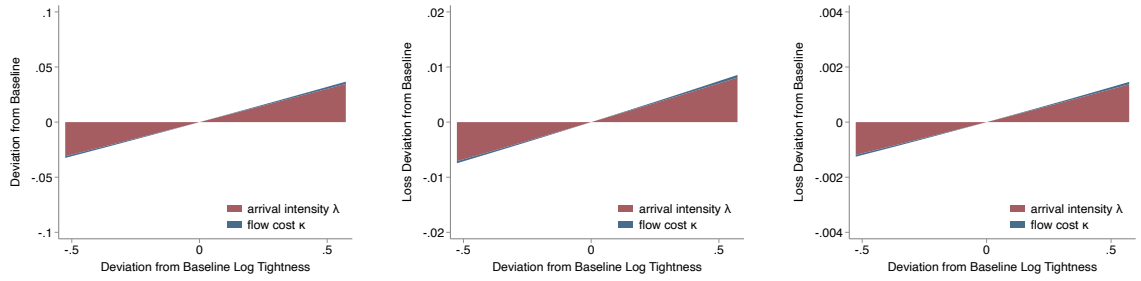
This calculation provides a scale conversion for the quality-side cost of early contact initiation after adjusting for post-initiation continuation and the contact-to-hire pipeline. Using the baseline Relative Timing Loss in Table 7, Equation (6.9) implies a wage-equivalent timing loss of approximately \$390 to \$781 per vacancy-year.

Recruiting pressure counterfactuals. Section 5 shows that the timing gradient is steeper in tighter labor markets. I use the estimated model to ask how the screening-value cost of timing changes as local tightness varies over its empirical support. Tightness enters the model through two channels: the congestion term in effective arrival intensity λ and the tightness loading in the shadow cost of delay κ. Figure 7 reports a continuous tightness counterfactual and decomposes the change in each outcome into the two channels using a Shapley decomposition. This decomposition is a model-implied diagnostic, not a separate nonparametric identification of the two channels.

The counterfactuals reinforce the reduced-form tightness evidence. As recruiting pressure rises, contact initiation moves earlier on the experimental margin and timing losses increase. In the preferred specification, most of the fitted tightness response operates through the effective-arrival-intensity channel rather than through the shadow-cost channel. This suggests that, within the model, tighter labor markets primarily change the applicant-arrival environment faced by vacancies, rather than operating only through a higher shadow cost of delay. I interpret this channel decomposition cautiously: it is the decomposition that best rationalizes the targeted timing moments under the maintained contact-initiation structure. Appendix E.2 discusses details of the Shapley decomposition and reports the

[23] Direct evidence on the conversion from correspondence-study contact to hire is limited. I benchmark several studies from related settings that document post-contact evidence to establish the pass-through rate range: Zuchuat et al. (2023) document a 25% job offer rate conditional on receiving an interview using Swiss job-search diaries covering over 600,000 applications. Brown, Setren and Topa (2016) report 23.6% offers per interview and 18.8% hires per interview using applicant-tracking data from a large U.S. financial-services firm. Cowgill and Perkowski (2024) report that recruiters in a two-sided audit study expect a 39% hire rate conditional on being offered an interview.

Figure 7: Labor Market Tightness and Timing Losses



(a) $\bar{P}^{\exp}(\widehat{\Theta})$ (b) Missed-Best Probability (c) Relative Timing Loss

Notes: The figure plots model-implied changes in experimental initiation probability and timing-loss measures as log tightness is shifted relative to its baseline value. Each panel decomposes the change into the effective-arrival-intensity channel and the shadow-cost-of-delay channel using a Shapley decomposition. The decomposition is based on the preferred specification and is computed on vacancies with eight experimental applications. Missed-Best Probability and Relative Timing Loss are defined in Section 6.4.

same exercise by wave group and for the full initiation-day distribution.

Takeaway. The counterfactuals quantify the screening-value cost of speed in the contact-initiation process. They do not imply that firms make mistakes: early contact may be privately optimal when the shadow cost of delay exceeds the expected screening-value gain from waiting. But they show that the timing structure of recruiting has economically meaningful consequences. Earlier contact initiation increases the probability that high-screening-value applications have not yet arrived, and tighter markets amplify this timing cost. At a given initiation date, better screening of applications already received cannot eliminate the timing loss from better applications that arrive later. Appendix Figure E.1 illustrates this point: reallocating initiation weight to the highest-screening-value applications within each submission-day bin leaves Relative Timing Loss largely unchanged. The next subsection discusses how this timing friction can generate applicant-side incentives to compete on speed and why such incentives matter for welfare and inequality.

6.5 Implications: Timing Frictions in Recruiting

The model clarifies the structural source of the timing friction: applications arrive sequentially, and firms often enter the contact phase before the experimental pool is complete. This creates a queue-position wedge in consideration probabilities.

Otherwise comparable applicants can face different probabilities of contact simply because they arrive at different points in the vacancy's recruiting process.

For firms, the model quantifies their tradeoff when choosing how quickly to move candidates into the contact phase. Initiating contact early may be optimal when vacancies are costly to leave open, when applicants may become unavailable, or when recruiters face internal capacity constraints. But it can mean missing higher-screening-value applications that arrive later.[24] These results highlight a time margin of firm competition. Firms compete not only through wages, amenities, and vacancy creation, but also through the speed with which they move applicants through the recruiting pipeline. Descriptive heterogeneity shows that the early application advantage varies with firm prominence, offer competitiveness, and recruiting organization.[25] These patterns provide external-validity support for the model and suggest that firms' incentives to compete on speed vary with effective arrival intensity and the shadow cost of delay.

For applicants, the same timing wedge creates a private return to speed. If contact probabilities depend on queue position, job seekers have incentives to reallocate effort toward speed: monitoring postings closely, applying quickly, and compressing preparation time. In equilibrium, the timing premium can therefore generate an arms race: applicants spend resources to move earlier in the queue, but the aggregate quality of matches need not improve. Because the ability to invest in speed is unevenly distributed, the timing premium can also have distributional consequences. Workers with caregiving responsibilities, inflexible schedules, weaker job-search networks, or less access to automated alerts may be less able to compete on this margin. This channel can generate disparities even when employers treat otherwise identical observed applications similarly, because the relevant constraint may operate before the application is submitted or observed.[26]

[24] Carrillo-Tudela, Gartner and Kaas (2023) find that, at the establishment level, faster hiring is associated with lower hiring standards and more generous wages.

[25] In Appendix Figure F.3, larger firm shares of local postings are associated with a weaker early application advantage in Waves 1 and 5, although the relationship is close to zero in Waves 2-4. Appendix Figure F.4 shows a steeper advantage at firms advertising lower salaries relative to comparable local entry-level postings. Appendix Figure F.5 shows a steeper advantage at firms with more centralized callback systems. I use the centralization measure constructed by Kline, Rose and Walters (2022), in which fewer distinct callback phone numbers per vacancy with at least one callback indicate greater centralization.

[26] Gender provides one example. Some correspondence experiment studies find small or no average gender gaps in employer contact conditional on submitted applications. A timing premium

At the market level, the results highlight within-vacancy recruiting timing as an employer-side margin often hidden inside aggregate matching functions. The reduced-form tightness gradients and the estimated recruiting-state shifter ξ_j suggest that this timing margin varies with labor-market conditions. These results provide a way to microfound the cyclicality of matching efficiency through changes in employer-side screening and contact timing. Aggregate search models often summarize variation not explained by unemployment and vacancies as matching efficiency (Barnichon and Figura, 2015), and micro evidence shows that firms adjust recruiting intensity and hiring standards over the cycle (Davis, Faberman and Haltiwanger, 2013; Carrillo-Tudela, Gartner and Kaas, 2023). The model estimates a micro analogue of this margin at an earlier stage of the hiring process: how quickly vacancies move from receiving applications to initiating contact.

Finally, the framework clarifies how new screening technologies may change, but not necessarily eliminate, timing frictions. More efficient screening may reduce the cost of ranking applications that have already arrived and may compress the initial screening stage. But as long as applications arrive sequentially and firms make decisions before later candidates appear, the queue-position wedge remains. In the extreme case in which job seekers also adopt technology that submits applications immediately after posting, the contact-stage timing game may unravel toward the posting date. Even then, as long as hiring remains sequential, similar timing wedges can reappear later in the pipeline: in interview scheduling, offer timing, candidate follow-up, or final selection. The model therefore provides a framework for studying how the timing of employer decisions shapes who is considered as recruiting technology and labor-market conditions change.

7 Conclusion

This paper studies how application timing affects employer consideration through firms' sequential screening. In many recruiting environments, workers apply sequentially and firms do not necessarily wait for the full applicant pool before be-

points to a different margin: if women face greater constraints in monitoring postings or submitting quickly, equal treatment conditional on employer observation need not imply equal access to consideration. Potential sources of such constraints include overpreparation or underconfidence (Aksoy, Exley and Kessler, 2024; Tekleselassie et al., 2025), marriage or household constraints (Bandiera, Jalal and Roussille, 2025), and differential domestic responsibilities (Philippe and Skandalis, 2026).

ginning the contact process. This creates a queue-position margin that is difficult to study in observational data because earlier applicants may differ in motivation, information, or quality. I use experimentally timed resume submissions to trace employer-side screening within vacancies, holding resume content fixed while varying when applications arrive.

The central empirical finding is a causal early application advantage. Earlier experimental submissions are more likely to receive employer contact, and the timing gradient is larger in tighter local labor markets. The mechanism evidence identifies contact initiation as the key change in the recruiting process. Contacts occur quickly, applications submitted after a vacancy enters the contact phase face lower contact probabilities, and contact initiation responds to the highest-screening-value application observed so far. In contrast, the evidence does not support explanations based primarily on applicant selection, firms waiting to compare the full experimental pool, or a direct premium on early submission, for example, because an early timestamp signals motivation.

To organize these facts, I develop a partial-equilibrium model of contact initiation. Firms choose when to enter the contact phase by solving an optimal stopping problem, trading off the option value of waiting for additional applications against a shadow cost of delay. The model separates the rate at which additional applicant opportunities arrive from the shadow cost of delay and maps experimental application outcomes into a contact-initiation policy. This structure allows me to quantify the screening-value cost of faster initiation while remaining explicit about the object being identified: a contact-initiation policy on the experimental margin, not the full hiring technology.

The counterfactuals show that the timing structure of recruiting creates a meaningful speed-quality tradeoff. Faster contact initiation raises the probability that the highest-screening-value experimental application has not yet arrived, and it increases the screening-value loss associated with moving before later applications enter the pool. The speed-loss frontier quantifies the screening value forgone when privately optimizing firms move faster in response to delay costs and applicant arrival conditions. At a given initiation date, better ranking among applications already in the pool cannot eliminate the timing loss from higher-screening-value applications that arrive later. This distinction is important: recruiting frictions arise not only from imperfect screening, but also from the timing structure of the recruit-

ing process itself.

The results have important implications for firms, applicants, and aggregate matching. For firms, the relevant margin is a choice over recruiting speed: moving quickly may be optimal when vacancies are costly to leave open, but it also reduces the option value of later applications. For applicants, the timing premium creates a private return to speed that can incentivize job seekers to monitor postings more frequently, apply quickly, or compress preparation time, even when these investments do not improve match quality. Therefore, the private return to early application may exceed its social return. The experiment does not measure worker-side constraints directly, but it shows that the labor market rewards the application speed margin on which such constraints could matter for both efficiency and distribution. At the market level, this paper identifies a time margin of firm competition that is often hidden inside aggregate matching functions. Firms compete not only through wages, amenities, vacancy creation, and recruiting intensity, but also through how quickly they move applicants through the recruiting pipeline. The tightness gradients and the model's recruiting-state shifter connect this within-vacancy margin to the cyclicality of matching efficiency.

From an institutional-design perspective, the findings suggest that rules governing application visibility and review timing can shape who receives consideration. Application windows, batching rules, timestamp visibility, recruiter dashboards, and information about vacancy status can all affect whether queue position translates into contact probability. The relevant challenge is not simply to make firms wait, but to reduce timing-driven consideration wedges while preserving firms' ability to move quickly when waiting is costly. This distinction is especially important as recruiting technology changes. Recent work shows that AI-powered tools can alter both worker search and employer screening (Le Barbanchon, Hensvik and Rathelot, 2026; Wiles, Munyikwa and Horton, 2025; Aka et al., 2025). My results suggest a complementary point: improved technology may compress particular stages of recruiting, but sequential timing remains a distinct friction that can persist or reappear elsewhere in the hiring pipeline.

The framework opens a broader research agenda. A first extension is to connect contact initiation to downstream hiring stages while allowing workers to choose both when and where to apply. A general-equilibrium model in which firms jointly choose wages, amenities, recruiting intensity, and contact speed would make tim-

ing an endogenous dimension of competition. Such a framework could study how market concentration and monopsony shape firms' willingness to wait, how workers sort across vacancies with heterogeneous queues and initiation policies, and how these choices feed back into matching efficiency and inequality. The descriptive heterogeneity by firm prominence, offer competitiveness, and recruiting organization documented in this paper motivates these questions, while leaving the equilibrium sorting mechanism for future work. A complementary direction is to study how attention, recall, and information acquisition evolve within recruiting pipelines. Firms may differ not only in when they initiate contact, but also in which applications they review, which previously observed candidates they recall, and how contact changes the value of acquiring further information. New technologies may alter each of these margins and reallocate timing frictions across stages of the hiring process.

More broadly, the paper provides a framework for treating within-vacancy timing as an endogenous component of matching. Standard search models often abstract from the sequence of evaluation, contact, interviews, and offers; the evidence here shows that this sequence shapes both who is considered and how market conditions translate into matches. Bringing empirically disciplined employer-side dynamics into search models offers a route to understanding not only how many matches are formed, but how workers are selected into them. A richer theory of matching should therefore explain not only whether workers and vacancies meet, but how the dynamics of employer decisions determine who ultimately matches.

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A Additional Empirical Results

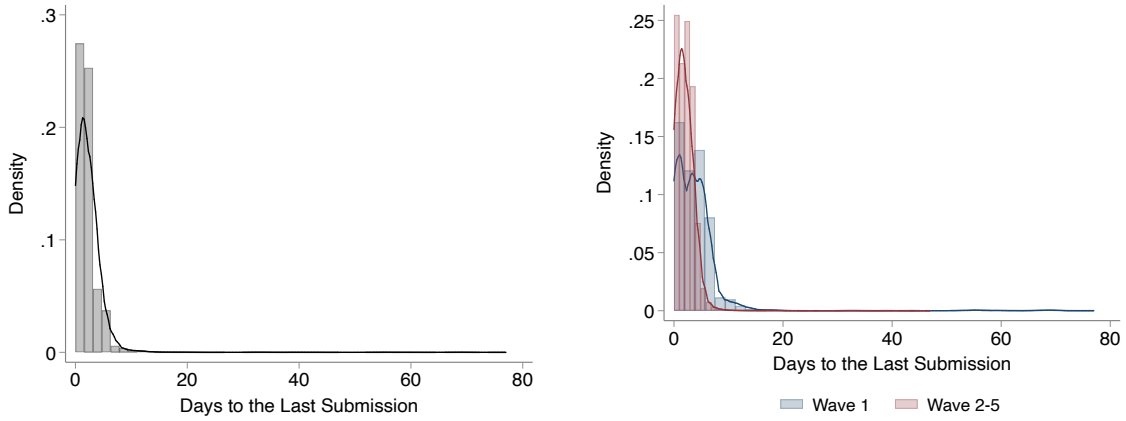
A.1 Timing Measures and Sample Construction

Table A.1: Application Counts by Realized Submission Order

Order	(1) Full	(2) (< 8)	(3) Wave 1	(4) Wave 2-5
1	11,114	1,285	2,063	9,051
2	11,031	1,202	2,040	8,991
3	10,682	853	1,883	8,799
4	10,606	777	1,879	8,727
5	10,278	449	1,748	8,530
6	10,204	375	1,734	8,470
7	9,899	70	1,604	8,295
8	9,829	0	1,590	8,239
N applications	83,643	5,011	14,541	69,102
N vacancies	11,114	1,285	2,063	9,051
N firms	108		85	

Note: This table summarizes the number of applications by realized submission order from 1 to 8. *Full* refers to the full sample; (< 8) refers to vacancies with fewer than eight submissions in the full sample; *Wave 1* refers to the Wave 1 subsample; and *Wave 2-5* refers to the Wave 2-5 subsample.

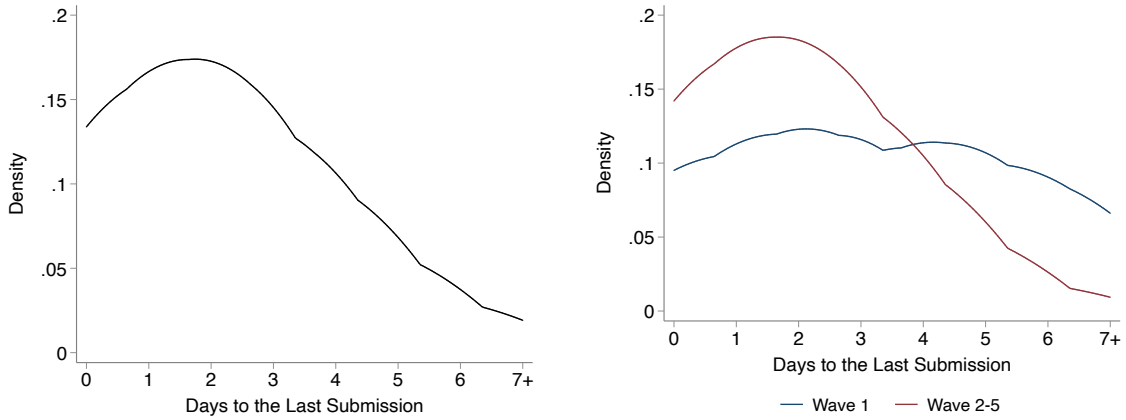
Figure A.1: Distribution of Days to the Last Submission $(D2L)$



(a) Full Sample (b) Wave 1 versus Wave 2-5

Notes: The variable $D2L$ is constructed as in Section 3.1. The distribution of all applications is depicted in the left panel; the distributions of applications in Wave 1 and Wave 2-5 are depicted separately in the right panel. In both panels, an Epanechnikov kernel is used, with the kernel half-width set to 1.

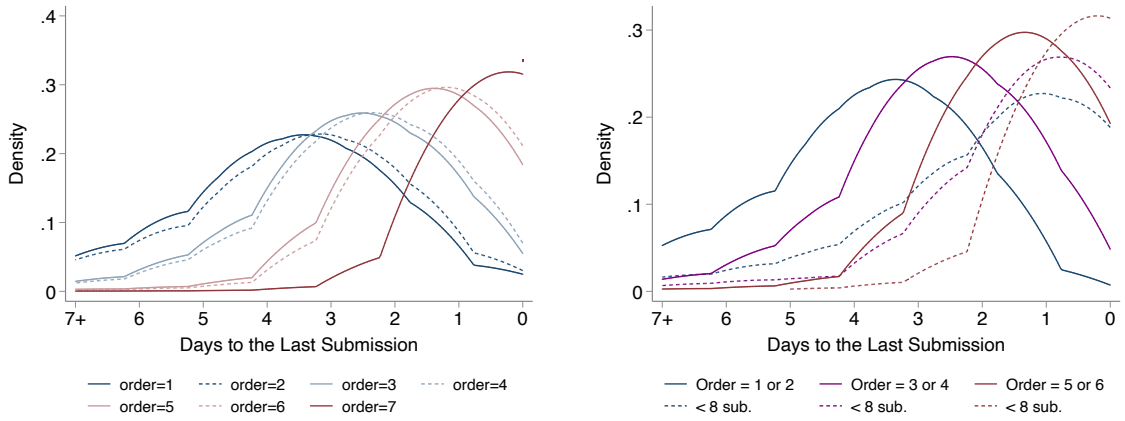
Figure A.2: Distribution of Days to the Last Submission, Top-Coded $(D2L^t)$



(a) Full Sample (b) Wave 1 versus Wave 2-5

Notes: The variable $D2L^t$ is constructed as in Section 3.1. The distribution of all applications is depicted in the left panel; the distributions of applications in Wave 1 and Wave 2-5 are depicted separately in the right panel. In both panels, an Epanechnikov kernel is used, with the kernel half-width set to 1.5.

Figure A.3: Distribution of $D2L^t$, by Submission Order



(a) Full Sample

(b) Complete versus Incomplete

Notes: The variable $D2L^t$ is constructed as in Section 3.1. In the left panel, applications are grouped by the within-vacancy orders of submission. In the right panel, applications with submission orders 1/2, 3/4, and 5/6 are grouped together. The solid lines correspond to vacancies with eight experimental submissions (*Complete*); the dashed lines correspond to vacancies with fewer than eight submissions (*Incomplete*). In both panels, an Epanechnikov kernel is used, with the kernel half-width set to 1.

Table A.2: Balance of Timing Measures on Randomized Resume Characteristics

Dep.Var.	(1) $early^1$	(2) $early^2$	(3) $early^3$	(4) $Order$	(5) $Order^{pair}$	(6) $D2L^t$
Specification A: with Vacancy FEs, standard errors clustered at the vacancy level						
Balanced	1.55	1.37	1.19	0.73	1.02	1.25
	[0.1228]	[0.1938]	[0.2967]	[0.6864]	[0.4212]	[0.2568]
Full	1.07	0.98	0.83	0.63	0.83	0.97
	[0.3831]	[0.4564]	[0.5857]	[0.7692]	[0.5876]	[0.4618]
Complete	0.99	0.99	0.82	0.53	1.11	0.92
	[0.4481]	[0.4481]	[0.5988]	[0.8540]	[0.3519]	[0.5040]
Incomplete	1.10	1.06	0.87	0.93	1.03	0.75
	[0.3566]	[0.3931]	[0.5478]	[0.5007]	[0.4142]	[0.6610]
Specification B: with Vacancy FEs, heteroskedasticity-robust standard errors						
Balanced	1.24	1.25	1.21	0.81	0.86	0.62
	[0.2647]	[0.2582]	[0.2823]	[0.6096]	[0.5593]	[0.7791]
Full	1.03	1.01	0.87	0.68	0.59	0.61
	[0.4130]	[0.4252]	[0.5497]	[0.7313]	[0.8043]	[0.7879]
Complete	0.93	0.93	0.83	0.59	0.56	0.51
	[0.4989]	[0.4989]	[0.5871]	[0.8066]	[0.8282]	[0.8676]
Incomplete	1.26	1.16	0.91	0.94	1.15	0.95
	[0.2520]	[0.3192]	[0.5191]	[0.4856]	[0.3255]	[0.4792]
Specification C: with wave and region FEs, standard errors clustered at the vacancy level						
Balanced	1.97**	1.66*	1.16	1.13	2.03**	1.38
	[0.0388]	[0.0938]	[0.3181]	[0.3395]	[0.0319]	[0.1929]
Full	1.33	1.19	0.90	0.81	1.39	1.09
	[0.2129]	[0.2943]	[0.5221]	[0.6086]	[0.1864]	[0.3692]
Complete	1.13	1.13	0.94	0.61	1.27	1.05
	[0.3383]	[0.3383]	[0.4911]	[0.7934]	[0.2454]	[0.4004]
Incomplete	0.55	0.82	0.52	1.61	1.69*	0.32
	[0.8377]	[0.5972]	[0.8627]	[0.1065]	[0.0860]	[0.9691]

Note: Each entry reports the joint F-statistic from a regression of the indicated timing measure on the randomized resume characteristics used by Kline, Rose and Walters (2022). p-values are reported in brackets. Each column refers to a constructed timing measure indicated by the column title as described in Section 3.1. $Order^{pair}$ refers to the pair order of submission i to vacancy $j(i)$. $early^2$ is defined such that the last four submissions to a vacancy are coded as late. For vacancies with no more than four submissions, all submissions are coded as late. In the full sample, 40,210 (48.07%) are coded as early submissions. $early^3$ is based on the designed pair orders and defined such that the last two pairs of submissions to a vacancy are coded as late. For vacancies with no more than two pairs of submissions (836 vacancies, 7.52%; 2,321 submissions, 2.77%), all submissions are coded as late. The four rows in each panel correspond to the balanced sample, the full sample, vacancies with eight submissions (*Complete*), and vacancies with fewer than eight submissions (*Incomplete*). *Balanced* refers to the balanced sample constructed by Kline, Rose and Walters (2022) (N = 65,368). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

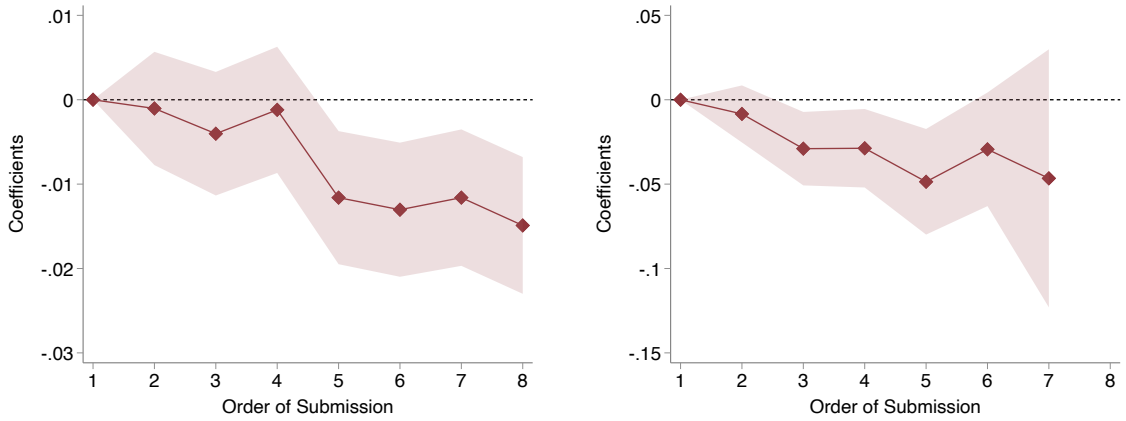
A.2 Robustness of the Early Application Advantage

Table A.3: Early Application Advantage, Incomplete Vacancies

	(1) Full	(2) Incomplete	(3) Full	(4) Incomplete
$early_i^1$	0.0116	0.0240		
	(0.0024)	(0.0127)		
$D2L_i^t$			0.0040	0.0087
			(0.0008)	(0.0035)
Black	-0.0205	-0.0125	-0.0205	-0.0125
	(0.0018)	(0.0066)	(0.0018)	(0.0066)
Female	-0.0021	-0.0006	-0.0021	-0.0007
	(0.0022)	(0.0080)	(0.0022)	(0.0080)
Over 40	-0.0043	0.0041	-0.0043	0.0034
	(0.0020)	(0.0083)	(0.0020)	(0.0083)
Pol. club	0.0059	0.0037	0.0060	0.0026
	(0.0051)	(0.0251)	(0.0051)	(0.0253)
Acad. club	0.0049	0.0136	0.0050	0.0136
	(0.0051)	(0.0175)	(0.0051)	(0.0174)
LGBTQ club	-0.0003	0.0152	-0.0002	0.0154
	(0.0036)	(0.0171)	(0.0036)	(0.0171)
Typical pron.	-0.0077	0.0056	-0.0078	0.0040
	(0.0051)	(0.0203)	(0.0051)	(0.0202)
Neutral pron.	-0.0086	0.0065	-0.0085	0.0061
	(0.0051)	(0.0217)	(0.0051)	(0.0216)
Assoc. degree	0.0021	0.0082	0.0021	0.0083
	(0.0020)	(0.0085)	(0.0020)	(0.0085)
Constant	0.2475	0.1680	0.2453	0.1668
	(0.0025)	(0.0097)	(0.0027)	(0.0093)
Vacancy FEs	✓	✓	✓	✓
N	83,643	5,011	83,643	5,011

Note: This table presents the effects of early submission on the contact probability. The outcome is employer contact within 30 days. Columns (1) and (2) estimate Equation (3.1) with $T_i = early_i^1$; columns (3) and (4) estimate the same specification with $T_i = D2L_i^t$. Columns titled "Full" include all vacancies, while columns titled "Incomplete" only include vacancies with fewer than eight experimental submissions. The construction of $early_i^1$ and $D2L_i^t$ is described in Section 3.1. All regressions include vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

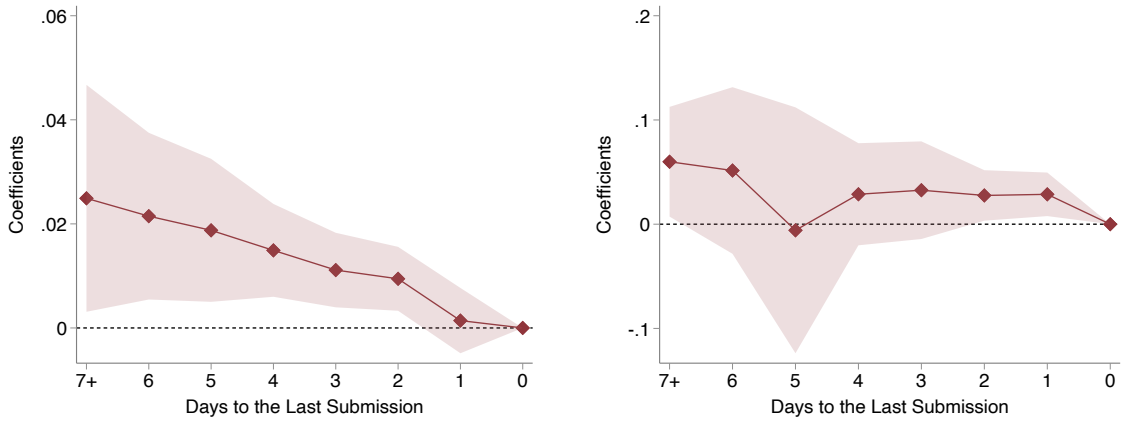
Figure A.4: Contact Gap and Order of Submission, *Complete* versus *Incomplete* Vacancies



(a) *Complete* Vacancies ($N = 78{,}632$) (b) *Incomplete* Vacancies ($N = 5{,}011$)

Notes: The figures plot the estimated coefficients $\hat{\beta}_k, k \in \{2, \cdots, 8\}$ of Equation (3.2) with the order of submission on the x-axis, controlling for resume characteristics. In the left panel, the sample consists of vacancies with eight submissions. In the right panel, the sample consists of vacancies with fewer than eight submissions. Each regression includes vacancy fixed effects and all resume characteristics. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

Figure A.5: Contact Gap and $D2L^t$, *Complete* versus *Incomplete* Vacancies



(a) *Complete* Vacancies ($N = 78{,}632$) (b) *Incomplete* Vacancies ($N = 5{,}011$)

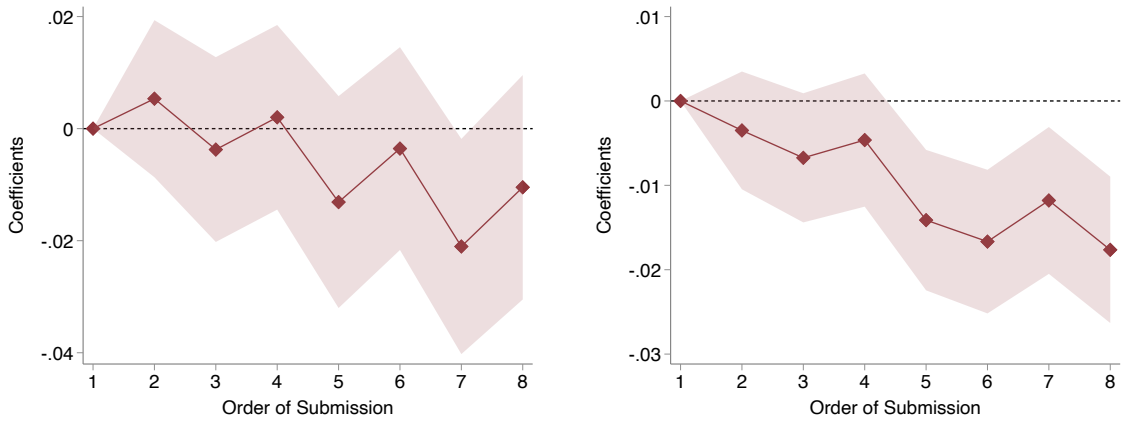
Notes: The figures plot the estimated coefficients $\hat{\delta}_j, j \in \{1, \cdots, 7\}$ of Equation (3.3) with days to the last submission $D2L^t$ on the x-axis, controlling for resume characteristics. The left panel shows results for complete vacancies. The right panel shows results for incomplete vacancies. Each regression includes vacancy fixed effects. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

Table A.4: Early Application Advantage, Wave 1 versus Wave 2-5

	(1)	(2)	(3)	(4)	(5)	(6)
	All		Wave 1		Wave 2-5	
$early_i^1$	0.0116		0.0128		0.0113	
	(0.0024)		(0.0061)		(0.0025)	
$D2L_i^t$		0.0040		0.0036		0.0043
		(0.0008)		(0.0013)		(0.0010)
Resume controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	83,643	83,643	14,541	14,541	69,102	69,102

Note: This table presents the effects of early submission on the contact probability. The outcome is employer contact within 30 days. Columns (1), (3) and (5) estimate Equation (3.1) with $T_i = early_i^1$; columns (2), (4) and (6) estimate the same specification with $T_i = D2L_i^t$. Columns titled "All" include all vacancies; columns titled "Wave 1" include vacancies sampled in Wave 1; and columns titled "Wave 2-5" include vacancies sampled in Waves 2 through 5. The construction of $early_i^1$ and $D2L_i^t$ is described in Section 3.1. All regressions include the full set of randomized resume characteristics and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

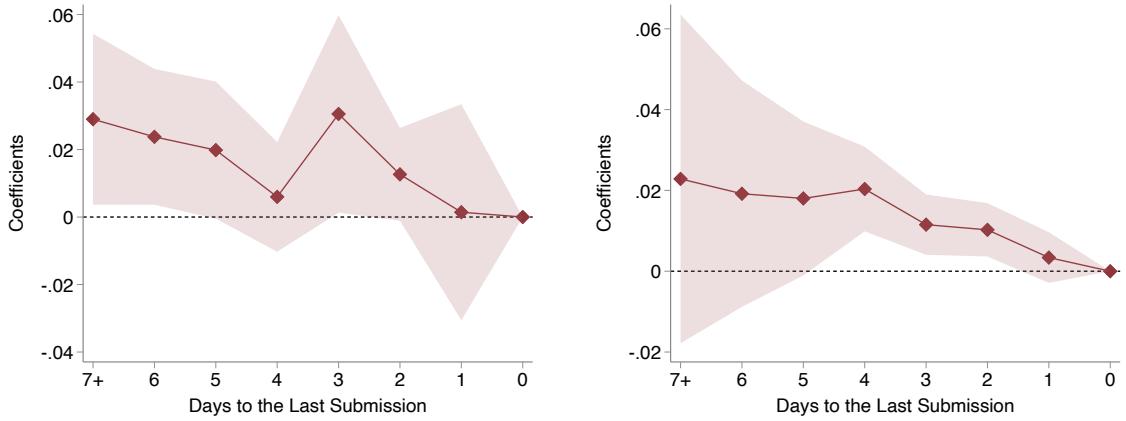
Figure A.6: Contact Gap and Order of Submission, Wave 1 versus Wave 2-5



(a) Wave 1 ($N = 14{,}541$) (b) Wave 2-5 ($N = 69{,}102$)

Notes: The figures plot the estimated coefficients $\hat{\beta}_k$, $k \in \{2, \ldots, 8\}$, from Equation (3.2), with submission order on the x-axis and controlling for resume characteristics. The left panel shows Wave 1 results. The right panel shows Wave 2-5 results. Each regression includes vacancy fixed effects and all resume characteristics. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

Figure A.7: Contact Gap and $D2L^t$, Wave 1 versus Wave 2-5



(a) Wave 1 ($N = 14{,}541$)

(b) Wave 2-5 ($N = 69{,}102$)

Notes: The figures plot the estimated coefficients $\hat{\delta}_j, j \in \{1, \cdots, 7\}$ of Equation (3.3) with days to the last submission $D2L^t$ on the x-axis, controlling for resume characteristics. The left panel shows Wave 1 results. The right panel shows Wave 2-5 results. Each regression includes vacancy fixed effects and all resume characteristics. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

Table A.5: Early Application Advantage under Alternative Timing Measures

	(1)	(2)	(3)	(4)
$early_i^2$	0.0135			
	(0.0023)			
$early_i^3$		0.0105		
		(0.0023)		
$\mathbf{1}(Order_i^{pair} = 2)$			-0.0036	
			(0.0027)	
$\mathbf{1}(Order_i^{pair} = 3)$			-0.0126	
			(0.0031)	
$\mathbf{1}(Order_i^{pair} = 4)$			-0.0127	
			(0.0032)	
$D2L_i^w$				0.0039
				(0.0008)
Resume controls $\mathbf{X}_i$	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓
N	83,643	83,643	83,643	83,643

Note: This table presents the robustness of the early application advantage reported in Table 1 to alternative timing measures. The outcome is employer contact within 30 days. $early^2$ is defined such that the last four submissions to a vacancy are coded as late. For vacancies with no more than four submissions, all submissions are coded as late. In the full sample, 40,210 (48.07%) are coded as early submissions. $early^3$ is based on the designed pair orders and defined such that the last two pairs of submissions to a vacancy are coded as late. For vacancies with no more than two pairs of submissions (836 vacancies, 7.52%; 2,321 submissions, 2.77%), all submissions are coded as late. $\mathbf{1}\left(Order_i^{pair} = p\right)$ is the pair-order indicator, equal to 1 if fictitious applicant i is assigned to the p-th pair of submissions to vacancy $j(i)$. $D2L_i^w$ is the winsorized version of $D2L_i$ at the top 1% level. All regressions include the full set of randomized resume characteristics and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

Table A.6: Early Application Advantage under Alternative Fixed Effect Specifications

	(1)	(2)	(3)	(4)	(5)	(6)
$early_i^1$	0.0116	0.0111	0.0114			
	(0.0024)	(0.0024)	(0.0023)			
$D2L_i^t$				0.0040	0.0039	0.0053
				(0.0008)	(0.0008)	(0.0010)
Resume controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓		✓	✓	
Day-of-week FEs		✓	✓		✓	✓
Wave FEs			✓			✓
Region FEs			✓			✓
N	83,643	83,643	83,643	83,643	83,643	83,643

Note: This table presents the effects of early submission on the contact probability with alternative fixed effects. The outcome is employer contact within 30 days. Columns (1)-(3) estimate Equation (3.1) with $T_i = early_i^1$; columns (4)-(6) estimate the same specification with $T_i = D2L_i^t$. The construction of $early_i^1$ and $D2L_i^t$ is described in Section 3.1. All regressions include the full set of randomized resume characteristics. Standard errors in parentheses are clustered at the vacancy level. The specification with region and wave fixed effects is supplementary rather than an alternative version of the within-vacancy design.

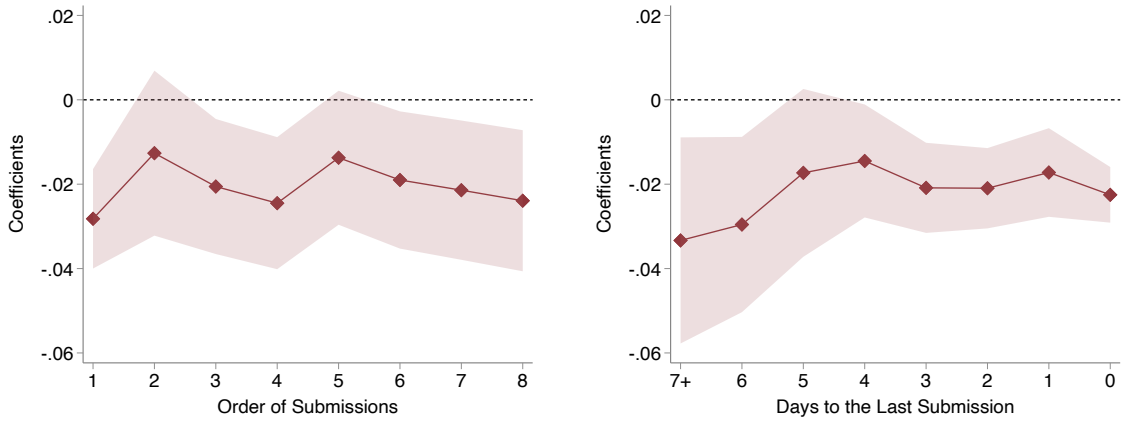
A.3 Early Application Advantage by Randomized Resume Characteristics

Table A.7: Early Application Interacted with Resume Characteristics

	(1) Late $early^1 = 0$	(2) Early $early^1 = 1$	(3) Diff. (2) − (1)
Black	-0.0148	-0.0157	-0.0009
	(0.0037)	(0.0036)	(0.0051)
Female	0.0012	0.0052	0.0040
	(0.0040)	(0.0041)	(0.0056)
Black × Female	-0.0089	-0.0123	-0.0034
	(0.0055)	(0.0055)	(0.0078)
Over 40	-0.0028	-0.0059	-0.0031
	(0.0028)	(0.0029)	(0.0040)
Pol. club	0.0046	0.0072	0.0026
	(0.0070)	(0.0074)	(0.0101)
Acad. club	0.0088	0.0012	-0.0076
	(0.0069)	(0.0072)	(0.0099)
LGBTQ club	0.0039	-0.0046	-0.0085
	(0.0051)	(0.0050)	(0.0071)
Typical pron.	-0.0043	-0.0112	-0.0069
	(0.0071)	(0.0073)	(0.0101)
Neutral pron.	-0.0106	-0.0067	0.0039
	(0.0072)	(0.0072)	(0.0102)
Assoc. degree	0.0044	-0.0001	-0.0045
	(0.0028)	(0.0029)	(0.0040)
Constant	0.2428	0.2585	0.0157
	(0.0035)	(0.0035)	(0.0051)
Vacancy FEs	✓	✓	
N	42,531	41,112	83,643
F-statistic for joint significance			0.58
p-value			0.8344

Note: This table presents the effects of interactions between early submission and resume characteristics on the probability of employer contact within 30 days. Column (1) displays results for applications with $early_i^1 = 0$. Column (2) displays results for applications with $early_i^1 = 1$. The construction of $early_i^1$, including the treatment of vacancies with at most four submissions, is described in Section 3.1. All regressions include vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level. The F-statistic and joint p-value come from a test of whether all differences in the reported coefficients other than the constant are zero. Sample: full sample.

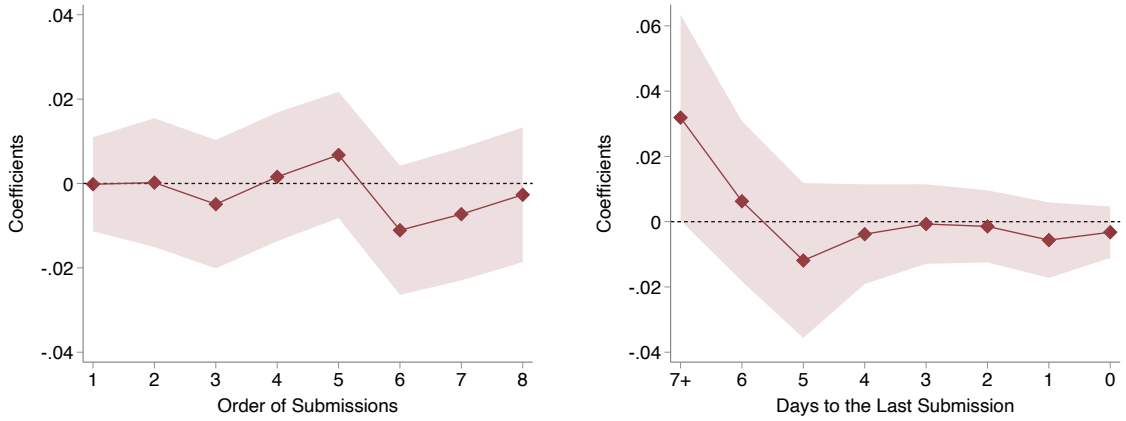
Figure A.8: Race Contact Gap Over Time



(a) $\mathbf{1}\left(Order_i = j\right) \times$ Black (b) $\mathbf{1}\left(D2L_i^t = d\right) \times$ Black

Notes: The figures plot how the race contact gap evolves over time. The left panel plots the estimated coefficients of $\mathbf{1}\left(Order_i = j\right) \times$ Black plus the estimated coefficient of Black for $j = 2, \ldots, 8$, and the coefficient of Black for $j = 1$. The right panel plots the estimated coefficients of $\mathbf{1}\left(D2L_i^t = d\right) \times$ Black plus the estimated coefficient of Black for $d = 1, \ldots, 7$, and the coefficient of Black for $d = 0$. Each regression includes vacancy fixed effects and all resume characteristics. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

Figure A.9: Gender Contact Gap Over Time



(a) $\mathbf{1}\left(Order_i = j\right) \times$ Female (b) $\mathbf{1}\left(D2L_i^t = d\right) \times$ Female

Notes: The figures plot how the gender contact gap evolves over time. The left panel plots the estimated coefficients of $\mathbf{1}\left(Order_i = j\right) \times$ Female plus the estimated coefficient of Female for $j = 2, \ldots, 8$, and the coefficient of Female for $j = 1$. The right panel plots the estimated coefficients of $\mathbf{1}\left(D2L_i^t = d\right) \times$ Female plus the estimated coefficient of Female for $d = 1, \ldots, 7$, and the coefficient of Female for $d = 0$. Each regression includes vacancy fixed effects and all resume characteristics. Standard errors are clustered at the vacancy level. The shaded area depicts 95% confidence intervals.

A.4 Additional Mechanism Evidence

Table A.8: Construction of Contact History, Example

Order	Date: Submission	Date: Contact	Contacted	NoC^{alt}	NoC	NoC^{ord}
1	10/8/19	10/8/19	1	1	0	0
2	10/8/19	10/8/19	1	1	0	1
3	10/12/19		0	3	2	2
4	10/12/19	10/12/19	1	2	2	2
5	10/14/19	10/15/19	1	3	3	3
6	10/14/19	10/15/19	1	3	3	4
7	10/16/19		0	5	5	5
8	10/16/19	10/20/19	1	5	5	5

Note: This table presents an example of the construction of the three contact history measures (vacancy ID: 80). NoC is calculated including other submissions contacted at least one day before the current submission; NoC^{alt} is calculated including all other submissions contacted no later than the day of the current submission; NoC^{ord} is the number of contacted submissions that were ordered before the current submission.

Table A.9: Number of Submissions Contacted before an Application

Order	N	NoC^{alt}	NoC	NoC^{ord}
1	11,114	0.0501	0	0
2	11,031	0.0935	0.0154	0.2451
3	10,682	0.2322	0.1127	0.4936
4	10,606	0.2626	0.1355	0.7362
5	10,278	0.4558	0.3070	0.9876
6	10,204	0.5025	0.3410	1.2249
7	9,899	0.7153	0.5359	1.4757
8	9,829	0.7703	0.5913	1.7177

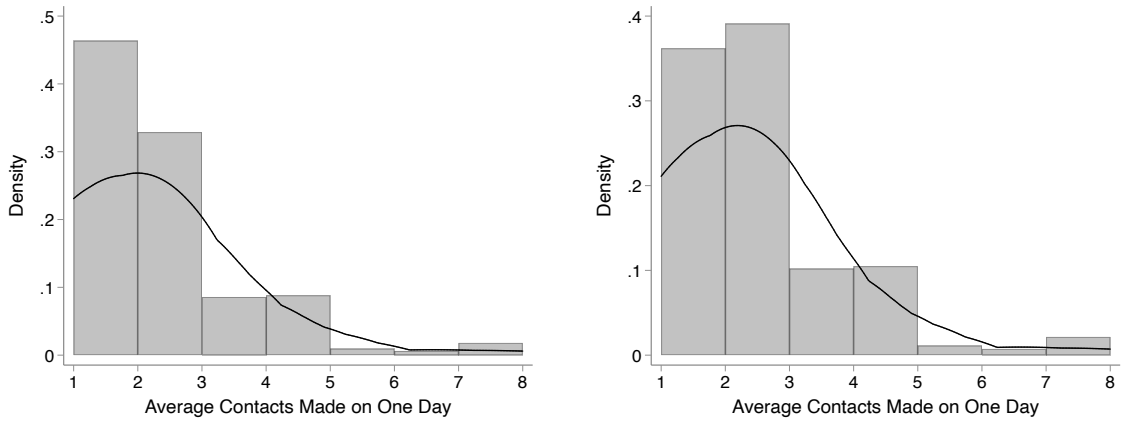
Note: This table summarizes the number of submissions already contacted when a submission was made, by the order of that submission. In the column labeled NoC, only other submissions contacted at least one day before the current submission are included. In the column labeled NoC^{alt}, other submissions contacted no later than the day of the current submission are included. In the column labeled NoC^{ord}, contacted submissions ordered before the current submission are included. Order is constructed with the date and time of actual submissions.

Table A.10: Number of Contacted Submissions and Timing Measures

	NoC_i^{alt}	NoC_i	$early^1$	$early^2$	$early^3$	$Order$	$D2L^t$
NoC_i^{alt}	1						
NoC_i	0.8843	1					
$early^1$	-0.1899	-0.2048	1				
$early^2$	-0.1859	-0.2006	0.9786	1			
$early^3$	-0.1850	-0.1993	0.9569	0.9710	1		
$Order$	0.2238	0.2385	-0.8164	-0.8076	-0.8006	1	
$D2L^t$	-0.1747	-0.1790	0.7169	0.7195	0.7146	-0.7406	1

Note: This table summarizes pairwise correlations between the constructed measures. NoC_i is the number of other submissions contacted at least one day before submission i. $early^1$, $early^2$, and $early^3$ are the three constructed early-submission indicators; NoC_i^{alt} refers to the number of other submissions already contacted no later than the day a submission was made; Order is constructed with the date and time of actual submissions; $D2L^t$ is the number of days before the last submission to a vacancy. All coefficients are significant at the 1% level.

Figure A.10: Distribution of Observed Contact Batching across Vacancies



(a) Vacancies w. ≥ 1 Contact (b) Vacancies w. ≥ 2 Contacts

Notes: The figures plot the distribution of vacancies by the average number of contacts made on a contact day. For each vacancy, this average is calculated as the total number of contacts divided by the number of distinct contact dates. The left panel plots the distribution for vacancies with at least one contact. The right panel plots the distribution for vacancies with at least two contacts. In both panels, an Epanechnikov kernel is used, with the kernel half-width set to 1.

Table A.11: Contact Initiation and Resume Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
		$\mathbf{1}(NoC > 0)$			$\mathbf{1}(NoC^{alt} > 0)$	
	$= 0$	$= 1$	$(2) - (1)$	$= 0$	$= 1$	$(5) - (4)$
Black	-0.0188	-0.0389	-0.0201	-0.0181	-0.0369	-0.0188
	(0.0019)	(0.0075)	(0.0077)	(0.0019)	(0.0063)	(0.0066)
Female	-0.0030	0.0002	0.0032	-0.0018	-0.0061	-0.0043
	(0.0022)	(0.0090)	(0.0092)	(0.0022)	(0.0077)	(0.0080)
Over 40	-0.0040	-0.0076	-0.0036	-0.0028	-0.0140	-0.0113
	(0.0020)	(0.0090)	(0.0093)	(0.0020)	(0.0080)	(0.0083)
Pol. club	0.0070	0.0036	-0.0035	0.0077	-0.0014	-0.0091
	(0.0052)	(0.0218)	(0.0224)	(0.0052)	(0.0190)	(0.0197)
Acad. club	0.0040	0.0121	0.0080	0.0044	0.0067	0.0023
	(0.0053)	(0.0201)	(0.0208)	(0.0052)	(0.0180)	(0.0188)
LGBTQ club	-0.0043	0.0302	0.0345	-0.0053	0.0253	0.0305
	(0.0037)	(0.0159)	(0.0164)	(0.0037)	(0.0133)	(0.0138)
Typical pron.	-0.0086	0.0014	0.0100	-0.0063	-0.0180	-0.0117
	(0.0053)	(0.0206)	(0.0212)	(0.0053)	(0.0179)	(0.0187)
Neutral pron.	-0.0091	0.0014	0.0105	-0.0093	-0.0085	0.0008
	(0.0051)	(0.0222)	(0.0227)	(0.0051)	(0.0189)	(0.0194)
Assoc. degree	0.0022	-0.0067	-0.0089	0.0009	0.0058	0.0049
	(0.0021)	(0.0087)	(0.0089)	(0.0021)	(0.0075)	(0.0078)
Constant	0.2641	0.0974	-0.1667	0.2686	0.1536	-0.1150
	(0.0034)	(0.0125)	(0.0127)	(0.0035)	(0.0116)	(0.0123)
Vacancy FEs	✓	✓		✓	✓	
Order FEs	✓	✓		✓	✓	
N	70,293	8,339	78,632	67,082	11,550	78,632
χ^2-statistics			1.41			1.77
p-value			0.1788			0.0677

Note: This table presents the effects of interactions between prior contact history and resume characteristics on the probability of employer contact within 30 days. NoC_i refers to the number of other submissions already contacted at least one day before an application was made; NoC_i^{alt} refers to the number of other submissions already contacted no later than the day when a submission was made. Column (3) displays the difference between Column (2) and Column (1). Column (6) displays the difference between Column (5) and Column (4). All regressions include vacancy fixed effects and submission order fixed effects. Standard errors in parentheses are clustered at the vacancy level. χ^2 statistics and joint p-values come from tests of the joint null that all reported coefficient differences other than the constant are zero. Sample: vacancies with 8 experimental applications.

Table A.12: Prior Contact History and Early Application Advantage

	(1)	(2)	(3)	(4)	(5)	(6)
	Equation (3.1)		Equation (3.2)		Equation (3.1)	
$\mathbf{1}(NoC > 0)$	-0.1733		-0.1776		-0.1760	
	(0.0086)		(0.0087)		(0.0087)	
$\mathbf{1}(NoC^{alt} > 0)$		-0.1257		-0.1285		-0.1273
		(0.0094)		(0.0094)		(0.0094)
$early^1$	-0.0099	-0.0017				
	(0.0022)	(0.0022)				
$\mathbf{1}\,(Order = 2)$			0.0018	0.0030		
			(0.0034)	(0.0034)		
$\mathbf{1}\,(Order = 3)$			0.0097	0.0066		
			(0.0037)	(0.0037)		
$\mathbf{1}\,(Order = 4)$			0.0144	0.0105		
			(0.0037)	(0.0037)		
$\mathbf{1}\,(Order = 5)$			0.0134	0.0054		
			(0.0039)	(0.0039)		
$\mathbf{1}\,(Order = 6)$			0.0132	0.0049		
			(0.0039)	(0.0039)		
$\mathbf{1}\,(Order = 7)$			0.0213	0.0104		
			(0.0039)	(0.0039)		
$\mathbf{1}\,(Order = 8)$			0.0194	0.0081		
			(0.0039)	(0.0040)		
$D2L^t$					-0.0039	-0.0011
					(0.0007)	(0.0008)
Controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632	78,632

Note: This table presents the effects of prior contact histories and early submission on the probability of employer contact within 30 days, controlling for resume characteristics. NoC_i refers to the number of other submissions already contacted at least one day before an application was made; NoC_i^{alt} refers to the number of other submissions already contacted no later than the day when a submission was made. $early^1$ is the constructed early submission dummy. For $j \in \{2, \cdots, 8\}$, $\mathbf{1}\,(Order = j)$ is the indicator for submission order j; $D2L^t$ is the number of days before the last submission to a vacancy. All regressions include resume characteristics $\mathbf{X}_i$ and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level. Sample: vacancies with 8 experimental applications.

Table A.13: Validation of the Screening-Value Index

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies			Vacancies w. ≥ 1 Contact		
Panel A: Across-vacancy validation						
Main index	1.0120			0.6084		
	(0.0182)			(0.0253)		
Alternative 1		1.1650			0.6988	
		(0.0214)			(0.0293)	
Alternative 2			1.1340			0.6743
			(0.0208)			(0.0285)
State-by-Month FEs	✓	✓	✓	✓	✓	✓
Panel B: With-vacancy validation						
Main index	0.1587			0.2807		
	(0.0404)			(0.0715)		
Alternative 1		0.1240			0.2525	
		(0.0282)			(0.0573)	
Alternative 2			0.1078			0.2060
			(0.0278)			(0.0533)
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	32,288	32,288	32,288

Note: This table reports regressions of employer contact within 30 days on the screening-value indices constructed as described in Section 4.3.1 and Appendix B. The main index, used in the main analysis, is generated by CatBoost with native categorical handling. Alternative 1 is generated by CatBoost with frequency-based grouping. Alternative 2 is generated by CatBoost with text-based grouping. Columns (1)-(3) show results for all vacancies with eight experimental applications; columns (4)-(6) show results for vacancies with eight applications and at least one contact. Regressions in Panel A include state-by-month fixed effects; regressions in Panel B include vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

Table A.14: Randomization over Time Validation with Screening-Value Index

Dep.Var.	(1)	(2)	(3)	(4)	(5)	(6)
	Main Index		Alternative 1		Alternative 2	
$early_i^1$	-0.0003		0.0004		0.0003	
	(0.0002)		(0.0003)		(0.0003)	
$D2L_i^t$		-0.0001		0.0002		0.0001
		(0.0001)		(0.0001)		(0.0001)
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632	78,632

Note: This table tests whether within-vacancy application timing predicts the screening-value indices constructed as described in Section 4.3.1 and Appendix B. Columns (1)-(2) use the index from the main analysis, generated with the CatBoost model's native handling of categorical variables, as the dependent variable. Columns (3)-(4) use the index generated by the CatBoost model with frequency-based grouping as the dependent variable. Columns (5)-(6) use the index generated by the CatBoost model with text-based grouping as the dependent variable. All regressions include vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level. The sample consists of vacancies with eight experimental applications.

Table A.15: Prior Screening Value and Contact Probability

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies			Vacancies w. ≥ 1 Contact		
Q_i^{best}	-0.0102		-0.0099	-0.0271		-0.0274
	(0.0050)		(0.0064)	(0.0130)		(0.0165)
Q_i^{mean}		-0.0129	-0.0007		-0.0309	0.0008
		(0.0098)	(0.0126)		(0.0233)	(0.0295)
Controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓	✓	✓
Day-of-Week FEs	✓	✓	✓	✓	✓	✓
N	39,300	39,300	39,300	16,142	16,142	16,142

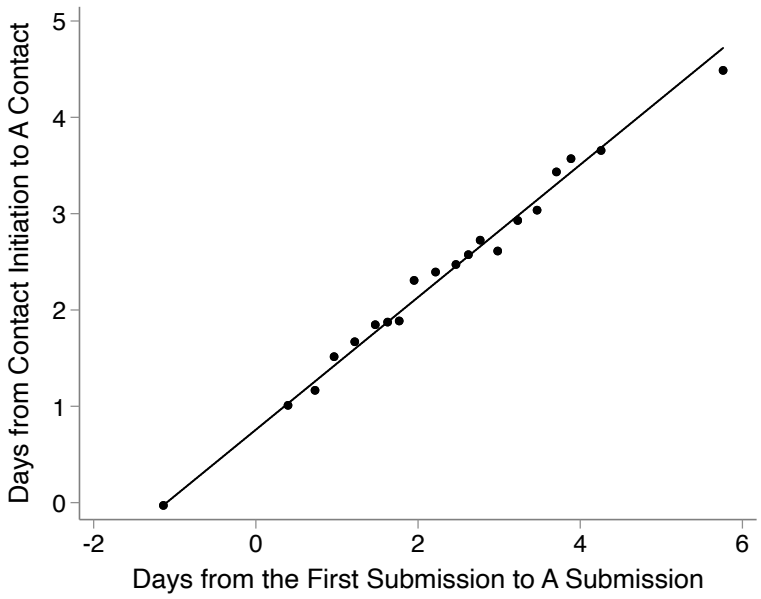
Note: This table estimates the correlation between the screening value of previous applications and the contact probability of the current application. Q_i^{best} represents the best-so-far screening-value index among applications submitted at least one day before submission i; Q_i^{mean} represents the mean-so-far screening-value index among applications submitted at least one day before submission i. In all columns, the dependent variable is an indicator for receiving contact within 30 days after submission. All regressions include vacancy fixed effects, order fixed effects, submission day-of-week fixed effects and the full set of randomized resume characteristics. Standard errors in parentheses are clustered at the vacancy level. The sample includes only the last four experimental applications to each vacancy; observations with no application submitted at least one day earlier are excluded because Q_i^{best} and Q_i^{mean} are undefined. Columns (1)-(3) include all vacancies with eight experimental applications; columns (4)-(6) include vacancies with at least one contact.

Table A.16: IV Estimates with Alternative Screening-Value Indices

	(1)	(2)	(3)	(4)
	All Vacancies		Vacancies w. ≥ 1 Contact	
	IV-2SLS	OLS	IV-2SLS	OLS
Panel A: CatBoost with frequency-based grouping				
$\mathbf{1}(NoC_i > 0)$	-0.6520	-0.1205	-0.5742	-0.1277
	(0.3981)	(0.0206)	(0.3469)	(0.0217)
First stage	0.0111		0.0318	
	(0.0034)		(0.0082)	
Kleibergen-Paap F	10.781		15.172	
Kleibergen-Paap LM	10.753		14.964	
Anderson-Rubin test p value	0.083		0.085	
Panel B: CatBoost with text-based grouping				
$\mathbf{1}(NoC_i > 0)$	-1.0857	-0.1204	-0.9264	-0.1275
	(0.8640)	(0.0206)	(0.6883)	(0.0217)
First stage	0.0065		0.0192	
	(0.0035)		(0.0085)	
Kleibergen-Paap F	3.406		5.112	
Kleibergen-Paap LM	3.405		5.083	
Anderson-Rubin test p value	0.112		0.113	
Resume Controls	✓	✓	✓	✓
Screening Value of i	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓
Day-of-Week FEs	✓	✓	✓	✓
N	39,300	39,300	16,142	16,142

Note: This table presents robustness checks for the IV estimates in Table 3 using alternative screening-value indices. Columns (1) and (3) report estimates from Equation (4.4) with Q_i^{best} as the instrument for contact initiation; columns (2) and (4) report the corresponding OLS estimates. In all columns, the dependent variable is the indicator of receiving contact within 30 days after submission. Panel A reports results using the screening-value index generated by a CatBoost model with frequency-based grouping; Panel B reports results using the index generated by a CatBoost model with text-based grouping. All regressions include vacancy fixed effects, order fixed effects, submission day-of-week fixed effects and the full set of randomized resume characteristics. Standard errors in parentheses are clustered at the vacancy level. The sample includes only the last four experimental applications to each vacancy; observations with no application submitted at least one day earlier are excluded because Q_i^{best} is undefined. Columns (1)-(2) include all vacancies with eight experimental applications; columns (3)-(4) include vacancies with at least one contact.

Figure A.11: Monotonicity of Contact Timing in Submission Timing



Notes: This figure plots contact timing against submission timing among contacted experimental applications. The x-axis is the number of days between the application's submission date and the first experimental submission date to the same vacancy. The y-axis is the number of days between the application's contact date and the first experimental contact date to the same vacancy. Points are binned into 20 equal-sized bins. The fitted line includes vacancy fixed effects.

Table A.17: Prior Contact History and Race/Gender

	(1)	(2)	(3)	(4)	(5)	(6)
		NoC			NoC^{alt}	
History	-0.0587	-0.0550	-0.0589	-0.0445	-0.0415	-0.0447
	(0.0029)	(0.0030)	(0.0032)	(0.0024)	(0.0024)	(0.0026)
Black	-0.0205	-0.0186	-0.0205	-0.0203	-0.0181	-0.0203
	(0.0018)	(0.0018)	(0.0018)	(0.0018)	(0.0018)	(0.0018)
History × Black		-0.0074			-0.0060	
		(0.0024)			(0.0018)	
Female	-0.0024	-0.0024	-0.0025	-0.0022	-0.0022	-0.0024
	(0.0022)	(0.0022)	(0.0022)	(0.0022)	(0.0022)	(0.0022)
History × Female			0.0005			0.0004
			(0.0031)			(0.0023)
Controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓	✓	✓
N	83,643	83,643	83,643	83,643	83,643	83,643

Note: This table presents the effects of prior contact histories on the probability of employer contact within 30 days, controlling for resume characteristics. In Columns (1)-(3), history is NoC_i, the number of other submissions contacted at least one day before application i was submitted. In Columns (4)-(6), history is NoC_i^{alt}, the number of other submissions contacted no later than the day application i was submitted. All regressions include submission order fixed effects and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

Table A.18: Race-Specific Contact History

	(1)	(2)	(3)	(4)	(5)	(6)
	History : NoC			History : NoC^{alt}		
Black	-0.0192	-0.0184	-0.0186	-0.0183	-0.0184	-0.0182
	(0.0018)	(0.0018)	(0.0018)	(0.0019)	(0.0018)	(0.0018)
History^{Black}	-0.0977		-0.0525	-0.0615		-0.0157
	(0.0058)		(0.0097)	(0.0041)		(0.0067)
History^{Black} $\times$ Black	-0.0162		-0.0035	-0.0330		-0.0455
	(0.0047)		(0.0099)	(0.0039)		(0.0095)
History^{White}		-0.0974	-0.0574		-0.0827	-0.0697
		(0.0056)	(0.0096)		(0.0050)	(0.0078)
History^{White} $\times$ Black		-0.0109	-0.0108		0.0093	0.0341
		(0.0047)	(0.0098)		(0.0037)	(0.0092)
Controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓	✓	✓
N	83,643	83,643	83,643	83,643	83,643	83,643

Note: This table presents the effects of race-specific prior contact histories on the probability of employer contact within 30 days, controlling for resume characteristics. In Columns (1)-(3), History refers to NoC_i, the number of other submissions already contacted at least one day before an application was made, by race (Black/White). In Columns (4)-(6), History refers to NoC_i^{alt}, the number of other submissions contacted no later than the day an application was made, by race (Black/White). All regressions include submission order fixed effects and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

Table A.19: Gender-Specific Contact History

	(1)	(2)	(3)	(4)	(5)	(6)
	History : NoC			History : NoC^{alt}		
Female	-0.0061	0.0008	-0.0032	-0.0061	0.0011	-0.0029
	(0.0022)	(0.0022)	(0.0022)	(0.0022)	(0.0022)	(0.0022)
History^{Female}	-0.0955		-0.0734	-0.0699		-0.0546
	(0.0057)		(0.0065)	(0.0046)		(0.0055)
History^{Female} $\times$ Female	0.0038		0.0087	0.0001		0.0078
	(0.0054)		(0.0063)	(0.0042)		(0.0053)
History^{Male}		-0.0803	-0.0447		-0.0626	-0.0347
		(0.0057)	(0.0064)		(0.0048)	(0.0056)
History^{Male} $\times$ Female		-0.0021	-0.0079		0.0010	-0.0077
		(0.0055)	(0.0064)		(0.0043)	(0.0054)
Controls $\mathbf{X}_i$	✓	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓	✓
Order FEs	✓	✓	✓	✓	✓	✓
N	83,643	83,643	83,643	83,643	83,643	83,643

Note: This table presents the effects of gender-specific prior contact histories on the probability of employer contact within 30 days, controlling for resume characteristics. In Columns (1)-(3), History refers to NoC_i, the number of other submissions already contacted at least one day before an application was made, by gender (female/male). In Columns (4)-(6), History refers to NoC_i^{alt}, the number of other submissions contacted no later than the day an application was made, by gender (female/male). All regressions include submission order fixed effects and vacancy fixed effects. Standard errors in parentheses are clustered at the vacancy level.

A.5 Additional Labor-Market Heterogeneity

Table A.20: Labor Market Condition Gradients: Contact Initiation

	(1)	(2)	(3)	(4)	(5)
	Pr (Contact Initiator)				
$D2F_i^t \times \log(\text{hires}_{m(i),t(i)})$	-0.0097				-0.0054
	(0.0052)				(0.0063)
	[0.0062]				[0.0072]
$D2F_i^t \times \log(\text{layoffs}_{m(i),t(i)})$		0.0006			-0.0062
		(0.0011)			(0.0014)
		[0.0012]			[0.0015]
$D2F_i^t \times \log(\text{quits}_{m(i),t(i)})$			-0.0166		0.0021
			(0.0047)		(0.0053)
			[0.0053]		[0.0056]
$D2F_i^t \times \log(\text{job opening}_{m(i),t(i)})$				-0.0278	0.0024
				(0.0055)	(0.0103)
				[0.0063]	[0.0118]
$D2F_i^t \times \log(\text{tightness}_{m(i),t(i)})$					-0.0368
					(0.0080)
					[0.0087]
Resume Controls	✓	✓	✓	✓	✓
Prior Market Condition	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632

Note: This table reports estimates of Equation (5.1). $D2F_i^t$ is the number of calendar days between application i's submission date and the first experimental submission date to the same vacancy, top-coded at seven. Larger values correspond to later submissions. In all columns, the dependent variable is the probability that application i is the contact initiator for the vacancy. When multiple experimental applications are contacted on the first contact date, each is assigned equal weight. In the vacancy's state and month of first experimental submission, $\text{tightness}_{m(i),t(i)}$ is vacancies per unemployed worker, $\text{hires}_{m(i),t(i)}$ is the rate of hires over employment, $\text{layoffs}_{m(i),t(i)}$ is the rate of layoffs and discharges over employment, $\text{quits}_{m(i),t(i)}$ is the rate of quits over employment, job $\text{opening}_{m(i),t(i)}$ is the job-opening rate, defined as job openings divided by employment plus job openings. All regressions include vacancy fixed effects, resume controls, and the interaction between $D2F_i^t$ and the three-month pre-period average of the corresponding market condition variable. In Column (5), the specification includes current and pre-period interactions for all five market condition variables jointly. Because these JOLTS series are highly correlated, I interpret the joint specification only as a robustness exercise rather than as evidence on the separate causal contribution of each series. Standard errors clustered at the vacancy level are shown in parentheses; standard errors clustered at the state-by-month level are shown in brackets. The sample consists of complete vacancies with eight experimental applications.

Table A.21: Labor Market Condition Gradients: Early Application Advantage

	(1)	(2)	(3)	(4)	(5)
			1 (Contact)		
$D2F_i^t \times \log(\text{hires}_{m(i),t(i)})$	-0.0039				-0.0129
	(0.0077)				(0.0097)
	[0.0075]				[0.0090]
$D2F_i^t \times \log(\text{layoffs}_{m(i),t(i)})$		-0.0020			-0.0050
		(0.0018)			(0.0022)
		[0.0018]			[0.0021]
$D2F_i^t \times \log(\text{quits}_{m(i),t(i)})$			-0.0058		0.0024
			(0.0070)		(0.0082)
			[0.0071]		[0.0080]
$D2F_i^t \times$ $\log(\text{job opening}_{m(i),t(i)})$				-0.0065	0.0070
				(0.0079)	(0.0151)
				[0.0070]	[0.0134]
$D2F_i^t \times \log(\text{tightness}_{m(i),t(i)})$					-0.0164
					(0.0114)
					[0.0108]
Resume Controls	✓	✓	✓	✓	✓
Prior Market Condition	✓	✓	✓	✓	✓
Vacancy FEs	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632

Note: This table reports estimates of Equation (5.1). $D2F_i^t$ is the number of calendar days between application i's submission date and the first experimental submission date to the same vacancy, top-coded at seven. Larger values correspond to later submissions. In all columns, the dependent variable is the indicator for employer contact within 30 days. In the vacancy's state and month of first experimental submission, $\text{tightness}_{m(i),t(i)}$ is vacancies per unemployed worker, $\text{hires}_{m(i),t(i)}$ is the rate of hires over employment, $\text{layoffs}_{m(i),t(i)}$ is the rate of layoffs and discharges over employment, $\text{quits}_{m(i),t(i)}$ is the rate of quits over employment, job $\text{opening}_{m(i),t(i)}$ is the job-opening rate, defined as job openings divided by employment plus job openings. All regressions include vacancy fixed effects, resume controls, and the interaction between $D2F_i^t$ and the three-month pre-period average of the corresponding market condition variable. In Column (5), the specification includes current and pre-period interactions for all five market condition variables jointly. Because these JOLTS series are highly correlated, I interpret the joint specification only as a robustness exercise rather than as evidence on the separate causal contribution of each series. Standard errors clustered at the vacancy level are shown in parentheses; standard errors clustered at the state-by-month level are shown in brackets. The sample consists of complete vacancies with eight experimental applications.

B Details of Machine Learning Predictions

This section discusses the contact propensity prediction model in detail. The purpose of the machine-learning (ML) component is to construct a resume-job contact propensity score that can be interpreted as an estimate of contact likelihood under a state-0 screening environment. This score is then used to quantify the foregone match quality from contacting early. To preserve a clean interpretation, the ML models exclude variables that directly encode the submission order and other timing-related variation. Instead, I build the model using only posting characteristics and resume attributes as predictors.

B.1 State-0 Environment and Feature Set

The ML models are designed to approximate contact propensity under an early-stage screening environment ("state-0"), before the recruiting process becomes endogenously shaped by prior contacts, time pressure, or other recruiting-state constraints. Specifically, I train the models on the first application submitted within each posting. This design has two advantages: First, it anchors the estimated mapping from resume and job features to contact outcomes to a common early point in the recruiting timeline and avoids implicitly conditioning the model on later-stage dynamics. Second, the model is able to predict how later submissions would be evaluated under the early-stage screening environment within the experiment, thereby creating a reasonable counterfactual to quantify the potential quality loss from rushing to contact. This yields a training and evaluation sample of 11,114 submissions, one per posting.

I use the following feature set as predictors:

- **Posting characteristics**, including state, firm ID, experimental wave, submission month, Standard Industrial Classification (SIC) code, three-digit Standard Occupational Classification (SOC3) category, and a hiring intermediary platform indicator.
- **Resume headline attributes**, including race, gender, age, education, political leaning indicators, and pronoun indicators.
- **Resume work-history details**: number of prior positions listed, and for up to three prior positions: (i) job type (1,512 categories), (ii) job title (23 categories), and (iii) duration in months.

Binary indicators are treated as numerical regressors. For numerical covariates (including binaries), missing values are imputed using the training-fold median; for binaries this coincides with imputing the modal value in the common case where the median is 0.

Categorical representation Since the job type of resume entries contains 1,512 categories, it can generate a large sparse matrix, which may lower the predictive performance of some algorithms. I therefore consider two encoding approaches to group rare categories and shrink the effective number of categories:

- **Frequency-based grouping.** I cap the number of distinct categories retained for high-cardinality type fields by keeping the K most frequent labels and mapping all other labels to an "Other" bucket. Specifically, for the latest resume entry (the most recent job prior to application submission), I one-hot encode the type variable with a maximum of 300 categories using infrequent-category handling, and the other resume entries are pooled and mapped similarly.
- **Text-based clustering.** I also use an unsupervised text-based approach for encoding. I construct a term frequency-inverse document frequency (TF-IDF) representation of the raw type labels and cluster the labels into 300 clusters using MiniBatch k-means. Each type label is replaced by its cluster ID, then encoded analogously to the frequency-based approach (the most recent entry separately, all other entries pooled). This provides a data-driven grouping based on lexical similarity rather than frequency alone.

B.2 Candidate Models and Evaluation Metrics

I considered three "workhorse" model classes as initial benchmarks:

- **Elastic-net logistic regression**: A linear model for binary outcomes (logistic regression) with combined L1/L2 regularization. It is fast and stable in sparse feature spaces, but relies on encoded inputs and can miss nonlinearities and complex interactions.
- **ExtraTrees classifier**: An ensemble of heavily randomized decision trees that captures nonlinearities and interactions with minimal feature engineering, but its probability outputs may be less well-calibrated and less interpretable.

- **CatBoost gradient boosting**: A gradient boosting algorithm designed for tabular problems, especially with many categorical predictors. It uses specialized categorical handling, but is more complex, less transparent, and requires careful validation.

I compare the three models using sparse encoded features to assess overall performance. Since CatBoost uses specialized categorical handling and boosting schemes, it handles high-dimensional categorical variables efficiently. I therefore also include a CatBoost model with the native, ungrouped categorical variables as predictors. It is attractive in this setting because it avoids high-dimensional one-hot expansions for high-cardinality categories, while still allowing flexible nonlinearities and interactions via gradient boosting. Models are evaluated with probability and ranking metrics computed on out-of-fold predictions for first applications.

Out-of-fold procedure A central design choice is to generate **out-of-fold (OOF)** predictions at the posting level. Let $\widehat{p}_{ij}$ denote the out-of-fold cross-fit predicted probability of contact from this model. Out-of-fold prediction ensures $\widehat{p}_{ij}$ is not mechanically inflated by in-sample overfitting and vacancy-level leakage. I implement 5-fold cross-fitting using GroupKFold:

1. Construct folds using only the first submission of each posting, with postings partitioned into 5 folds.
2. For each fold f, train the model on order-1 applications from postings in the training folds.
3. Score applications in held-out postings.
4. Concatenate predictions across folds to calculate model performance statistics.

B.3 Model Selection

I first evaluate which model performs the best in predicting contact outcomes for the first submissions of each posting. Table B.1 summarizes performance on first applications (order $= 1$) using OOF predictions. The CatBoost model with native handling of categorical variables delivers the strongest overall ranking and probability performance among the candidate models. Figure B.1 plots the ROC curve

and the precision-recall curve of the four candidate models. It further shows that the native CatBoost model consistently outperforms the other models. I therefore select CatBoost with native handling of categorical variables as the main specification.

Table B.1: Candidate Models' Out-of-Fold Performance on First Submissions

Model	ROC-AUC	Log Loss	Brier	Average Precision
CatBoost (native categories)	0.758	0.474	0.154	0.525
ExtraTrees (encoded)	0.743	0.491	0.160	0.511
CatBoost (encoded)	0.740	0.484	0.158	0.504
Elastic-net logit (encoded)	0.734	0.491	0.161	0.488

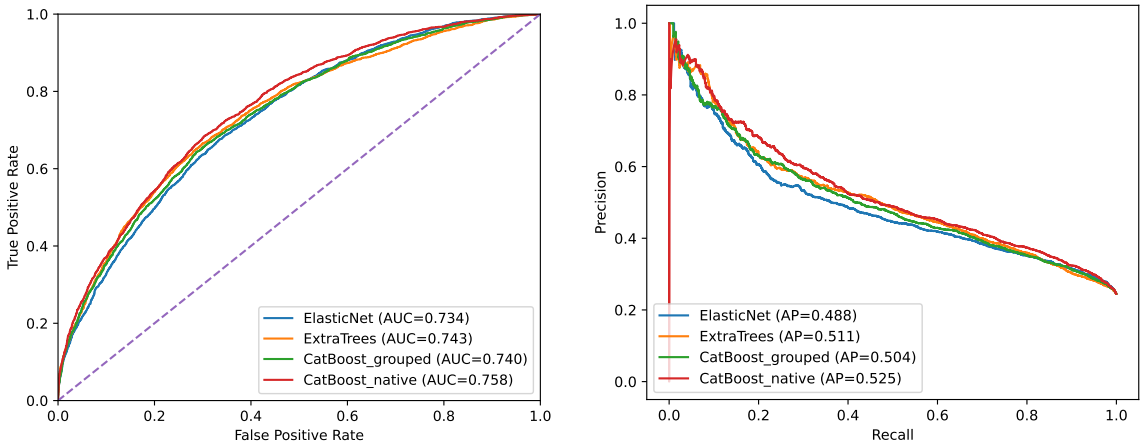
Note: This table compares four predictive models used to construct a contact propensity score. Elastic-net is a penalized logistic regression estimated with an elastic-net penalty (L1/L2), ExtraTrees is an ensemble of extremely randomized decision trees, and CatBoost is a gradient-boosted tree model. The three models that use categorical variables encoded with frequency-based grouping are noted with "(encoded)". CatBoost (native) uses CatBoost's native handling of categorical predictors. Performance is measured on the state-0 sample consisting of the first application in each posting, and all reported statistics are computed from posting-level out-of-fold (OOF) predictions using 5-fold GroupKFold cross-fitting. The four reported metrics are ROC-AUC (ranking quality), log loss (probability accuracy), Brier score (mean squared error of predicted probabilities), and average precision (area under the precision-recall curve). Lower values are better for log loss and Brier; higher values are better for ROC-AUC and average precision.

Since the model uses the first submissions of each posting as the training and evaluation sample, one concern is that the predictive power comes mainly from the job-side predictors. To assess whether the candidate models leverage resume information in prediction, I perform qualitative diagnostics of within-posting score dispersion. Specifically, I use the model trained on first submissions to generate out-of-fold predictions for later submissions and plot the distribution of within-posting score ranges and top-1 versus top-2 gaps. Figure B.2 plots the empirical cumulative distribution function (ECDF) across postings for the two dispersion measures. It shows that the native CatBoost model generates prediction scores that vary meaningfully within postings. This feature of the model significantly benefits downstream comparisons of "best available" applicants within a posting.

B.4 Hyperparameter Tuning

I tune the native CatBoost model using the same posting-level OOF design described in Section B.3. Tuning targets ROC-AUC with log loss as a tie-breaker, favoring better-calibrated probabilities when AUC differences are small.

Figure B.1: Candidate Models' Out-of-Fold Performance on First Submissions



(a) ROC Curve (b) Precision-Recall Curve

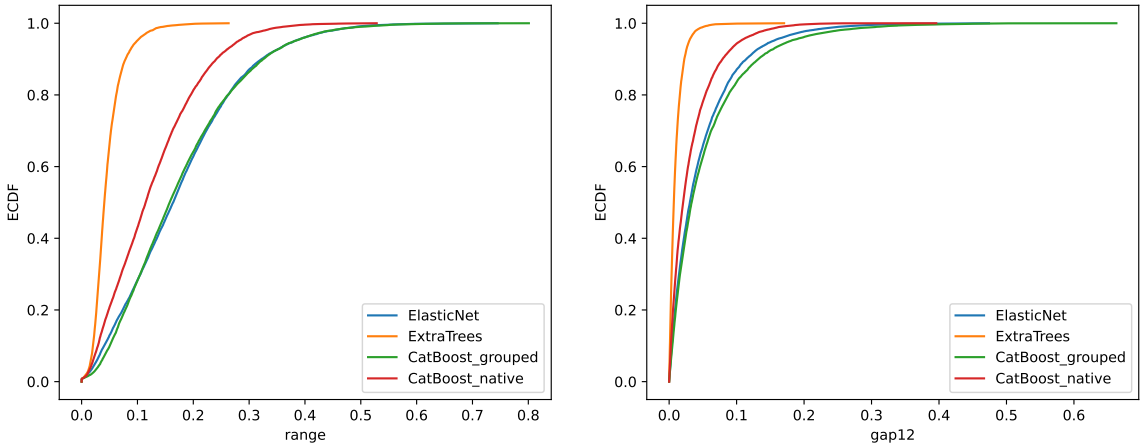
Notes: The figures plot the ROC curve in Panel B.1a and the precision-recall curve in Panel B.1b for the four candidate models as described in Section B.2. "AP" denotes average precision. The numbers in the legends represent the area under the curve in both panels.

Due to computational limitations, I perform the hyperparameter tuning via a two-stage grid search design. The initial grid search is over a moderately sized set of parameter values: depth $\in \{4, 6, 8\}$, learning rate $\in \{0.03, 0.06\}$, ℓ_2 regularization $\in \{1, 3, 10\}$, and iteration count $\in \{800, 1{,}200\}$. The best-performing configuration is concentrated in the "shallow + regularized" region, consistent with the high-cardinality categorical structure of the feature set. I then refine around the Stage 1 best-performing parameter region with a second grid that varies learning rate, ℓ_2 regularization, iterations, and minimum leaf size, with depth fixed at 4. The final tuned configuration is depth$= 4$, learning rate$= 0.04$, ℓ_2 regularization$= 20$, iterations$= 1{,}200$, and minimum number of training samples in a leaf$= 50$.

This tuned configuration improves the OOF performance on first-application contact prediction consistently. The ROC-AUC increases to 0.763 from 0.758, the average precision increases to 0.534 from 0.525, and both log loss and the Brier score decrease moderately. I proceed to use this configuration as the finalized version.

Robustness As robustness checks on the feature representation, I re-estimate CatBoost using the grid-search optimal parameters but replacing the native categorical inputs with (i) the frequency-based grouped encoding and (ii) the TF-

Figure B.2: Candidate Models' Dispersion Performance within Postings



(a) Predicted Score Range

(b) Top-1 versus Top-2 Score Gap

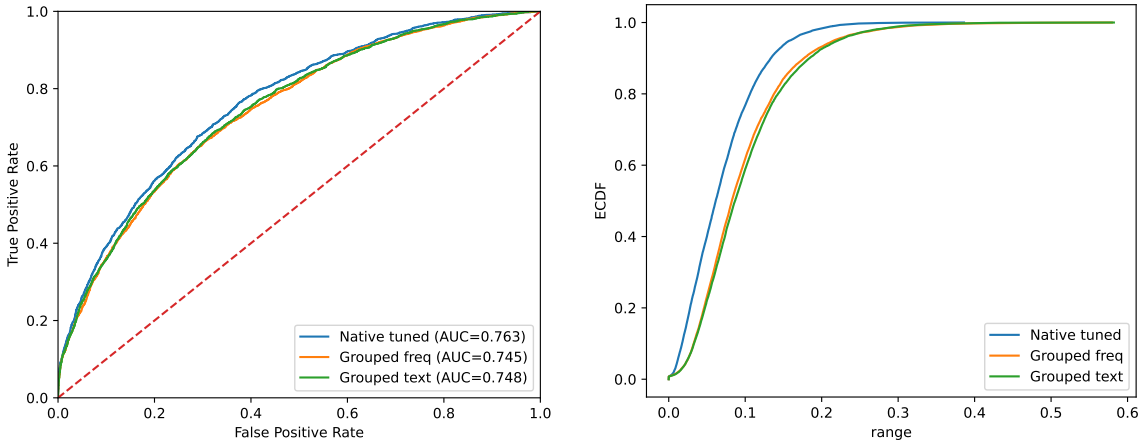
Notes: The figures plot the within-posting score ranges in Panel B.2a and the top-1 versus top-2 gaps in Panel B.2b for the four candidate models as described in Section B.2. In both panels, the y-axis is the empirical cumulative distribution function (ECDF). At any x-value, it shows the fraction of job postings whose within-posting dispersion statistic is less than or equal to x.

IDF clustering-based encoding described in Section B.1. Model comparisons are performed using the same out-of-fold evaluation protocol. Figure B.3 shows that the native CatBoost model outperforms the two encoded models in prediction but does not preserve as much within-posting dispersion. I use the predictions of the two encoded models in robustness checks.

B.5 Model Outputs and Reproducibility

I use the tuned native CatBoost model and the two encoded CatBoost models to predict the state-0 contact probability of later submissions. I use posting-level out-of-fold predictions to ensure that each posting's applications (orders 1-8) are scored using a model trained only on other postings. This yields uniformly out-of-sample predictions at the posting level, mitigates posting-level leakage through job characteristics, and provides a cleaner object for inference and for constructing within-posting counterfactual comparisons. Algorithm 1 summarizes the procedure. All reported downstream analyses use the score generated through this algorithm to avoid in-sample artifacts and posting-level leakage.

Figure B.3: CatBoost Out-of-Fold Performance: Native versus Encoded



(a) ROC Curve

(b) Predicted Score Range

Notes: Panel B.3a plots the ROC curve (the numbers in the legend report the area under the curve). Panel B.3b plots within-posting score ranges; the y-axis is the empirical cumulative distribution function (ECDF) as in Figure B.2. The blue line represents the tuned native CatBoost model, the orange line represents the tuned encoded CatBoost model with frequency-based grouping, and the green line represents the tuned encoded CatBoost model with text-based grouping as discussed in Section B.1.

Algorithm 1 OOF-by-posting prediction (GroupKFold cross-fitting)

Require: Dataset $\mathcal{D} = \{(X_{ij}, Y_{ij}, j)\}$ with posting ID j, order i, number of folds F, outcome $Y_{ij} = \mathbf{1}(\text{contacted})$.

Ensure: Out-of-fold predictions $\widehat{p}_{ij}^{OOF}$ for all applications.

Define state-0 sample $\mathcal{D}_0 \leftarrow \{(X_{ij}, Y_{ij}, j) \in \mathcal{D} : i = 1\}$.
Partition postings in $\mathcal{D}_0$ into folds $\{\mathcal{J}_1, \ldots, \mathcal{J}_F\}$ using GroupKFold by posting ID.
Initialize $\widehat{p}^{OOF}$ for all observations in $\mathcal{D}$.
for $f = 1, \ldots, F$ **do**
    Train on $\mathcal{T}_f \leftarrow \{(X_{ij}, Y_{ij}, j) \in \mathcal{D}_0 : j \notin \mathcal{J}_f\}$.
    Score $\mathcal{S}_f \leftarrow \{(X_{ij}, Y_{ij}, j) \in \mathcal{D} : j \in \mathcal{J}_f\}$ (all orders).
    Compute preprocessing parameters on $\mathcal{T}_f$ (e.g., medians for numeric imputation; `MISSING` category for categorical NA).
    Fit model $\widehat{m}_f$ on $\mathcal{T}_f$ and set $\widehat{p}_{ij}^{OOF} \leftarrow \widehat{m}_f(X_{ij})$ for all $(i, j) \in \mathcal{S}_f$.
end for
return $\widehat{p}^{OOF}$.

C Theoretical Appendix

This appendix presents the theoretical details that support Section 6. Section C.1 clarifies the screening-value object and the normalization of its scale. Section C.2 proves the reservation rule and comparative statics used in the main text. Sections C.3-C.5 show that the threshold logic is robust to post-contact continuation, non-stationary arrivals, and selective or batch review. Section C.6 sketches how the contact-initiation subgame could be embedded in a broader search environment.

C.1 Screening Value Microfoundation and Normalization

The main text works directly with a screening value v_{ij} for application i to vacancy j. This subsection gives a simple microfoundation for that object and explains why its location and scale are normalized in estimation.

Latent match quality and screening signals. Each application is characterized by observable resume attributes $\mathbf{x}_{ij}$ and vacancy attributes $\mathbf{z}_j$. Let q_{ij} denote the latent, job-specific match quality that would be realized if applicant i were hired for vacancy j. At the time of application, the employer does not observe q_{ij} directly. Reviewing the application generates a one-dimensional signal a_{ij} that is informative about match quality. For example,

$$q_{ij} = \mu(\mathbf{x}_{ij}, \mathbf{z}_j) + \nu_j^v + u_{ij}, \qquad a_{ij} = q_{ij} + \varepsilon_{ij},$$

where ν_j^v is a vacancy-level value component and u_{ij} and ε_{ij} are idiosyncratic components. The screening value is the employer's posterior value from entering the contact phase with this application as the current best candidate,

$$v_{ij} \equiv \mathbb{E}\left[S(q_{ij}) \mid a_{ij}, \mathbf{x}_{ij}, \mathbf{z}_j, \nu_j^v\right], \tag{C.1}$$

where $S(q)$ is the firm's payoff from eventually filling the vacancy with a match of quality q. The main text uses v_{ij} directly. It should be interpreted as a revealed-preference screening value, not as realized productivity or total match surplus.

Payoff from filling a vacancy. A standard bargaining environment rationalizes why the firm's payoff is monotone in match quality. Suppose match output is

$$y_{ij} = \chi_j + q_{ij},$$

where χ_j is a vacancy-specific productivity component. If wages are set by Nash bargaining with worker outside option b_i and worker bargaining weight $\eta \in (0, 1)$, then

$$w_{ij} = b_i + \eta(y_{ij} - b_i),$$

and the firm's payoff from filling the vacancy is

$$y_{ij} - w_{ij} = (1 - \eta)(\chi_j - b_i) + (1 - \eta)q_{ij}.$$

Thus, conditional on the vacancy and market environment, the firm's payoff is a positive affine transformation of match quality. Because the contact-initiation rule depends on rankings and threshold crossings, the main text normalizes the payoff scale and works directly with v_{ij}.[27]

Affine non-identification. The first-contact timing data identify the contact initiation rule only up to affine transformations of latent value. To see this, let the reservation equation in the stationary model be

$$\delta r + \kappa = \lambda \mathbb{E}\big[(v - r)^+\big], \tag{C.2}$$

where r is the reservation value, $\delta > 0$ is the discount rate, λ is the arrival rate, and κ is the flow shadow cost of delaying contact.

Lemma C.1 (Affine non-identification of screening value)**.** *Fix λ and δ, and suppose $v \sim F$. For any $a \in \mathbb{R}$ and $b > 0$, define $v' = a + bv$, $r' = a + br$, and $\kappa' = b\kappa - \delta a$. Then (r, κ, F) satisfies* (C.2) *if and only if (r', κ', F') satisfies*

$$\delta r' + \kappa' = \lambda \mathbb{E}\left[(v' - r')^+\right], \tag{C.3}$$

[27] Equivalently, one can carry $(1 - \eta)$ explicitly and treat b_i as a shifter in the opportunity cost of delay. In this study, wages are not observed for the matched hires and match quality is measured only up to a monotone transformation. As a result, the surplus-share factor $(1 - \eta)$ is not separately identified. The normalization is therefore convenient and without loss for the timing and screening decisions studied here.

where F' is the distribution of v'. Moreover, $\Pr(v' \geq r') = \Pr(v \geq r)$, so the transformed economy implies the same threshold-crossing probability and contact-initiation hazard.

Proof. Using the definitions above,

$$\delta r' + \kappa' = \delta(a + br) + (b\kappa - \delta a) = b(\delta r + \kappa).$$

Also, $(v' - r')^+ = b(v - r)^+$, so

$$\lambda \mathbb{E}\left[(v' - r')^+\right] = \lambda b \mathbb{E}\left[(v - r)^+\right].$$

Thus (C.3) holds if and only if (C.2) holds. Since $b > 0$, $\Pr(v' \geq r') = \Pr(v \geq r)$. That is, the transformed primitives imply the same contact-initiation hazard. ■

Lemma C.1 implies that timing moments alone do not separately identify the location and scale of the screening-value distribution from the level of the shadow cost of delay.

Corollary C.1 (Normalized reservation equation)**.** *Suppose $v = \mu + \sigma\tilde{v}$, with $\sigma > 0$ and normalized distribution $\tilde{v} \sim F_0$. Let $\tilde{r} = (r - \mu)/\sigma$. Then* (C.2) *is equivalent to*

$$\delta\tilde{r} + \frac{\delta\mu + \kappa}{\sigma} = \lambda \mathbb{E}\left[(\tilde{v} - \tilde{r})^+\right]. \tag{C.4}$$

Thus the identified delay-cost object is the normalized composite $(\delta\mu + \kappa)/\sigma$, not μ, σ, and κ separately.

Proof. Substitute $r = \mu + \sigma\tilde{r}$ and $v = \mu + \sigma\tilde{v}$ into (C.2) and divide by σ. ■

In the empirical implementation, I normalize the latent screening-value distribution and interpret estimated cost parameters in normalized screening-value units. Counterfactual losses are therefore first reported in screening-index units. Dollar magnitudes, when reported, are back-of-the-envelope calculations rather than direct estimates of realized productivity, wages, or firm profits.[28]

[28]The normalization argument applies to constant location and scale shifts of latent value. If applicant quality varies systematically with application timing, no constant affine transformation can absorb $\mu(d)$ into the flow cost term. I leave timing-dependent quality as an extension for future work.

C.2 Proofs for the Contact-Initiation Rule

C.2.1 Auxiliary results

Lemma C.2 (Stopping at arrivals)**.** *There is no loss from restricting attention to stopping times that occur at application arrival times, after at least one application has arrived.*

Proof. Between application arrivals, the best-so-far screening value is constant. Delaying a planned contact time from an interval interior to the next arrival weakly increases delay costs and discounting while providing no new information. Hence any optimal stopping rule can be modified to stop at an arrival time without lowering the payoff. ■

Let $W(v)$ denote the value function at an arrival time when the current best screening value is v. The Bellman equation in the main text is

$$W(v) = \max\left\{v, -\frac{\kappa}{\lambda+\delta} + \frac{\lambda}{\lambda+\delta}\mathbb{E}\big[W(\max\{v, v'\})\big]\right\}, \qquad v' \sim F_v. \tag{C.5}$$

By Lemma C.2, decisions can be analyzed at the sequence of arrival times. At each arrival n, after observing v_n and updating $v_n^{\max} \equiv \max\{v_{n-1}^{\max}, v_n\}$ for $n \geq 1$ and $v_0^{\max} \equiv -\infty$, the firm chooses whether to stop and realize $v_n^{\max}$ or continue.

Lemma C.3 (Finiteness and Lipschitz property)**.** *Under $\delta > 0$, $\lambda > 0$, $\kappa \geq 0$, and $\mathbb{E}[v^+] < \infty$, the Bellman equation (C.5) admits a finite solution W. Moreover, W is increasing and 1-Lipschitz: for all $v_2 \geq v_1$,*

$$0 \leq W(v_2) - W(v_1) \leq v_2 - v_1.$$

Proof. The Bellman operator associated with (C.5) is monotone and maps increasing 1-Lipschitz functions into increasing 1-Lipschitz functions. From $W_0(v) = v$, value iteration therefore preserves monotonicity and the Lipschitz bound. The sequence is bounded above by $v^+ + B$ for a finite constant B proportional to $(\kappa + \lambda\mathbb{E}[v^+])/\delta$, and hence converges pointwise to a finite fixed point. The fixed point inherits monotonicity and the Lipschitz property. ■

C.2.2 Proof of Proposition 6.1

Proof. By Lemma C.2, it is enough to consider arrival-time stopping rules. Let $C(v)$ denote the continuation value in (C.5) and define the gain from waiting relative to

initiating contact as

$$H(v) \equiv C(v) - v.$$

Continuing is optimal if and only if $H(v) \geq 0$.

First, H is strictly decreasing. For any $\varepsilon > 0$, the Lipschitz property implies

$$C(v + \varepsilon) - C(v) \leq \frac{\lambda}{\lambda + \delta}\varepsilon.$$

Therefore,

$$H(v + \varepsilon) - H(v) \leq \left(\frac{\lambda}{\lambda + \delta} - 1\right)\varepsilon < 0, \tag{C.6}$$

that is, $H(\cdot)$ is strictly decreasing.

Second, $H(v) \to +\infty$ as $v \to -\infty$: a feasible continuation strategy is to wait for one more arrival and then stop. This strategy yields the continuation payoff

$$J(v) \equiv -\frac{\kappa}{\lambda + \delta} + \frac{\lambda}{\lambda + \delta}\left(\mathbb{E}[\max\{v, v'\}]\right).$$

Because $C(v)$ is the optimal continuation value, $C(v) \geq J(v)$ for all v. Moreover, $\mathbb{E}[\max\{v, v'\}] \geq \mathbb{E}[v']$ for all v. $J(v)$ is therefore bounded below by a finite constant while the stopping payoff v tends to $-\infty$. Thus

$$H(v) = C(v) - v \geq J(v) - v \to +\infty, \quad \text{as } v \to -\infty.$$

Third, $H(v) \to -\infty$ as $v \to +\infty$ because H decreases at least linearly by the inequality (C.6) and is finite at any fixed value of v. By Lemma C.3, W is 1-Lipschitz and continuous. Therefore, both $C(v)$ and $H(v)$ are continuous. $H(v)$ takes both positive and negative values, and is strictly decreasing. Therefore, there exists a unique $r \in \mathbb{R}$ such that $H(r) = 0$. $H(v) \geq 0$ if and only if $v \leq r$. The optimal policy is then: continue if $v < r$ and stop if $v \geq r$. Equivalently, the optimal stopping time is the first arrival time at which the best-so-far value crosses r. ■

At the cutoff, the firm is indifferent between stopping and waiting. Substituting $W(\max\{r, v'\}) = \max\{r, v'\}$ into the Bellman equation yields the reservation condition

$$\delta r + \kappa = \lambda \mathbb{E}\left[(v - r)^+\right]. \tag{C.7}$$

This is the equation used in the main text.

C.3 Post-Contact Continuation

The main model focuses on the decision to enter the contact phase. This does not require the firm to stop all recruiting activity literally after first contact. The next result shows that a monotone post-contact pipeline preserves the threshold structure of the initiation decision.

Lemma C.4 (Robustness to post-contact processing)**.** *Suppose that when the firm initiates contact, the expected present discounted value of entering the post-contact pipeline is $G(v)$, where v is the current best screening value and $G : \mathbb{R} \to \mathbb{R}$ is weakly increasing. Then the optimal contact-initiation policy is still a cutoff rule: there exists a threshold r such that it is optimal to initiate contact if and only if $v \geq r$.*

Proof. With post-contact payoff $G(v)$, the arrival-time Bellman equation becomes

$$W(v) = \max\left\{G(v), -\frac{\kappa}{\lambda+\delta} + \frac{\lambda}{\lambda+\delta}\left(\mathbb{E}\left[W(\max\{v, v'\})\right]\right)\right\}, \tag{C.8}$$

where v is the current best screening value at an arrival and $v' \sim F_v$ is the next screened draw. Let $y \equiv G(v)$ and $y' \equiv G(v')$. Since G is weakly increasing, $G(\max\{v, v'\}) = \max\{y, y'\}$. Define the generalized inverse $G^{-1}(y) \equiv \inf\{v : G(v) \geq y\}$ and the transformed value function $\widetilde{W}(y) \equiv W(G^{-1}(y))$.[29] (C.8) is equivalent to

$$\widetilde{W}(y) = \max\left\{y, -\frac{\kappa}{\lambda+\delta} + \frac{\lambda}{\lambda+\delta}\left(\mathbb{E}\left[\widetilde{W}(\max\{y, y'\})\right]\right)\right\}, \tag{C.9}$$

which has the same form as (C.5). Proposition 6.1 therefore implies a cutoff in y, and monotonicity of G maps this cutoff back into a cutoff in v. ■

Lemma C.4 covers post-contact processes such as contacting backup candidates, moving several applicants to interviews, or continuing to screen after outreach begins, as long as the value of entering the post-contact pipeline is monotone in the best-so-far screening value. These features affect the level and interpretation of the contact-phase value, but not the main threshold logic for initiation.[30]

[29] If G is strictly increasing then G^{-1} is the usual inverse.

[30] For example, if the first contacted candidate ultimately converts into a hire with probability

C.4 Nonstationary Arrival Profiles

The main text uses a stationary arrival environment for tractability. This subsection allows applications to arrive according to a deterministic posting-age profile,

$$\lambda(d) = \lambda\rho(d), \qquad d \geq 0,$$

where $\lambda > 0$ is a scale parameter and $\rho(d) \geq 0$ is common across vacancies. The screening-value distribution remains stationary.

Let $W(d, v)$ be the value at posting age d when the best-so-far value is v. In the continuation region, $W(d, v)$ satisfies

$$0 = -\kappa - \delta W(d, v) + \partial_d W(d, v) + \lambda\rho(d) \left(\mathbb{E}[W(d, \max\{v, v'\})] - W(d, v)\right), \quad \text{(C.10)}$$

where $v' \sim F_v$ and κ, δ are constant within the vacancy spell. Under the same single-crossing logic as in the stationary model, the optimal policy is characterized by a time-varying reservation value $r(d)$. Assuming smooth pasting at the boundary, the reservation condition is

$$\delta r(d) + \kappa = \lambda\rho(d)\mathbb{E}\big[(v - r(d))^+\big]. \quad \text{(C.11)}$$

Proposition C.1 (Monotonicity of the reservation function)**.** *Suppose ρ is differentiable, $\rho'(d) \leq 0$, F_v has a continuous density, and the smooth-pasting condition holds. Let $p(r) = \Pr(v \geq r)$ and $G(r) = \mathbb{E}[(v - r)^+]$. Then*

$$r'(d) = \frac{\lambda\rho'(d)G(r(d))}{\delta + \lambda\rho(d)p(r(d))} = \frac{\rho'(d)\left(\delta r(d) + \kappa\right)}{\left[\delta + \lambda\rho(d)p(r(d))\right]\rho(d)}. \quad \text{(C.12)}$$

In particular, if $\rho'(d) \leq 0$, then $r'(d) \leq 0$ whenever $G(r(d)) > 0$.

Proof. Differentiate the boundary condition (C.11) with respect to d:

$$\delta r'(d) = \lambda\rho'(d)G(r(d)) + \lambda\rho(d)G'(r(d))r'(d).$$

$\alpha \in (0, 1]$, and failure yields a normalized continuation value of zero, then $G(v) = \alpha v$ and the initiation policy remains a reservation rule in v with an appropriately rescaled cutoff. Accordingly, in the main text I normalize v to be the firm's expected payoff from entering the contact stage without loss for the timing and comparative statics here.

Using $G'(r(d)) = -p(r(d))$ and rearranging yields

$$(\delta + \lambda\rho(d)p(r(d)))\, r'(d) = \lambda\rho'(d)G(r(d)),$$

which is the first equality in (C.12). The second equality follows from (C.11). The sign statements follow because $\delta + \lambda\rho(d)p(r(d)) > 0$. ■

Thus, when application inflow declines with posting age, the option value of waiting falls and the firm becomes weakly less selective over the recruiting spell. The main model abstracts from this time dependence because posting dates are not observed for the experimental vacancies; the extension shows that the core option-value versus delay-cost mechanism is preserved.

C.5 Selective and Batch Review

The model used for estimation assumes that arrival and review coincide at the contact-initiation margin. This subsection discusses how the logic extends when firms choose which applications to review or review applications in batches.

Selective review. Suppose arrived applications can be reviewed at cost $c_R \geq 0$, after which the firm observes screening value v. For an unreviewed application with observable attributes, define a review index ζ by

$$\mathbb{E}[(v - \zeta)^+ \mid \text{observables}] = c_R. \tag{C.13}$$

In a static set of arrived applications with independent values, Weitzman's Pandora rule implies that the firm reviews applications in descending order of ζ and stops reviewing when the best revealed value exceeds the highest remaining index. With ongoing arrivals, selective review adds a review-order margin to the contact-initiation margin. It changes which applications are processed before initiation, but it preserves the same economic force: additional applications are valuable only through the option that they exceed the current best revealed value.

Batch review. Suppose the firm reviews applications at review times separated by an interval Δ. If arrivals during the interval follow a Poisson process with rate

λ, the best value in the batch, M_Δ, has distribution

$$\Pr(M_\Delta \leq x) = \exp\{-\lambda\Delta[1 - F_v(x)]\}. \tag{C.14}$$

The expected improvement from waiting Δ when the current best is v is

$$\mathbb{E}[(M_\Delta - v)^+] = \int_v^\infty (1 - \exp\{-\lambda\Delta[1 - F_v(x)]\})\, dx. \tag{C.15}$$

This object is increasing and concave in Δ.

A reduced-form dynamic program at review times is:

$$W_\Delta(v) = \max\{v,\ C_\Delta(v)\}, \tag{C.16}$$

with continuation value

$$C_\Delta(v) = -\int_0^\Delta e^{-\delta s}\kappa \mathrm{d}s - e^{-\delta\Delta}c_0 + e^{-\delta\Delta}\mathbb{E}\left[W_\Delta\left(\max\{v, M_\Delta\}\right)\right], \tag{C.17}$$

where $c_0 \geq 0$ is the fixed setup cost to initiate a review session.

Batch review therefore creates a frequency tradeoff. The firm chooses the review interval Δ. Longer intervals economize on review sessions but delay contact and yield diminishing marginal information gains. The immediate-review model in the main text is the limiting case in which contact-stage review is sufficiently frequent that the initiation decision can be approximated at arrivals ($\Delta \to 0$ with $c_0 = 0$).

These extensions are not estimated. They are included to clarify that same-day batching or selective review does not by itself invalidate the contact-initiation model. The key empirical object is still whether the vacancy has entered the contact phase, and the key theoretical force remains the tradeoff between the option value of additional applicants and the shadow cost of waiting. In the data, there is no clear evidence supporting batch review.

C.6 Embedding the Subgame in a Search Environment

This subsection sketches how the within-vacancy contact-initiation problem can be embedded in a broader search environment. The sketch is not used for estimation. It clarifies the equilibrium objects that are held fixed in the partial-equilibrium

analysis.

Consider a market m and calendar period t with unemployment U_{mt}, vacancies V_{mt}, tightness $\theta_{mt} = V_{mt}/U_{mt}$, and congestion $c_{mt} = U_{mt}/V_{mt}$. Workers choose an application intensity and route applications across active vacancies. In an undirected benchmark with application intensity ξ_{mt}, the arrival rate to an individual vacancy is

$$\lambda_{mt} = \xi_{mt} \frac{U_{mt}}{V_{mt}} = \xi_{mt} c_{mt}. \tag{C.18}$$

Directed search or taste-based firm attractiveness can be represented by routing weights that scale this per-vacancy inflow.

Conditional on these market aggregates and a posted vacancy, the firm solves the contact-initiation subgame estimated in the main text. If initiating contact leads to a hire with probability $h(v^{\max})$, the vacancy-filling hazard can be written schematically as

$$q_{mt} = \underbrace{\lambda_{mt} \Pr(v \geq r_{mt})}_{\text{contact-initiation hazard}} \times \underbrace{\mathbb{E}[h(v^{\max}) \mid \text{initiation}]}_{\text{post-contact conversion}}. \tag{C.19}$$

The experiment disciplines the firm's within-vacancy timing and selection behavior holding resume content fixed, which is sufficient to identify the recruiting policy rule conditional on (θ_{mt}, ξ_{mt}). However, closing equilibrium requires additional objects that are not identified by the experimental data, including worker application costs, routing preferences, vacancy posting costs, wage setting, and post-contact conversion outcomes. For this reason, the paper estimates only the within-vacancy recruiting subgame and interprets the counterfactuals as partial-equilibrium changes holding the market environment fixed.

D Estimation Appendix

This appendix describes the implementation of the method of simulated moments (MSM) estimator used in Section 6.3. The main text defines the contact-initiation model, the mapping to the experimental initiation margin, and the fit to the central level moments. Appendix D.1 defines the estimation sample and empirical initiation weights. Appendix D.2 gives the full parameterization of effective arrival intensity, the shadow cost of delay, and the recruiting-state shifter. Appendix D.3 defines the moment vector and its sampling uncertainty. Appendix D.4 describes the MSM objective, numerical implementation, and bootstrap. Appendix D.5 reports the identification discussion, normalizations, and full parameter estimates. Appendix D.6 reports the full model fit and robustness checks. I use the notation of Section 6.3: for vacancy j, $m(j)$ is the state of the vacancy, $t(j)$ is the month in which the first experimental application is submitted, and $d_{ij} = D2F_i^t$ is application i's relative submission-day bin.

D.1 Estimation Sample and Experimental Initiation Weights

The estimation sample is restricted to vacancies with the full set of eight experimental applications. This restriction aligns the model with the experimental tracer design and avoids mixing the contact-initiation process with cases in which the experimental sequence is truncated because a vacancy closes or becomes unavailable before all tracer applications are sent. Let $\mathcal{J}$ denote the set of complete vacancies. Each vacancy contributes eight application-level observations.

For each experimental application i in vacancy j, timing is measured by

$$d_{ij} \equiv D2F_i^t \in \{0, 1, \ldots, 7\},$$

where $D2F_i^t$ is the number of calendar days between application i's submission date and the first experimental submission date in the vacancy, top-coded at seven or more days. Let n_{jd} be the number of experimental tracers submitted in bin d, and let $N_{j,<d} \equiv \sum_{\tau<d} n_{j\tau}$ be the number of tracers submitted before bin d. For a parameter vector Θ and simulation draw s, the model implies an effective arrival intensity λ_{js} and a threshold-crossing probability $p_{js} = p(\lambda_{js}, \kappa_j)$. The probability

that experimental initiation is assigned to bin d is

$$P_{jds}(\Theta) = \underbrace{\exp(-\lambda_{js} p_{js} d)}_{\substack{\text{no acceptable outside arrival} \\ \text{before bin } d}} \cdot \underbrace{(1-p_{js})^{N_{j,<d}}}_{\substack{\text{no acceptable tracer} \\ \text{before bin } d}} \cdot \underbrace{[1-(1-p_{js})^{n_{jd}}]}_{\substack{\text{at least one acceptable} \\ \text{tracer in bin } d}}. \tag{D.1}$$

Conditional on initiation being assigned to bin d, the model splits the mass equally across tracers submitted in that bin:

$$\omega_{ijs}(\Theta) = \frac{P_{j,d_{ij},s}(\Theta)}{n_{j,d_{ij}}}. \tag{D.2}$$

The residual mass

$$\omega_{js}^{\text{none}}(\Theta) = 1 - \sum_{i=1}^{8} \omega_{ijs}(\Theta) \tag{D.3}$$

is the probability that initiation is not assigned to one of the eight experimental tracers. This residual state may include an organic first contact, no contact during the observation window, vacancy closure, or other unobserved outcomes. It disciplines how much initiation mass leaves the experimental set, and should not be interpreted as a primitive vacancy-filling outcome.

The empirical analogue uses the first experimental contact date within each vacancy. Let C_{ij}^{first} equal one if application i is contacted on the first date on which any experimental application to vacancy j receives employer contact. If multiple experimental applications are contacted on that date, I split the initiation mass equally across them:

$$\omega_{ij}^{\text{data}} = \begin{cases} C_{ij}^{\text{first}} / \sum_{i'=1}^{8} C_{i'j}^{\text{first}}, & \text{if } \sum_{i'=1}^{8} C_{i'j}^{\text{first}} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{D.4}$$

The empirical none state is

$$\omega_{j}^{\text{data,none}} = 1 - \sum_{i=1}^{8} \omega_{ij}^{\text{data}}. \tag{D.5}$$

Thus $\omega_{j}^{\text{data,none}} = 1$ for vacancies with no experimental contact initiator. This construction is the same equal-weighted contact-initiator outcome used in Section 5.

The wave-group notation follows the estimation sample. Let $h(j) \in \{1, 2, 3, 4\}$

denote the experimental wave group. Wave groups 1 and 2 correspond to the first two experimental waves. Wave group 3 pools Waves 3 and 4 because they overlap substantially in calendar time and have similar reduced-form timing patterns. Wave group 4 corresponds to Wave 5. The wave grouping is used only for empirical level moments and for the wave-specific recruiting-state specification described below.

D.2 Parameterization of Arrival Intensity and Delay Costs

The estimation parameterizes the two primitives of the contact-initiation model: the effective arrival intensity λ_j and the shadow cost of delay κ_j. The effective arrival intensity is not an observed organic application count. It is the model-implied rate at which additional applicant opportunities arrive and can preempt initiation on the experimental tracers.

Effective arrival intensity. Let $\theta_j \equiv \theta_{m(j),t(j)}$ be labor-market tightness in the state of vacancy j and the month of the first experimental submission, measured as vacancies per unemployed worker, and let $c_j \equiv 1/\theta_j$ denote congestion. For simulation draw s,

$$\lambda_{js} = \exp\left(\log \xi_j + \alpha_c \widetilde{\log c_j} + \nu^{\lambda}_{js}\right), \qquad \nu^{\lambda}_{js} \sim \mathcal{N}\left(-\sigma^2_\nu/2, \sigma^2_\nu\right). \tag{D.6}$$

The term $\widetilde{\log c_j}$ is centered within each wave group, so ξ_j captures shifts in the recruiting environment net of within-wave congestion differences. The log-normal shock integrates over unobserved vacancy-level heterogeneity in effective arrival intensity and is normalized so that $\mathbb{E}[\exp(\nu^{\lambda}_{js})] = 1$. The congestion term follows the urn-ball logic of directed-search application technologies: expected applications per vacancy scale with application intensity and workers per vacancy (Wright et al., 2021; Birinci, See and Wee, 2025). I estimate the congestion elasticity α_c rather than imposing it.

I use two specifications for the recruiting-state shifter ξ_j. In the wave-specific specification,

$$\log \xi_j = \log \xi_{h(j)}. \tag{D.7}$$

In the macro-disciplined specification,

$$\log \xi_j = \xi_0 + \sum_{\ell \in \{\text{HIR,QUR,LDR}\}} \xi_\ell Z_{\ell,t(j)} + \sum_{\ell \in \{\text{HIR,QUR,LDR}\}} \xi_\ell^{\Delta} Z^{\Delta}_{\ell,m(j),t(j)}. \tag{D.8}$$

$Z_{\ell,t(j)}$ is the standardized aggregate month value of JOLTS series ℓ[31], and $Z^{\Delta}_{\ell,m(j),t(j)}$ is the standardized state-month deviation from the aggregate month mean. These series summarize broader recruiting conditions not captured by tightness alone. I do not interpret the individual ξ_ℓ coefficients as separate causal effects.

Shadow cost of delaying contact. The shadow cost of delay κ_j is parameterized as

$$\log \kappa_j = \log \kappa_0 + \exp\left(\kappa_{\theta,0} + \kappa_{\theta,1}\widetilde{\log \xi_j}\right) z(\log \theta_j), \tag{D.9}$$

where $z(\log \theta_j)$ is the globally standardized log tightness measure and $\widetilde{\log \xi_j}$ is the centered recruiting-state shifter. The exponential loading disciplines the tightness channel in the direction suggested by Section 5: tighter labor markets raise the shadow cost of waiting and therefore encourage earlier contact initiation. Because κ_j is a composite shadow cost, this parameterization should not be interpreted as separating vacancy carrying costs, applicant outside-option risk, recruiter capacity, or organizational incentives.

D.3 Moment Vector and Sampling Uncertainty

The main estimator uses a deliberately lean moment set. The moments are built from the empirical initiation weights $\omega_{ij}^{\text{data}}$ and their model analogues $\bar{\omega}_{ij}(\Theta) \equiv S^{-1}\sum_{s=1}^{S}\omega_{ijs}(\Theta)$.

Level moments. For each wave group h and day bin d, the empirical level moment is the average initiation mass assigned to applications in bin d:

$$g_{hd}^{\text{data}} = \frac{1}{N_h}\sum_{j:h(j)=h}\sum_{i=1}^{8}\omega_{ij}^{\text{data}}\mathbf{1}\{d_{ij}=d\}, \tag{D.10}$$

[31] HIR denotes hires rate, QUR denotes quits rate, LDR denotes layoffs and discharges rate.

where N_h is the number of complete vacancies in wave group h. The residual-none moment is

$$g^{\text{data}}_{h,\text{none}} = \frac{1}{N_h} \sum_{j:h(j)=h} \omega^{\text{data,none}}_j. \tag{D.11}$$

The model analogues replace $\omega^{\text{data}}_{ij}$ and $\omega^{\text{data,none}}_j$ with $\bar{\omega}_{ij}(\Theta)$ and $\bar{\omega}^{\text{none}}_j(\Theta)$. Because the day-bin masses and the residual none state sum to one within each wave group, one level moment ($d = 7$) per wave group is omitted.

Tightness-gradient moment. The second moment family is the pooled tightness-gradient moment from Section 5. In the data, I estimate the contact-initiation specification using $\omega^{\text{data}}_{ij}$ as the dependent variable. The targeted coefficient is the interaction between $D2F^t_i$ and local log tightness as in Column (2) of Table 6. In the model, I estimate the same auxiliary regression using $\bar{\omega}_{ij}(\Theta)$ as the dependent variable, with the same controls and clustering level as in the corresponding reduced-form specification. This moment disciplines how the contact-initiation policy shifts with recruiting pressure.

I also use grouped tightness moments as robustness checks. Within each wave group, vacancies are split into bottom and top terciles of local tightness. The first grouped specification uses the high-minus-low differences in $\mathbb{E}[\omega^{\text{data,none}}_j]$ and $\mathbb{E}[\omega^{\text{data}}_{ij} \mid d_{ij} \leq 1]$, computed within each wave group and averaged across wave groups using vacancy-count weights. The second grouped specification adds the high-minus-low difference in $\mathbb{E}[\omega^{\text{data}}_{ij} \cdot d_{ij}]$. These grouped moments replace the linear auxiliary regression with transparent extensive and intensive timing summaries.

Bootstrap uncertainty. I estimate the sampling uncertainty of g^{data} using a 500-draw vacancy-cluster bootstrap stratified by wave group. Each bootstrap replication resamples vacancies and preserves the eight experimental applications within each vacancy. Duplicated vacancies receive distinct bootstrap identifiers so each resampled copy enters as a separate cluster. Let $G^{(b)}$ be the moment vector in bootstrap replication $b = 1, \ldots, B$. The bootstrap covariance matrix is $\widehat{\Sigma} = \text{Var}_b(G^{(b)})$, and $\widehat{se}_m$ denotes the bootstrap standard error of moment m. The empirical moments and their bootstrap standard errors are summarized in Table D.1.

Table D.1: Targeted Empirical Moments

Level moments: Contact-initiation day mass g_{hd}^{data} and $g_{h,\text{none}}^{\text{data}}$								
	$d=0$	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$	$d=6$	None
$h(j)=1$	0.1628	0.0478	0.0345	0.0294	0.0263	0.0179	0.0234	0.6384
	(0.0088)	(0.0051)	(0.0042)	(0.0038)	(0.0036)	(0.0030)	(0.0034)	(0.0118)
$h(j)=2$	0.2120	0.0768	0.0580	0.0446	0.0123	0.0035	0.0026	0.5867
	(0.0088)	(0.0050)	(0.0041)	(0.0037)	(0.0022)	(0.0011)	(0.0011)	(0.0111)
$h(j)=3$	0.1552	0.0766	0.0670	0.0503	0.0174	0.0039	0.0021	0.6255
	(0.0051)	(0.0035)	(0.0032)	(0.0029)	(0.0018)	(0.0009)	(0.0006)	(0.0072)
$h(j)=4$	0.2465	0.1131	0.0818	0.0629	0.0134	0.0018	0.0005	0.4792
	(0.0083)	(0.0060)	(0.0048)	(0.0045)	(0.0021)	(0.0009)	(0.0005)	(0.0107)
Tightness moments								
Main: regression moments							-0.0291	
							(0.0037)	
Tightness-tercile-based moments, basic								
$\mathbb{E}[\omega_j^{\text{data,none}} \mid \text{high}] - \mathbb{E}[\omega_j^{\text{data,none}} \mid \text{low}]$							-0.0639	
							(0.0117)	
$\mathbb{E}[\omega_{ij}^{\text{data}} \mid d_{ij} \leq 1, \text{high}] - \mathbb{E}[\omega_{ij}^{\text{data}} \mid d_{ij} \leq 1, \text{low}]$							0.0294	
							(0.0153)	
Tightness-tercile-based moments, shape								
$\mathbb{E}[\omega_{ij}^{\text{data}} \cdot d_{ij} \mid \text{high}] - \mathbb{E}[\omega_{ij}^{\text{data}} \cdot d_{ij} \mid \text{low}]$							-0.0767	
							(0.0509)	

Note: This table reports the empirical moments targeted in the MSM procedure. The moments are constructed as described in Section 6.3 and Appendix D.3. Level moments refer to the day-specific contact-initiation mass, g_{hd}^{data} and $g_{h,\text{none}}^{\text{data}}$ for wave group h and day bin d. The column labeled "None" reports the residual probability that contact initiation is not assigned to an experimental tracer. The main estimation results are based on the regression-based tightness moment as introduced in Table 6. All estimates are based on the sample of vacancies with eight experimental applications. Bootstrapped standard errors are reported in parentheses.

D.4 MSM Objective, Optimization, and Bootstrap

Let $g(\Theta)$ denote the model-implied analogue of the empirical moment vector g^{data}. The preferred estimator uses diagonal bootstrap weighting:

$$\widehat{\Theta}_{\text{diag}} = \arg\min_{\Theta} \sum_{m=1}^{M} \left(\frac{g_m(\Theta) - g_m^{\text{data}}}{\widehat{se}_m} \right)^2 . \tag{D.12}$$

Diagonal weighting avoids placing excessive weight on small differences among highly correlated day-bin moments. As a robustness check, I also estimate a full-covariance specification,

$$\widehat{\Theta}_{\text{full}} = \arg\min_{\Theta} \left(g(\Theta) - g^{\text{data}} \right)' \widehat{\Sigma}^{-1} \left(g(\Theta) - g^{\text{data}} \right), \tag{D.13}$$

implemented using a ridge-stabilized Cholesky whitening rather than explicit inversion.

For a given Θ, I compute simulated moments using common random numbers. The Gaussian arrival-heterogeneity draws are generated once and held fixed across objective evaluations. For each vacancy and simulation draw, I construct λ_{js} from Equation (D.6), construct κ_j from Equation (D.9), evaluate the threshold-crossing probability $p_{js} = p(\lambda_{js}, \kappa_j)$, compute the tracer weights $\omega_{ijs}(\Theta)$ and none mass, average them over simulation draws, and aggregate them into $g(\Theta)$ using the same definitions as in the data.

To speed computation, I precompute $p(\lambda, \kappa)$ on a grid in $(\log \lambda, \log \kappa)$ by solving the reservation condition in Equation (6.4). During estimation, a bivariate interpolator maps each simulated (λ_{js}, κ_j) pair into a threshold-crossing probability. Values outside the grid are clipped to the grid boundary, and probabilities are clipped to $[10^{-12}, 1 - 10^{-12}]$ for numerical stability. I choose grid bounds to make clipping negligible in the preferred estimates.

Optimization uses a bound-constrained nonlinear least-squares routine with trust-region reflective steps and common simulation draws. I test multiple starting values to avoid local optima. Parameter uncertainty is computed by vacancy-cluster bootstrap: in each bootstrap replication, I recompute the empirical moments, reestimate the model with the same simulation and interpolation objects, and recompute fitted and counterfactual objects. The main text reports the diagonally weighted macro-disciplined specification. Wave-specific, full-covariance, and grouped-tightness specifications are reported as robustness checks below.

D.5 Identification, Normalizations, and Parameter Estimates

This subsection explains how the targeted moments discipline the model and reports the full set of parameter estimates. The discussion is deliberately interpretive: the model separates effective arrival intensity, threshold-crossing probabilities, and the shadow cost of delay through the maintained contact-initiation structure rather than through one-to-one empirical variation.

Identification and normalizations. The model is disciplined by the experimental contact-initiation moments described in Table D.1: the distribution of equal-

weighted experimental initiators across $D2F_i^t$ bins, the residual none state, and the tightness-gradient moment. These moments jointly discipline the frequency and timing with which contact initiation is assigned to experimental tracers. Because the estimation targets timing moments rather than realized productivity or wages, the latent quality distribution is normalized and match-quality objects are constructed only after estimation using the ML screening index. Table D.2 reports the estimates for both the preferred macro-disciplined and benchmark wave-specific specifications.

The resulting decomposition into effective arrival intensity λ, threshold crossing probabilities, and the shadow cost of delay κ is model-implied under the maintained sequential contact-initiation structure. It is not a nonparametric decomposition. The experiment does not observe the full organic applicant pool, the true first contact among all applicants, or downstream outcomes. The model therefore identifies a contact-initiation policy on the experimental tracer margin, not a complete hiring technology.

The scale of screening value is also normalized. The MSM moments use the timing and incidence of contact initiation, not realized productivity or wages. The ML screening index used in the counterfactuals provides a revealed-preference ranking of applications, but it is not a direct measure of match surplus. The parameter estimates below should be interpreted with these normalizations in mind.

Recruiting-state shifter ξ_j. ξ_j summarizes variation in the effective arrival environment not captured by tightness alone. It is related to the matching-efficiency residual in aggregate search models: conditional on tightness, some periods generate more matches than others. In the contact-initiation model, ξ_j plays an analogous role at an earlier stage of the hiring process by shifting the effective arrival intensity relevant for assigning initiation to experimental tracers. I therefore interpret ξ_j as a contact-initiation analogue of matching efficiency, not as an estimate of aggregate matching efficiency itself.

The macro-disciplined specification parameterizes ξ_j using the hires rate, quits rate, and layoffs and discharges rate. These series are motivated by work linking recruiting intensity, worker turnover, and matching efficiency to job-filling rates (Petrongolo and Pissarides, 2001; Davis, Faberman and Haltiwanger, 2012; Faber-

Table D.2: Parameter Estimation Results and Model Fitting

	(1) Wave (diag)	(2) Wave (full)	(3) Macro (diag)	(4) Macro (full)
Panel A: λ and κ parameters				
α_c	0.860	0.845	0.401	0.449
$\log \sigma_\nu$	0.708	0.690	0.252	0.274
$\log \kappa_0$	-4.522	-4.577	-4.335	-4.375
$\kappa_{\theta,0}$	-3.847	-4.000	-3.954	-4.000
$\kappa_{\theta,1}$	1.289	2.000	1.258	1.933
Panel B: ξ parameters				
$\log \xi_1$	2.451	2.338		
$\log \xi_2$	2.162	2.080		
$\log \xi_3$	2.457	2.379		
$\log \xi_4$	1.518	1.419		
ξ_0			1.159	1.159
ξ_{HIR}			-0.507	-0.523
ξ_{LDR}			-0.994	-1.071
ξ_{QUR}			-0.444	-0.472
$\xi^{\Delta}_{\text{HIR}}$			-0.382	-0.377
$\xi^{\Delta}_{\text{LDR}}$			1.355	1.212
$\xi^{\Delta}_{\text{QUR}}$			-0.592	-0.607
Panel C: Model fitting summary				
Q_{diag}	73.695	75.014	57.029	59.455
Q_{full}	76.786	75.709	61.407	61.680
Q_{level}	73.679	75.007	56.977	59.451
Q_θ	0.016	0.007	0.053	0.004
N(submissions)		78,632		
N(vacancies)		9,829		

Note: This table reports parameter estimates and MSM objectives for four specifications. Columns (1) and (2) use wave-group shifters for $\log \xi_j$; columns (3) and (4) use the macro-disciplined recruiting-state shifter. Columns (1) and (3) use diagonal bootstrap weighting; columns (2) and (4) use the ridge-regularized full bootstrap covariance matrix. Panel A reports parameters governing effective arrival intensity and the shadow cost of delay. Panel B reports recruiting-state parameters. In the wave-specific specifications, $\{\log \xi_h\}_{h=1}^4$ are wave-group shifters. In the macro-disciplined specifications, the ξ coefficients load on standardized JOLTS series and state-month deviations. Panel C reports the diagonal and full-covariance MSM criteria. The Q_{level} and Q_θ rows decompose Q_{diag} into contributions from the level moments and the tightness-gradient moment; each block contribution is the sum of squared standardized residuals within that moment family. Individual ξ coefficients are components of a fitted recruiting-state shifter and should not be interpreted as separate causal effects. Bounds are imposed for numerical stability.

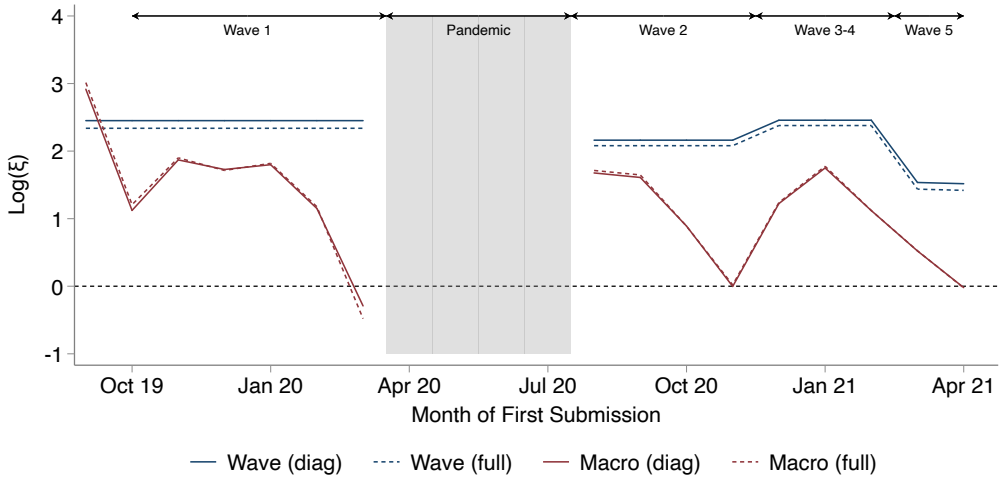
man, 2020).[32] I use these variables to span a low-dimensional index of recruiting conditions, not to estimate separate causal effects of each flow series. The individ-

[32]Recent work also emphasizes the role of recruiting intensity, worker turnover, mismatch, and congestion in shaping aggregate matching efficiency and job-filling rates; see, for example, Forsythe and Weinstein (2025); Mercan, Schoefer and Sedláček (2024); Afrouzi et al. (2026); Birinci, Mercan and See (2025); Cederlöf et al. (2025).

ual coefficients in Table D.2 are reported for transparency, but the relevant object for the model is the fitted index ξ_j.

Figure D.1 plots month averages of the estimated $\log \xi_j$ under the wave-specific and macro-disciplined specifications. The wave-specific specification captures discrete shifts across experimental regimes, while the macro-disciplined specification produces a smoother path driven by observed JOLTS series. The figure emphasizes relative movements over time.

Figure D.1: Estimated Recruiting-State Shifter $\log \xi_j$



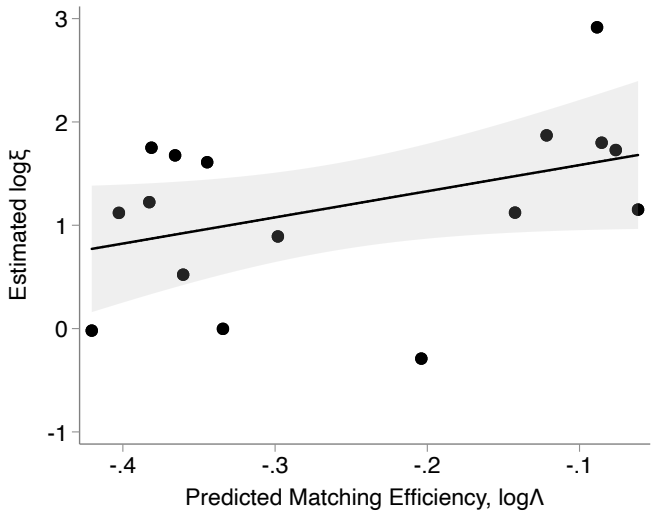
Notes: The figure plots monthly averages of the implied recruiting-state shifter $\log \xi_j$ under the wave-specific and macro-disciplined specifications. For comparability, all four series are computed over vacancies observed in each month. The levels of $\log \xi_j$ are not separately meaningful because of scale normalizations, so the figure emphasizes relative movements over time.

As a descriptive validation exercise, I compare the estimated ξ_j to an external national matching-efficiency proxy constructed from aggregate JOLTS hires, vacancies, and unemployment. Under Cobb-Douglas matching, the vacancy yield $q_t \equiv H_t/V_t$ satisfies $\log q_t = \log \Lambda_t + \alpha_c \log(U_t/V_t)$. Accordingly, I construct

$$\widehat{\log \Lambda}_t \equiv \log\left(H_t/V_t\right) - \widehat{\alpha}_c \log\left(U_t/V_t\right),$$

where $\widehat{\alpha}_c$ is the estimated congestion elasticity from the macro-disciplined specification (Column (3) of Table D.2). Figure D.2 plots the macro-disciplined $\log \xi_t$ against this proxy. The positive comovement supports interpreting ξ_j as a shift of the recruiting state rather than a purely mechanical artifact of the MSM fit. This comparison is descriptive and is not used in estimation.

Figure D.2: External Validity of Estimated Recruiting-State Shifter $\log \xi_j$



Notes: The figure plots the estimated monthly shifter $\log \xi_t$ from the macro-disciplined specification against the tightness-adjusted matching-efficiency proxy $\widehat{\log \Lambda}_t$. The vacancy yield is $q_t \equiv H_t/V_t$, computed from national monthly JOLTS hires H_t and job openings V_t; congestion is U_t/V_t, where U_t is national monthly unemployment. The elasticity $\widehat{\alpha}_c$ is taken from the preferred macro-disciplined MSM estimate with diagonal weighting (Column (3) of Table D.2). The line shows a fitted bivariate relationship; the shaded band reports robust 95% confidence intervals.

Table D.3 decomposes the fitted variation in $\log \xi_j$ into the additive components implied by the macro-disciplined specification. Each row reports the mean and cross-vacancy standard deviation of a component's contribution, evaluated at the estimated coefficients over the vacancy-level sample. Because the mapping into $\log \xi_j$ is linear in the macro covariates, the table summarizes which observed macro series shape the fitted recruiting-state shifter. It should be interpreted as a descriptive decomposition of the fitted index, not as a variance decomposition or a causal interpretation of the individual components.

D.6 Full Model Fit and Robustness

This subsection reports fit diagnostics and robustness checks for the structural estimation. Figure 5 in the main text summarizes the level-moment fit visually. The additional figures and tables below document the fit to the full targeted moment vector and the robustness of the preferred specification.

Figure D.3 plots the full set of simulated moments against their empirical targets for the preferred macro-disciplined specification and for the more flexible

Table D.3: Recruiting-State Shifter $\log \xi_j$: Component Contributions

	Macro (diag)		Macro (full)	
	S.D.	mean	S.D.	mean
Hires Rate	0.3496	-0.1738	0.3605	-0.1791
ΔHIR	0.3514	0.0593	0.3475	0.0587
Layoffs and Discharges Rate	0.5785	0.2266	0.6233	0.2442
ΔLDR	1.3009	-0.1110	1.1640	-0.0994
Quits Rate	0.3650	-0.1561	0.3883	-0.1661
ΔQUR	0.6228	0.0630	0.6379	0.0645

Note: This table reports the mean and cross-vacancy standard deviation of each macro term's contribution to the fitted recruiting-state shifter $\log \xi_j$ under the macro-disciplined specification. Hires rate, layoffs and discharges rate, and quits rate denote the national monthly components of the JOLTS-based rate series after the standardization used in estimation. The Δ rows denote state-month deviations from the national monthly component. Each contribution equals the estimated coefficient multiplied by the corresponding standardized macro regressor, evaluated at the vacancy's state and first-submission month. Standard deviations are in $\log \xi$ units. The table is a descriptive decomposition of the fitted index; it is not a variance decomposition and should not be interpreted as evidence on separate causal effects of the macro series.

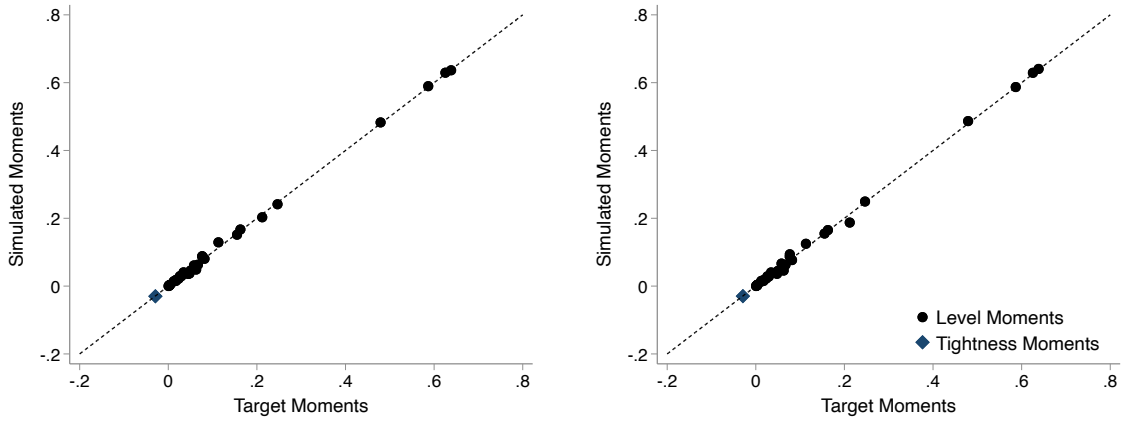
wave-specific specification. Both specifications fit the targeted moments well, indicating that the macro-disciplined recruiting-state shifter preserves the fit of the wave-group specification while using observed labor-market series to discipline variation in ξ_j.

Table D.4 lists the ten moments with the largest contributions to the diagonal MSM objective in the preferred specification. The table is diagnostic rather than substantive: it shows which pieces of the targeted moment vector are least well matched and whether misfit is concentrated in economically central moments. Because the level moments are precisely estimated, small absolute deviations can generate relatively large standardized contributions. I therefore emphasize both the absolute differences and the standardized differences.

Table D.5 documents the specification-selection path for the macro-disciplined recruiting-state shifter. The exercise is included for transparency: it shows how the preferred HIR-QUR-LDR specification was chosen within a common moment set and weighting scheme. I do not interpret the relative performance of individual JOLTS series as evidence on separate causal effects.

Finally, I assess robustness to alternative tightness moment choices. I reestimate the preferred macro-disciplined specification using the two grouped tightness-moment sets described in Appendix D.3. I then simulate the main moment vector from those estimates and compare the simulated moments to the empirical targets.

Figure D.3: Model Fitting: Target and Simulated Moments



(a) Macro (diag) (b) Wave (diag)

Notes: The panels plot simulated moments against empirical targets for the macro-disciplined and wave-specific specifications with diagonal weighting. The 45-degree line indicates perfect fit. Moments are computed using identical definitions in the data and simulation and correspond to the moment families described in Section 6.3 and Appendix D.3.

Table D.4: Top 10 Mismatched Moments in Macro-Disciplined Specification

Rank	Moment	Difference	z-score	Contribution z^2
1	Level $h = 3, d = 1$	0.0114	3.28	10.77
2	Level $h = 4, d = 3$	-0.0138	-3.08	9.48
3	Level $h = 4, d = 1$	0.0165	2.77	7.65
4	Level $h = 2, d = 3$	-0.0085	-2.28	5.18
5	Level $h = 1, d = 1$	-0.0108	-2.13	4.56
6	Level $h = 2, d = 1$	0.0105	2.11	4.45
7	Level $h = 3, d = 3$	-0.0044	-1.49	2.22
8	Level $h = 1, d = 2$	0.0061	1.48	2.18
9	Level $h = 3, d = 6$	-0.0009	-1.41	1.99
10	Level $h = 3, d = 2$	-0.0038	-1.21	1.47

Note: The table lists the ten targeted moments with the largest squared standardized residuals under the preferred macro-disciplined specification with diagonal weighting. The difference is the simulated moment minus the empirical target. The z-score divides this difference by the bootstrap standard error of the empirical moment. The contribution column reports the squared z-score, which is the contribution to the diagonal MSM objective.

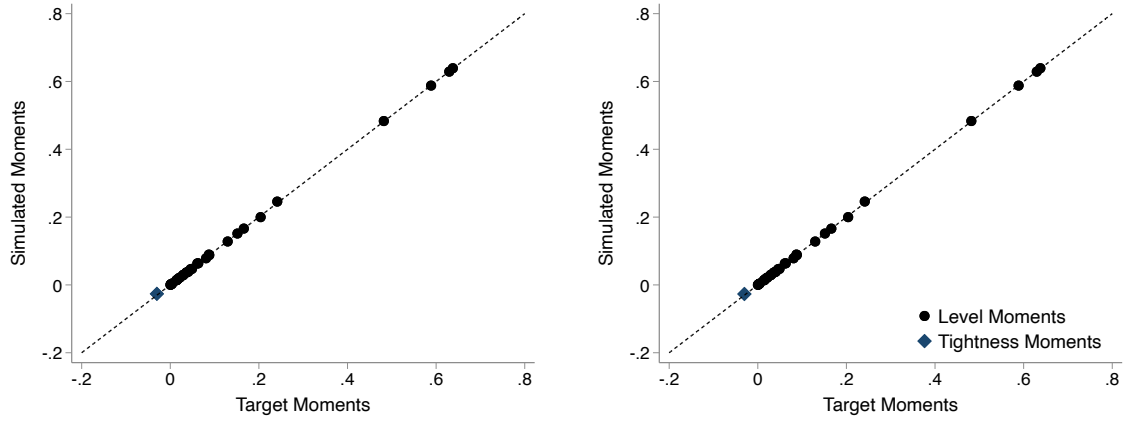
Figure D.4 shows that the fitted core moments are stable across these alternative choices.

Table D.5: Model Selection for Parameterization of ξ_j

Round 1	Hires	Quits	Job Opening	Layoffs	Total Separation
$Q_{\text{diag}}/2$	47.970	78.825	85.719	102.642	147.995
Round 2	Hires	+ Quits	+ Opening	+ Layoffs	+ Quits + Layoff
$Q_{\text{diag}}/2$		44.002	45.538	47.552	35.977
Round 3	Hires(Δ)	+ Quits(Δ)	+ Layoffs(Δ)	+ Quits(Δ) + Layoffs(Δ)	
$Q_{\text{diag}}/2$		42.198	33.531	28.192	

Note: The table reports one-half of the diagonal MSM criterion, $Q_{\text{diag}}/2$, for candidate parameterizations of the recruiting-state shifter ξ_j, holding the targeted moment vector and weighting scheme fixed. Round 1 enters each aggregate monthly JOLTS series separately. Round 2 augments the hire-rate specification with additional aggregate series. Round 3 compares hires rate + quits rate, hires rate + layoffs and discharges rate, and hires rate + quits rate + layoffs and discharges rate specifications, with each included series entering through both its aggregate monthly component and its state-month deviation. Lower values indicate better fit to the targeted moments. The preferred macro-disciplined specification uses hires rate, quits rate, and layoffs and discharges rate with state-month deviations. The table is a model-selection diagnostic and is not used to interpret the individual JOLTS series as causal drivers of contact initiation.

Figure D.4: Model Selection Robustness: Regression and Grouped Tightness Moments



(a) Alt. Tightness Moment Set 1 (b) Alt. Tightness Moment Set 2

Notes: The panels plot simulated moments against empirical targets for the main moment vector, using models reestimated with two alternative grouped tightness-moment sets. The 45-degree line indicates perfect fit. Alternative tightness-moment set 1 targets high-minus-low differences in $\mathbb{E}[\omega_j^{\text{data,none}}]$ and $\mathbb{E}[\omega_{ij}^{\text{data}} \mid d_{ij} \leq 1]$. Alternative tightness-moment set 2 adds the high-minus-low difference in $\mathbb{E}[\omega_{ij}^{\text{data}} \cdot d_{ij}]$. All high-minus-low differences are formed by comparing top and bottom tightness terciles within each wave group and averaging across wave groups using vacancy counts. The three families of moments are discussed in Section 6.3 and Appendix D.3.

E Counterfactual and Screening-Index Robustness

This appendix provides technical details and robustness checks for the counterfactual exercises in Section 6.4. Appendix E.1 defines the timing-loss measures. Appendix E.2 discusses the Shapley decomposition used in the tightness counterfactual exercises. Appendix E.3 reports additional counterfactual results and robustness to alternative screening-value indices.

E.1 Construction of Loss Measures

This subsection gives the formulas underlying the counterfactual objects in Section 6.4. Let I_j denote the eight experimental applications to vacancy j, let A_{ij} be the ML screening-value index for application i, and let $d_{ij} = D2F_i^t$ be the experimental day bin. For a parameter estimate $\widehat{\Theta}$, let $\omega_{ij}(\widehat{\Theta})$ denote the model-implied initiation weight assigned to tracer i. The model-implied initiation mass on day d is

$$P_{jd}(\widehat{\Theta}) \equiv \sum_{i \in I_j} \omega_{ij}(\widehat{\Theta}) \mathbf{1}\{d_{ij} = d\},$$

and the probability that the initiating contact is among tracers is

$$P_j^{\text{exp}}(\widehat{\Theta}) \equiv \sum_d P_{jd}(\widehat{\Theta}) = \sum_{i \in I_j} \omega_{ij}(\widehat{\Theta}).$$

For a sample of J vacancies, let $\bar{P}^{\text{exp}}(\widehat{\Theta}) \equiv J^{-1} \sum_{j=1}^{J} P_j^{\text{exp}}(\widehat{\Theta})$ denote the sample-average experimental-initiation probability.

Define the highest screening value among all experimental applications and the highest screening value available by day d as

$$A_j^{\max} = \max_{i \in I_j} A_{ij}, \qquad A_j(d) = \max_{i \in I_j : d_{ij} \le d} A_{ij}.$$

If initiation is assigned to tracer i on day d_{ij}, the screening-value gap can be decomposed as

$$A_j^{\max} - A_{ij} = \underbrace{A_j^{\max} - A_j(d_{ij})}_{\text{timing component}} + \underbrace{A_j(d_{ij}) - A_{ij}}_{\text{within-pool value gap}}.$$

The timing component is the gap between the full-pool maximum and the highest

screening value available at initiation. The within-pool value gap is the difference between the highest screening value available at initiation and the screening value of the tracer assigned initiation weight.

The Missed-Best Probability and Relative Timing Loss are computed by aggregating the corresponding application-level objects using the model-implied initiation weights $\omega_{ij}(\widehat{\Theta})$. These objects are conditional on initiation being assigned to an experimental tracer. Specifically, the *Missed-Best Probability* is

$$\text{Missed-Best Probability} = \frac{\sum_j \sum_{i \in I_j} \omega_{ij}(\widehat{\Theta}) \mathbf{1}\{A_j(d_{ij}) < A_j^{\max}\}}{\sum_j P_j^{\exp}(\widehat{\Theta})}, \tag{E.1}$$

and the *Relative Timing Loss* scales the expected timing component by the expected screening value of the experimental application receiving initiation weight:

$$\text{Relative Timing Loss} = \frac{\sum_j \sum_{i \in I_j} \omega_{ij}(\widehat{\Theta}) \left[A_j^{\max} - A_j(d_{ij})\right]}{\sum_j \sum_{i \in I_j} \omega_{ij}(\widehat{\Theta}) A_{ij}}. \tag{E.2}$$

E.2 Tightness Counterfactuals and Shapley Decomposition

This subsection describes the channel decomposition used for the tightness counterfactuals. Let $Y(\lambda, \kappa)$ denote a counterfactual target, such as $\bar{P}^{\exp}(\widehat{\Theta})$, Missed-Best Probability, or Relative Timing Loss. For a counterfactual environment x that shifts primitives from (λ_0, κ_0) to (λ_x, κ_x), define

$$Y_{00} \equiv Y(\lambda_0, \kappa_0), \qquad Y_{10} \equiv Y(\lambda_x, \kappa_0), \qquad Y_{01} \equiv Y(\lambda_0, \kappa_x), \qquad Y_{11} \equiv Y(\lambda_x, \kappa_x).$$

A one-at-a-time decomposition depends on the order in which channels are shifted. I use a two-player Shapley decomposition that averages marginal contributions across the two possible orders:

$$\phi_\lambda(Y) \equiv \frac{1}{2}\left[(Y_{10} - Y_{00}) + (Y_{11} - Y_{01})\right], \quad \phi_\kappa(Y) \equiv \frac{1}{2}\left[(Y_{01} - Y_{00}) + (Y_{11} - Y_{10})\right].$$

By construction, $\phi_\lambda(Y) + \phi_\kappa(Y) = Y_{11} - Y_{00}$. The decomposition is diagnostic: it describes how the fitted model attributes changes across the effective-arrival-intensity and shadow-cost channels under the maintained structure.

In Figure 7 and Figure E.2, I use continuous dose-response counterfactuals that trace the target as the counterfactual lever varies smoothly. Specifically, I vary

each lever x over a grid of deviations from baseline labor market tightness and recompute the total effect $Y_{11}(x) - Y_{00}$ and the channel contributions $\phi_\lambda(Y; x)$ and $\phi_\kappa(Y; x)$.

E.3 Additional Counterfactual Results and Robustness Checks

This subsection reports additional counterfactual results and robustness checks for Section 6.4. Confidence intervals are computed by re-estimating the model on bootstrap moment draws and recomputing the post-estimation screening-value objects for each draw. The bootstrap propagates estimation uncertainty in the contact-initiation model while holding the ML screening index fixed.

Table E.1 reports baseline timing losses by wave group. The estimates are stable across groups: roughly one-half of experimental initiations occur before the highest-screening-value experimental application has arrived, and Relative Timing Loss ranges from about 6 to 9 percent. Table E.2 reports selected counterfactuals from the speed-loss frontier in Figure 6.

Figure E.2 repeats the tightness decomposition in Figure 7 separately by wave group. Figure E.1 reports the within-day reallocation benchmark for the full sample and separately by wave group. Figure E.3 compares the loss measures using the preferred out-of-fold screening-value index from Section 4.3.1 with those using two alternative CatBoost specifications: frequency-based categorical encodings and text-based categorical encodings. The comparison holds the estimated contact-initiation model fixed and varies only the screening-value index used to rank experimental applications.

Within-day reallocation. Figure E.1 reports a stylized reallocation benchmark that holds each vacancy's initiation mass fixed within every submission-day bin. Full reallocation assigns that mass to the highest-screening-value experimental applications within the same bin. In the full sample, Relative Timing Loss falls from 0.0656 to 0.0620 but remains positive. The numerator of Relative Timing Loss is unchanged because the timing component is common to all experimental applications in a given vacancy-day bin. The ratio falls because reallocation raises the expected screening value in its denominator. Missed-Best Probability is also unchanged. Panel (b) shows the same qualitative pattern in every wave group.

Table E.1: Baseline Timing Losses by Wave Group

	$\bar{P}^{\exp}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Wave Group 1	0.3635	0.5352	0.0915
$N^{\text{vacancy}} = 1{,}590$	[0.3420, 0.3813]	[0.5232, 0.5505]	[0.0889, 0.0955]
Wave Group 2	0.4104	0.5155	0.0645
$N^{\text{vacancy}} = 2{,}064$	[0.3893, 0.4324]	[0.4913, 0.5319]	[0.0605, 0.0675]
Wave Group 3	0.3706	0.5108	0.0650
$N^{\text{vacancy}} = 4{,}157$	[0.3581, 0.3845]	[0.5017, 0.5200]	[0.0630, 0.0675]
Wave Group 4	0.5172	0.5119	0.0591
$N^{\text{vacancy}} = 2{,}018$	[0.4974, 0.5375]	[0.5018, 0.5255]	[0.0573, 0.0612]

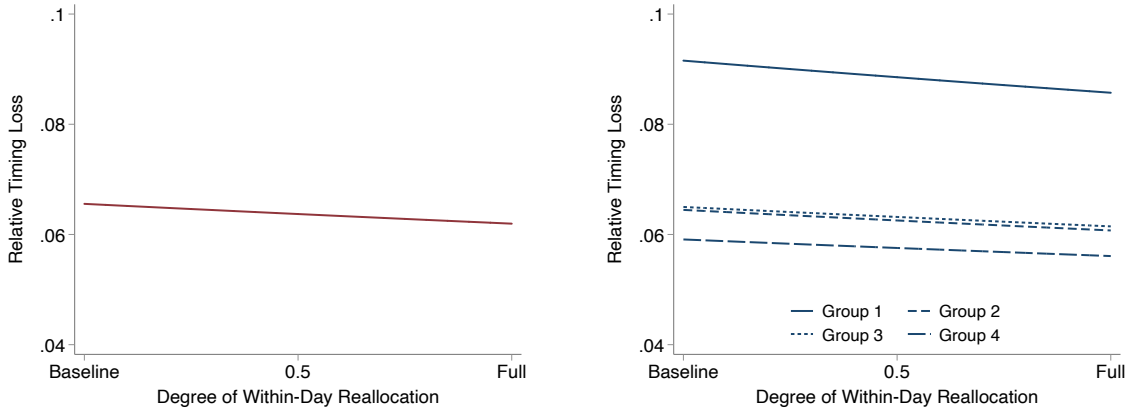
Note: The table reports the model-implied probability of experimental contact initiation and the timing-loss measures defined in Section 6.4, separately by wave group. $\bar{P}^{\exp}(\widehat{\Theta})$ is the sample average probability that initiation is assigned to one of the experimental applications. Missed-Best Probability is the probability that the highest-screening-value experimental application has not yet arrived when contact initiation occurs, defined in Equation (E.1). Relative Timing Loss refers to the timing component in Equation (6.8) scaled by the expected screening value of the application receiving initiation weight, defined in Equation (E.2). Brackets report 500-draw vacancy-clustered bootstrap 95% confidence intervals. The sample consists of vacancies with eight experimental submissions.

Table E.2: Selected Counterfactuals on the Speed-Loss Frontier

Counterfactual	Expected Initiation-Day Bin	$\bar{P}^{\exp}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	1.213	0.4079	0.5156	0.0656
$\kappa_j \times 0.5$	1.326	0.3726	0.4937	0.0620
$\kappa_j \times 2$	1.044	0.4579	0.5482	0.0711
Mean Log Tightness to p_{10}	1.248	0.3754	0.5081	0.0643
Mean Log Tightness to p_{90}	1.182	0.4359	0.5221	0.0667

Note: This table reports selected counterfactuals from Figure 6, computed on vacancies with eight experimental applications under the preferred specification. $\bar{P}^{\exp}(\widehat{\Theta})$ is the sample-average probability that initiation is assigned to an experimental application. Expected initiation timing and both loss measures condition on experimental initiation; the final day bin is top-coded at seven. The loss measures are defined in Appendix E.1. The κ_j counterfactuals multiply each vacancy's shadow cost of delay by the indicated factor. The tightness counterfactuals add the same constant to every vacancy's log tightness so that the sample mean equals the 10th or 90th percentile of the baseline vacancy-level log-tightness distribution. Each counterfactual recomputes the contact rule and initiation weights while holding experimental application schedules and the screening-value index fixed. Entries are point estimates; confidence intervals are not reported.

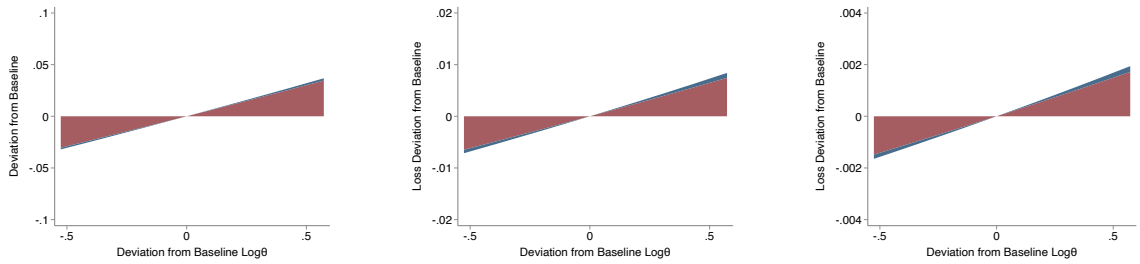
Figure E.1: Timing Loss under Within-Day Reallocation



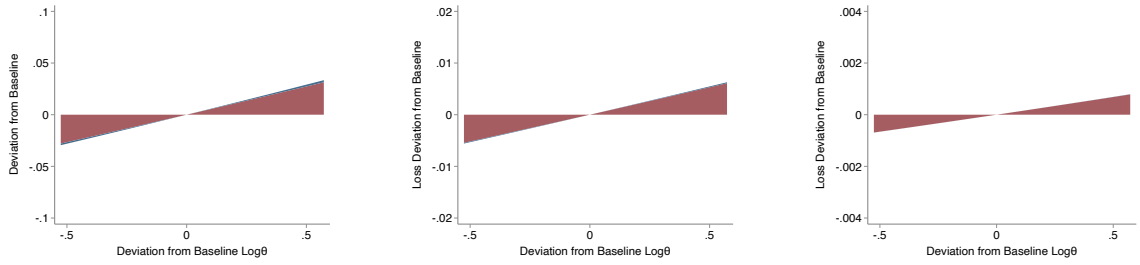
(a) Full Sample (b) By Wave Group

Notes: The figure plots Relative Timing Loss against the degree of within-day reallocation, $\alpha \in [0, 1]$, for the full sample in Panel (a) and separately by wave group in Panel (b). Application-level initiation weights are $\omega_{ij}(\alpha) = (1-\alpha)\omega_{ij}^{\text{base}} + \alpha\omega_{ij}^{\text{top}}$, where $\omega_{ij}^{\text{base}}$ denotes the baseline weight. The weights ω_{ij}^{top} preserve each vacancy-day bin's total baseline initiation mass and assign that mass equally among applications tied for the highest screening-value index within the same submission-day bin. "Baseline" corresponds to $\alpha = 0$ and "Full" to $\alpha = 1$. The experimental-initiation probability and initiation-day distribution remain fixed. Relative Timing Loss is defined in Equation (E.2). Wave Group 3 pools Waves 3 and 4; Wave Group 4 corresponds to Wave 5. The sample consists of vacancies with eight experimental applications under the preferred specification.

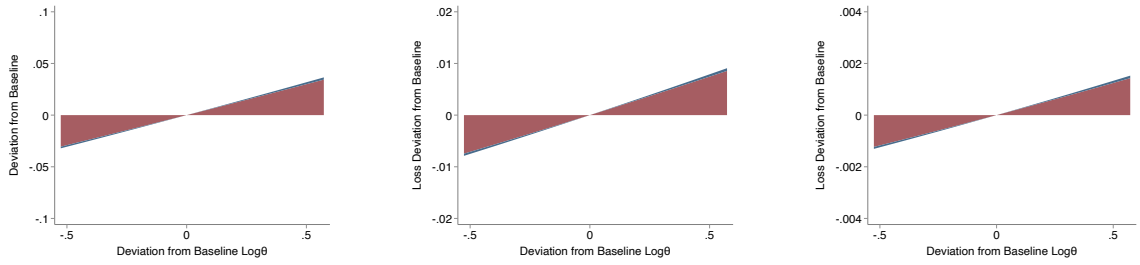
Figure E.2: Labor Market Tightness and Timing Losses by Wave Group



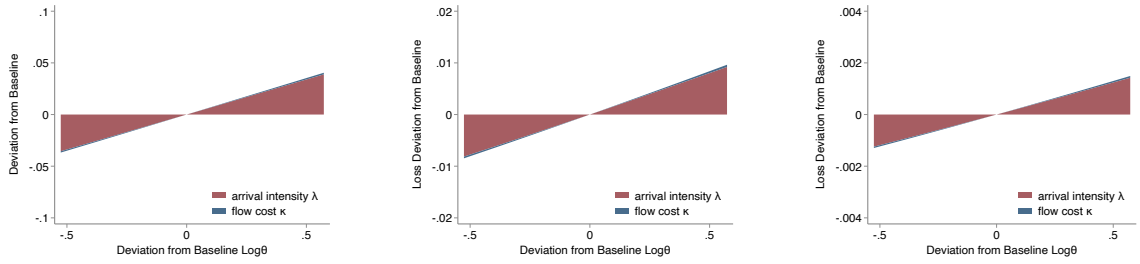
(a) $P_j^{\exp}(\widehat{\Theta})$, Wave Group 1 (b) MBP, Wave Group 1 (c) RTL, Wave Group 1



(d) $P_j^{\exp}(\widehat{\Theta})$, Wave Group 2 (e) MBP, Wave Group 2 (f) RTL, Wave Group 2



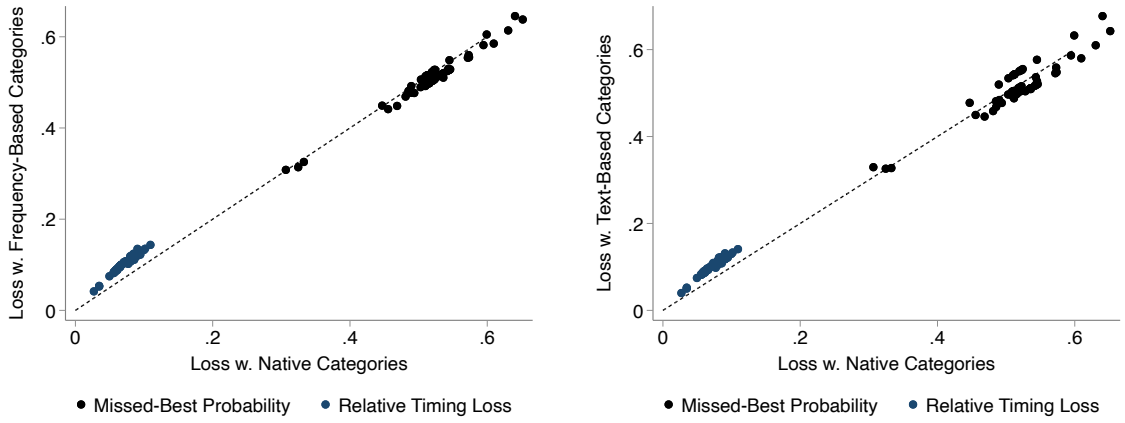
(g) $P_j^{\exp}(\widehat{\Theta})$, Wave Group 3 (h) MBP, Wave Group 3 (i) RTL, Wave Group 3



(j) $P_j^{\exp}(\widehat{\Theta})$, Wave Group 4 (k) MBP, Wave Group 4 (l) RTL, Wave Group 4

Notes: The figure repeats the Shapley decomposition in Figure 7 separately by wave group. The left column reports changes in the probability that initiation is assigned to an experimental application. The middle column reports changes in Missed-Best Probability. The right column reports changes in Relative Timing Loss. Wave Group 3 pools Waves 3 and 4 of the experiment. All panels use vacancies with eight experimental applications.

Figure E.3: Loss-Measure Robustness across Screening-Value Indices



(a) Frequency-based categories (b) Text-based categories

Notes: The figure compares counterfactual loss measures computed using the preferred screening-value index with loss measures computed using alternative out-of-fold CatBoost indices. Panel (a) uses a CatBoost model with frequency-based categorical encodings. Panel (b) uses a CatBoost model with text-based categorical encodings. The contact-initiation model and initiation weights are held fixed. Loss measures are defined in Section 6.4 and Appendix E.1. The ML models are described in Appendix B.

F Model-Guided Reduced-Form Heterogeneity

This appendix reports additional reduced-form heterogeneity in the daily early application advantage. These results are not used to estimate the structural model. Instead, they serve as descriptive external-validity checks for the model's interpretation of the timing gradient. In the model, the strength of the early application advantage depends on two broad forces: the effective arrival intensity of additional applicant opportunities, λ, and the shadow cost of delaying contact, κ. The covariates studied here proxy for environments that may shift these forces, including applicant queue thickness, offer competitiveness, evaluation costs, and firm recruiting organization.

For application i submitted to vacancy $j(i)$, I estimate

$$Y_i = \alpha_0 + \delta D2L_i^t + \phi \left(D2L_i^t \times Z_i \right) + \boldsymbol{\gamma} \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i, \tag{F.1}$$

where Y_i is an indicator for employer contact, $D2L_i^t$ is the number of days between application i's submission date and the last experimental submission to the vacancy, top-coded at seven, and Z_i is the market-, firm-, or vacancy-level covariate of interest. As in Specification (3.1), larger values of $D2L_i^t$ correspond to earlier experimental submissions, so a more positive coefficient on $D2L_i^t \times Z_i$ indicates a stronger early application advantage in environments with higher Z_i. All specifications include vacancy fixed effects and randomized resume controls, with standard errors clustered at the vacancy level. When a pre-period control is used, the figure notes describe it explicitly. For figures that report multiple covariates together, continuous covariates are standardized and indicator variables are left in levels.

F.1 Market Structure

Figure F.1 relates the daily early application advantage to local posting-market concentration. Concentration is measured using Herfindahl-Hirschman indices constructed from Revelio Labs job posting data at the state and state-by-industry levels. Higher concentration is associated with a smaller timing gradient, especially in the more granular state-by-industry measures. One interpretation is that concentrated posting markets reduce the competitive pressure firms face to contact quickly, lowering the shadow cost of delay. This interpretation is consistent

with work using posting data to measure local labor-market concentration (Azar, Marinescu and Steinbaum, 2022) and with recent evidence linking labor market concentration to firm hiring strategies (Dodini and Willén, 2025; Schubert et al., 2025). I treat the evidence as descriptive, since concentration may also be correlated with firm composition, job types, and local applicant behavior.

Figure F.3 examines a related queue-thickness proxy: the firm's share of postings in the relevant local market-month. I use posting share as a descriptive measure of firm prominence in the local recruiting market. Prominent employers may face thicker applicant queues, increasing the option value of waiting and reducing the early application advantage. Because posting-share coverage is incomplete, Figure F.2 compares the geographic and time-series coverage of the posting-share sample to the full experimental sample. The relationship between posting share and the timing gradient is negative outside the immediate post-pandemic recovery and close to zero during Waves 2-4, suggesting that the relevance of firm prominence varies with broader recruiting conditions. The results are similar when posting shares are defined at the state-month and state-by-industry-month levels. These patterns are consistent with directed application behavior (Cai, Gautier and Wolthoff, 2025) and granular search models (Jarosch, Nimczik and Sorkin, 2024), but I interpret them only as descriptive evidence on queue thickness.

Taken together, the market structure results suggest that concentration and firm prominence may operate through different margins. Concentration is more naturally linked to the shadow cost of delay, by changing the competitive pressure firms face when deciding how quickly to contact. Firm posting share is more naturally linked to effective arrival intensity, by proxying for the thickness of the applicant queue facing prominent employers. Due to data limitations, I leave a fuller analysis of market structure and recruiting timing for future work.

F.2 Offer Competitiveness

Figure F.4 relates the timing gradient to posting-based measures of offer competitiveness. Using Revelio Labs job posting data, I construct "markdown" measures for salary and posting duration as log differences between the local-market average for comparable entry-level jobs (O*NET Job Zones 1–2) and the firm's average posting characteristic. For remote-friendliness, I use the market-minus-firm differ-

ence in posting shares. Larger salary shortfalls are associated with a steeper early application advantage, while the other shortfall measures are weaker. A natural interpretation is that less competitive offers attract thinner or less stable inflows of qualified applicants, increasing the value of early arrivals. These correlations may also reflect broader differences in firm wage-setting power, amenities, or vacancy quality. They should be read as descriptive evidence on offer competitiveness. I leave identifying the causal effects of compensation or amenities on recruiting timing for future work.

F.3 Recruiting Environment Correlates

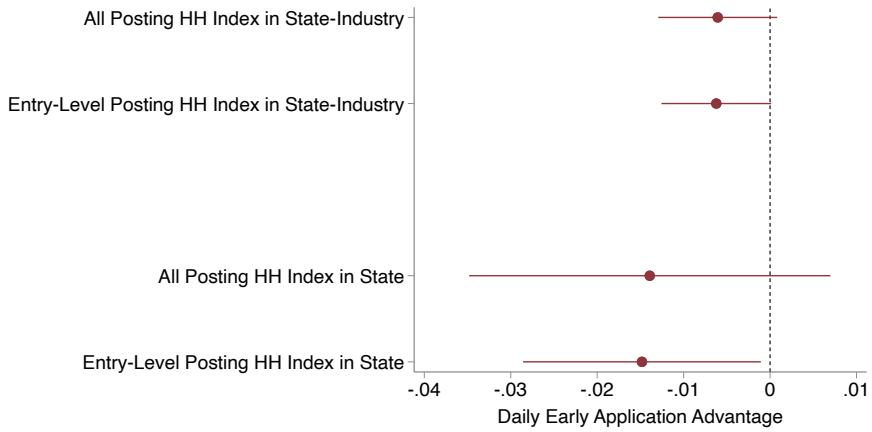
Figure F.5 summarizes additional correlates motivated by the λ-versus-κ interpretation. The figure reports three specifications: univariate regressions, a market-only multivariate specification that jointly includes the market covariates, and a full multivariate specification that includes all covariates on a common sample. Continuous covariates are standardized, so coefficients are comparable across variables, whereas indicator variables are left in levels.

The market covariates proxy for applicant-pool thickness and evaluation complexity. Timing gradients are smaller in thicker or more deliberative environments, such as industries with higher employment and higher analytical-skill intensity. The firm covariates proxy for recruiting organization, capacity, and vacancy stakes. Timing gradients are steeper for more centralized recruiting systems and for some measures associated with higher urgency, while they are weaker for larger or more capital-intensive firms. These patterns provide descriptive support for the interpretation that the early application advantage varies systematically with the recruiting environment rather than being a fixed feature of the experimental design.

Overall, the results are consistent with the mechanism emphasized in the main text. Timing gradients are weaker in environments consistent with thicker applicant queues, greater recruiting slack, or more deliberative evaluation, and stronger in environments consistent with thinner queues, weaker offer competitiveness, or greater organizational urgency. Because the covariates are not experimentally assigned and are measured with varying degrees of coverage, I use these results only as descriptive external-validity evidence. They complement the structural model by showing that the early application advantage varies in directions suggested by

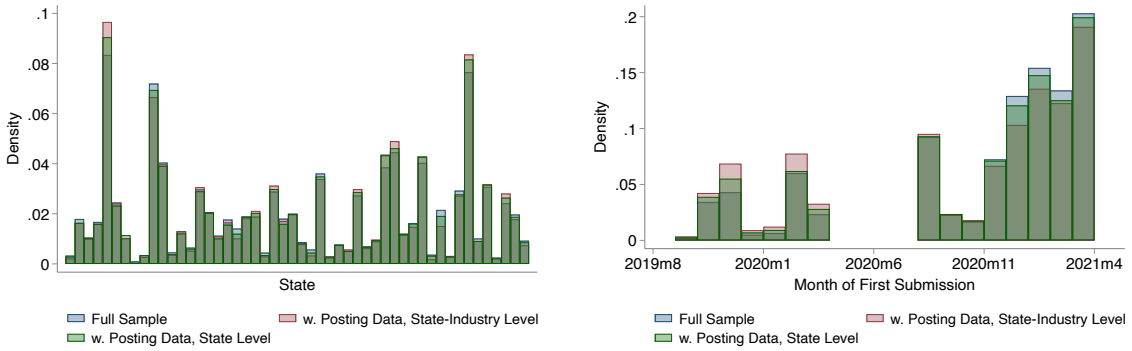
the effective-arrival-intensity and shadow-cost-of-delay channels.

Figure F.1: Daily Early Application Advantage and Market Concentration



Notes: The figure plots heterogeneity in the daily early application advantage by local posting-market concentration. Concentration is measured by the Herfindahl-Hirschman index of postings constructed from Revelio Labs data at the state and state-by-industry levels. The regressions follow Equation (F.1), include vacancy fixed effects and randomized resume controls, and cluster standard errors at the vacancy level. Error bars denote 95% confidence intervals.

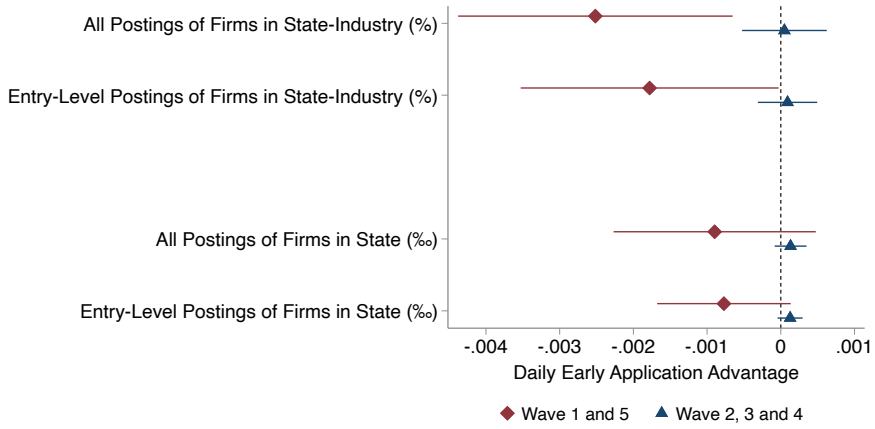
Figure F.2: Coverage of Firm Posting-Share Measures



(a) Across States (b) By First Submission Month

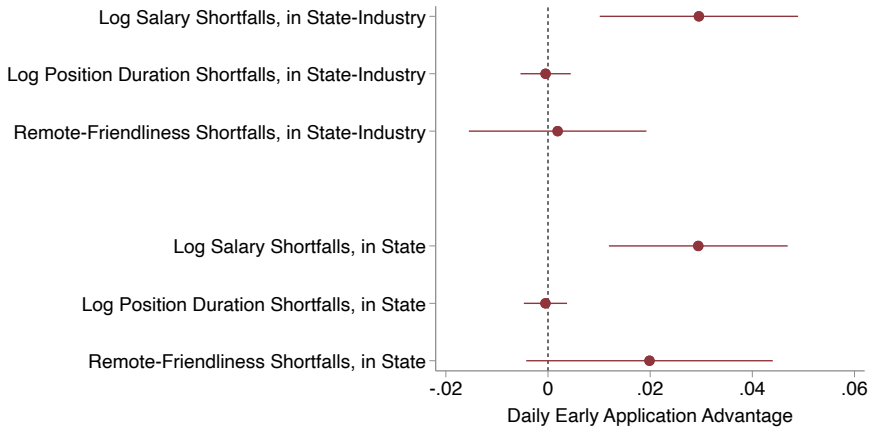
Notes: The figures show whether nonmissing posting-share coverage is systematically concentrated in particular states or time periods. This figure compares the distribution of vacancies in the full experimental sample to the subsamples for which Revelio-based firm posting shares are observed. "Posting Data, State Level" restricts to vacancies for which the firm posting share is defined at the state-month level. "Posting Data, State-Industry Level" restricts to vacancies for which the posting share is defined at the state-industry-month level. Posting-share measures are missing when the relevant market-month posting universe is empty or firm-market posting coverage is unavailable.

Figure F.3: Recruiting Prominence and the Daily Early Application Advantage



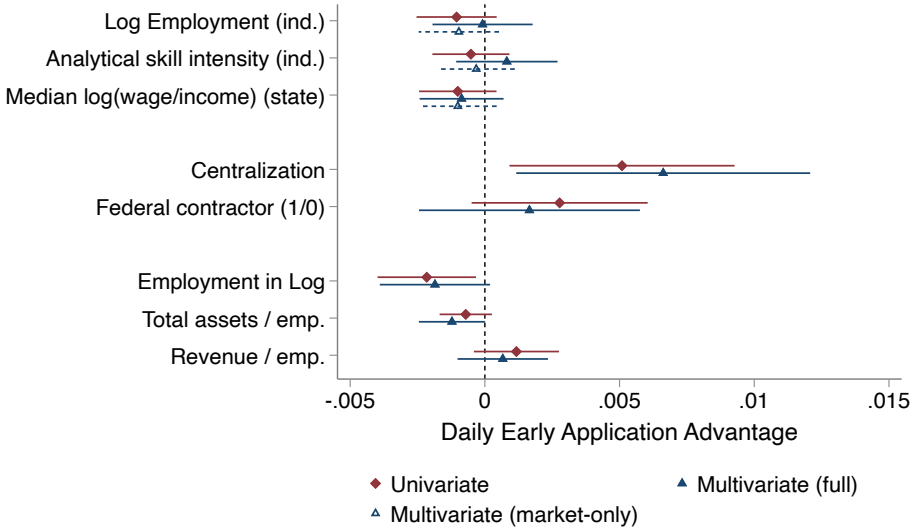
Notes: This figure plots estimates of how the daily early application advantage varies with firm posting shares constructed from Revelio Labs job postings. Each marker reports the coefficient on $D2L_i^t \times Share_{m,t}$ in a variant of Equation (F.1), where $Share_{m,t}$ is the firm's share of postings in the indicated market and month of first submission. "Entry-level" restricts postings to O*NET Job Zones 1-2. Diamonds report estimates for Waves 1 and 5; triangles report estimates for Waves 2-4. All specifications include vacancy fixed effects and the full set of randomized resume controls. Error bars denote 95% confidence intervals based on vacancy-clustered standard errors.

Figure F.4: Daily Early Application Advantage and Offer Competitiveness



Notes: The figure reports heterogeneity in the daily early application advantage by posting-based offer competitiveness measures. The salary and posting-duration measures are log differences between the local-market average and the firm's average posting feature; the remote-friendliness measure is the market-minus-firm difference in posting shares. Measures are constructed from Revelio Labs job posting data. Regressions include vacancy fixed effects and resume controls; standard errors are clustered at the vacancy level. Error bars denote 95% confidence intervals.

Figure F.5: Daily Early Application Advantage and Additional Model-Guided Correlates



Notes: The figure reports heterogeneity in the daily early application advantage by additional market and firm covariates motivated by the contact-initiation model. Continuous covariates are standardized; indicator variables are left in levels. The state income covariate is the log wage-to-income ratio, defined as the log wage-income proxy minus log median total income. The figure reports univariate estimates, a market-only multivariate specification, and a full multivariate specification estimated on a common sample. All regressions include vacancy fixed effects and resume controls; standard errors are clustered at the vacancy level. Error bars denote 95% confidence intervals.

G Complementary Data and Variable Construction

This appendix documents the industry-, state-, and firm-level covariates introduced in Section 2.3 and used in the descriptive heterogeneity analyses. Unless otherwise stated, each sampled vacancy is assigned the value corresponding to its industry, state, or firm in the month of the first experimental submission. Pre-period measures are constructed from months or years preceding that date. Continuous covariates marked *(std.)* are standardized to have mean zero and standard deviation one in the analysis; summary tables report the underlying unstandardized values.

G.1 Industry-Level Covariates

Industry classification. Industries are defined using two-digit North American Industry Classification System (NAICS) codes. The data released by Kline, Rose and Walters (2022) report two-digit Standard Industrial Classification (SIC) codes. For firms covered by Compustat, I use the reported six-digit NAICS code and aggregate it to the two-digit level. For firms without Compustat coverage, I map SIC-2 codes to NAICS-2 industries using the 2017 NAICS–SIC crosswalk.

Industry task measures. Following Kline, Rose and Walters (2022), I combine occupation-level identifiers from the Occupational Information Network (O*NET) database with projected occupation-by-industry employment weights from the original replication materials. I first construct occupation-level indices and then aggregate them to the NAICS-2 level using projected employment as weights. To better match the entry-level vacancies in the experiment, I restrict the aggregation to occupations in O*NET Job Zones 1 and 2, which typically require little or some preparation.

I construct five time-invariant indices from the corresponding O*NET descriptors:

- **Analytical**: the average of mathematical reasoning, mathematics knowledge, and mathematics skill.
- **Customer interaction**: the average of establishing and maintaining interpersonal relationships, working directly with the public, customer and personal service, and contact with others.

- **Social**: the average of social perceptiveness, coordination, persuasion, and negotiation.
- **Routine**: the average of degree of automation and importance of repeating the same tasks.
- **Service**: the average of assisting and caring for others and service orientation.

Time-varying industry conditions. I use two additional sets of industry-level series. Seasonally adjusted monthly employment comes from the Bureau of Labor Statistics (BLS) Current Employment Statistics. Quarterly indices of total compensation and wages and salaries for private-industry workers come from the BLS Employment Cost Index. I assign each quarterly observation to the three months in that quarter. For each vacancy, I construct three- and six-month averages ending before the month of the first experimental submission. Employment enters the analysis in logs.

G.2 State-Level Covariates

The state-level covariates capture pre-experiment income, labor-market institutions and demographic composition, and monthly recruiting conditions.

Income composition. I use annual state-level measures from the Income Distributions and Dynamics in America (IDDA).[33] I average each measure over 2015-2019 to obtain a pre-experiment state characteristic.

The income measures are the log of median total income and the log of a constructed wage-income proxy equal to median total income less median non-wage income.[34]

Minimum wages and state employment composition. I use the U.S. Department of Labor history of state minimum wages for nonfarm employment. When a state

[33]The variables are constructed from household income and earnings records based on Form 1040. See Kondo et al. (2023) for details.

[34]Because the difference between two medians need not equal the median of the individual-level difference, this variable should be interpreted as a state-level wage-income proxy rather than a literal median wage.

Table G.1: Summary of State-Level Income Covariates

	Mean	SD	Min	Max
Income and Wage				
Median Total Income (log)	11.3232	0.1630	11.0154	11.6375
Wage-Income Proxy (log)	11.2806	0.1557	10.9964	11.5965

Note: The table reports 2015–2019 averages for the 51 state-level units represented in the experiment. Both measures are constructed from IDDA Form 1040 data.

does not set its own minimum wage, I assign the federal minimum wage. Each vacancy is assigned the applicable minimum wage in its state and month of first experimental submission.

I use the BLS Expanded State Employment Status tables to construct the Black and female shares of employment in each state. Each vacancy is assigned the corresponding share measured in the year before the first experimental submission.

State-month labor-market conditions. I combine seasonally adjusted state series from the Job Openings and Labor Turnover Survey (JOLTS) with state unemployment from the Local Area Unemployment Statistics (LAUS). The variables are:

- **Job-opening rate**: job openings on the last business day of the month divided by employment plus job openings.
- **Hires rate**: additions to payroll during the month divided by employment.
- **Total-separation rate**: separations from payroll during the month divided by employment.
- **Quits rate**: voluntary separations other than retirements and transfers, divided by employment.
- **Layoffs and discharges rate**: employer-initiated involuntary separations divided by employment.
- **Tightness**: the number of job openings divided by the number of unemployed workers.

For each vacancy, I assign the value for the vacancy's state and the month of the first experimental submission. I also construct three-month pre-submission averages when they are used as controls. The JOLTS and unemployment series are seasonally adjusted. Details of the state-level JOLTS estimation procedure are available from the BLS Handbook of Methods.

G.3 Firm-Level Covariates

For the 108 sampled firms, I assemble measures of job-posting market structure, offer competitiveness, human-resource organization, and financial characteristics. The main proprietary source is Revelio Labs (Revelio Labs, 2026), which aggregates publicly available information from professional profiles and job postings.[35]

Firm crosswalk and posting coverage. Firm names are available in the replication materials for Kline, Rose and Walters (2024). I manually collect stock-market tickers for publicly traded firms and match the experimental firms to Revelio identifiers in two steps: exact ticker matching where possible and fuzzy name matching otherwise. I cross-check the two procedures and manually audit uncertain matches to produce a final firm-level crosswalk.

Experimental firms that cannot be matched to Revelio are coded as missing for posting-based measures. Among matched firms, a firm-market-month with no observed posting is assigned a posting share of zero. Market-months with no postings in the Revelio universe are dropped. Appendix Figure F.2 documents geographic and time-series coverage of the posting-share measures.

Posting-market structure and firm prominence. Using Revelio job postings, I define local markets at the state-month and state-by-NAICS-2-industry-month levels. Let s_{fmt} denote firm f's share of postings in market m and month t. I construct:

- **Firm posting share**: s_{fmt}, measured either in the state-month market or the state-by-industry-month market.
- **Posting concentration**: $HHI_{mt} = \sum_f s_{fmt}^2$, measured as the HH (Herfindahl-Hirschman) index of posting shares within the market.

For each measure, I also construct an entry-level version that restricts the posting universe to O*NET Job Zones 1 and 2. Each sampled vacancy is assigned the value for its state, industry where applicable, and month of first experimental submission. I also construct three-month pre-submission averages when required by a specification.

[35] Revelio Labs, Data Dictionary documentation (accessed February 24, 2026): https://www.data-dictionary.reveliolabs.com/data.html.

Offer competitiveness. I construct posting-based measures of firms' advertised offers relative to comparable entry-level postings in the same local market. For each feature x, the level gap is

$$G^x_{fmt} = \bar{x}_{mt} - \bar{x}_{fmt},$$

where $\bar{x}_{mt}$ and $\bar{x}_{fmt}$ denote the local-market and firm averages. For advertised salary and posting duration, the regressions use $\log \bar{x}_{mt} - \log \bar{x}_{fmt}$ to reduce right skewness. For remote-friendliness, I use the market-minus-firm difference in posting shares. Positive salary and remote-friendliness gaps indicate less competitive advertised offers, while the posting-duration gap captures relative vacancy persistence. Comparable postings are restricted to O*NET Job Zones 1 and 2, and local markets follow the state-month or state-by-industry-month definition used in the corresponding specification.

Human-resource organization. I use three firm-level measures:

- 1{**federal contractor**}: an indicator for federal-contractor status.
- **Labor-law violations per employee** *(std.)*: the number of violations related to employment recorded since 2000 in the Good Jobs First Violation Tracker, divided by Compustat employment.
- **Callback centralization**: following Kline, Rose and Walters (2022), the underlying centralization index is measured as the number of distinct phone numbers used by the firm to contact applicants divided by the total number of vacancies with at least one callback times -1. It is used without modification.

Financial characteristics. Following Kline, Rose and Walters (2022), I use annual Compustat data averaged over 2010-2018:

- **Log employment** *(std.)*: the natural log of average employment.
- **Total assets per employee** *(std.)*: total assets divided by employment.
- **Gross profit per employee** *(std.)*: gross profit divided by employment.
- **Total revenue per employee** *(std.)*: total revenue divided by employment.

Firms without Compustat coverage are omitted from specifications using these variables.

Table G.2: Summary of Firm-Level Covariates

	Mean	SD	Median	No. (firms)
Human resource management				
$\mathbf{1}$(federal contractor)	0.6667	0.4736	1.0000	108
Violations per emp.	0.0004	0.0007	0.0002	94
Callback centralization	-1.1204	0.3685	-1.0842	100
Financial performance				
log(employment)	11.2432	1.0384	11.0745	94
Total assets per emp. ($M)	0.8484	2.1732	0.1979	94
Gross profit per emp. ($M)	0.1042	0.0806	0.0819	94
Total revenue per emp. ($M)	0.3563	0.3828	0.2613	94

Note: The table reports unstandardized firm-level covariates. "No. firms" is the number of experimental firms with non-missing values. Callback centralization is defined so that higher values indicate that a smaller set of phone numbers is used across contacted vacancies. Continuous variables marked *(std.)* in the text are standardized only when used in regressions.